THERMODYNAMICS AND CHEMICAL KINETICS FOR COMBUSTION

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Chapter 1

Thermodynamics

1.1 Introduction

Thermodynamics describes the state of a system and its most favorable evolution. Thermodynamics enables to predict the final state (temperature, pressure, composition) of a system, given an initial state. The path followed by the system between the initial and the final states as well as the rate at which the evolution is taking place are not described by thermodynamics. The thermodynamic description of the system can be made using a number of thermodynamic properties, such as the enthalpy and the entropy, which describe the macroscopic states of the system [1]. Thermodynamics is of primary importance for combustion science because it characterizes the energy content of reactive mixtures, the energy released by individual reactions, and the direction of reactions.

This first chapter corresponds to a short review of the basic concepts of thermodynamics. It recalls the most important definitions, laws, and quantities of thermodynamics. A number of examples and applications of thermodynamics to combustion are also given. The description of the use of thermodynamics in combustion modeling is briefly introduced.

1.2 State of matter and equations of state of gas

1.2.1 State of matter

As a first approximation, it is possible to define four different states of matter :

— Solid

— Liquid

— Gas

— Plasma

These different states differ from each other by some physical properties such as degree of freedom of the individual molecules, degree of organization, density, structure, etc

In the present course, we focus on gases because the very large majority of chemical reactions that take place in combustion processes happen in the gas phase.

A gas is characterized by the large separation between the molecules and the limited impact of intermolecular forces on the path of the molecules between collisions. The molecules of a gas always have translational motion. A particular case is the gas at rest for which the mean velocity is zero.

A gas can be characterized by its

— Pressure

— Temperature

— Volume

Pressure corresponds to the force exerted by the gas molecules on the wall of a container. It has unit of Pascal which has the following equivalence [2] :

$$1\,Pa = 1\,Nm^{-2} = 1\,kg.m^{-1}.s^{-2}. \tag{1.1}$$

The temperature corresponds to the property that indicates the direction of the energy flow through a thermally conducting, rigid wall [2]. It has unit of Kelvin, degree Celsius, or degree Fahrenheit. It is noted that Fahrenheit is usually not used in the scientific field. The

following correspondence applies

$$T\,in\,K = T\,in\,C^0 + 273.15. \tag{1.2}$$

1.2.2 Perfect gas

If the intermolecular forces are neglected, the gas is referred to as "perfect gas" and the relationship between pressure (P), temperature (T), and volume (V) is given by the perfect gas law (or equation of state for perfect gas)

$$PV = nRT, \tag{1.3}$$

where n is the number of mole ; and R is the perfect gas constant : R=8.314 $JK^{-1}mol^{-1}$.

For a mixture of gases, Dalton's law applies [2]

"*The pressure exerted by a mixture of gases is the sum of the pressures that each one would exert if it occupied the container alone.*"

This law can be expressed mathematically as

$$P = \sum_{i=1}^{N} P_i, \tag{1.4}$$

where P_i is the partial pressure of gas i ; and N is the number of gases in the mixture.

The mole fraction, x_i, is given by

$$x_i = \frac{n_i}{\sum_{i=1}^{N} n_i}, \tag{1.5}$$

while the partial pressure of a gas in a mixture is given by

$$P_i = x_i P. \tag{1.6}$$

1.2.3 Real gas

The perfect gas law describes well the behavior of gas at low pressure. This is because at low pressure, the distance between molecule is large since the gas molecules occupy all

the volume available to them. However, as the pressure increases, the distance between the gas molecules becomes smaller and the interaction between the molecules cannot be neglected anymore. Figure 1.1 shows the evolution of the potential energy as a function of the separation distance between two molecules. At intermediate pressure, the distance between the molecules may become on the order of few molecular diameters. Under such conditions, the long-range intermolecular forces are mostly attractive and make the gas more compressible than a perfect gas. At high pressure, the distance between the molecules may become shorter than the molecular diameter. Under such conditions, the short-range intermolecular forces are mostly repulsive and make the gas less compressible than a perfect gas.

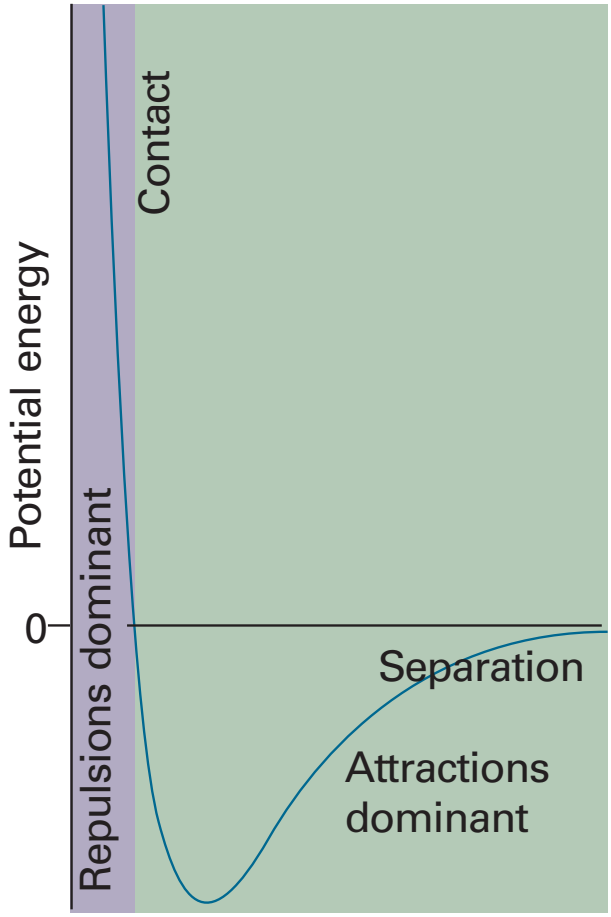


Figure 1.1: Evolution of the potential energy as a function of the separation distance between two molecules. Taken from [2].

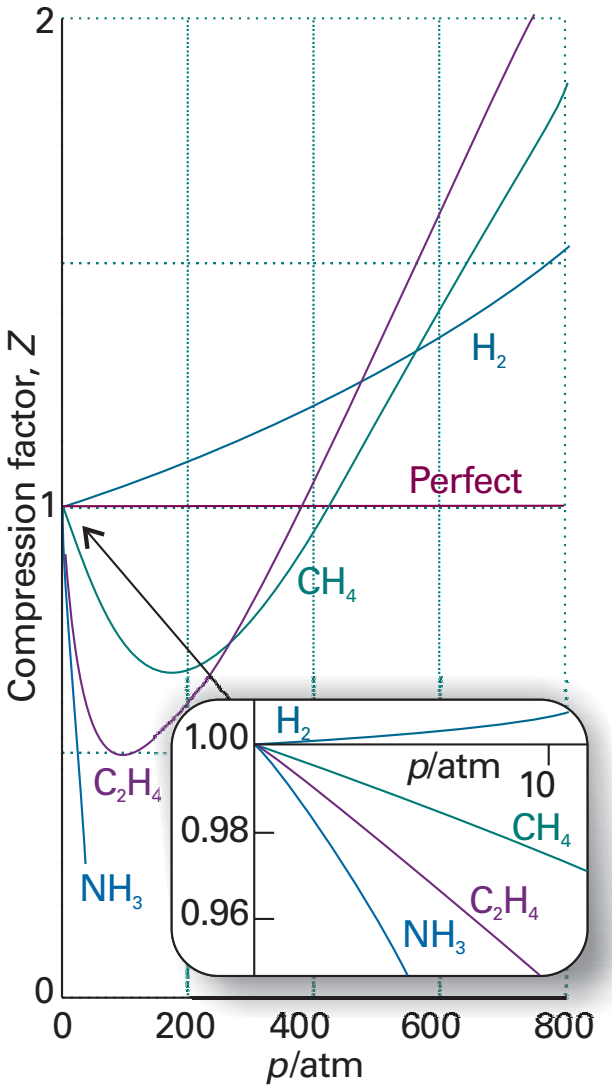


Figure 1.2: Evolution of the compression factor of different gases as a function of pressure at 273.15 K.. Taken from [2].

Since the compressibility, that is the propensity of a gas to be compressed, of a real gas is different from the one of a perfect gas, it is useful to define a compressibility factor or compression factor (Z) as

$$Z = \frac{V_m}{V_m^0} = \frac{RT}{PV_m^0}, \tag{1.7}$$

where V_m is the molar volume given by

$$V_m = \frac{V}{n}; \tag{1.8}$$

and V_m^0 is the molar volume of a perfect gas under similar pressure and temperature.

Figure 1.2 shows the evolution of the compression factor as a function of pressure for several gases. The value of Z as well as its evolution with pressure depends on which gas is considered. It is noted that ammonia (NH_3) demonstrates strong departure form the perfect gas behavior even at relatively low pressure ($< 1MPa$).

In order to account for the effect of intermolecular forces at intermediate and high pressure, it is required to modify the equation of state for perfect gas.
The repulsive interactions may be taken into account by considering the molecules as impenetrable spheres. Consequently, the volume available for the molecules to move in a given volume is smaller because part of this volume is occupied by the other molecules.
The attractive forces can be taken into account by considering that the pressure is reduced since the attractive forces counter-act collision at the wall of the container by reducing their frequency and the intensity (or force) of the collisions.
Mathematically, a state equation for real gas can be expressed as

$$P = \underbrace{\frac{RT}{V_m - b}}_{\text{Repulsion correction}} - \underbrace{\frac{a}{V_m^2}}_{\text{Attraction correction}} \quad ; \tag{1.9}$$

where a and b are constants which account for the attractive and the repulsive intermolecular forces, respectively. They are specific of each gas.

Equation 1.9 corresponds to the so-called "van der Waals equation". Although it includes the physics of the behavior of real gas through the introduction of the two empirical parameters a and b, the VDW equation is too simplistic to enable a quantitatively description of the real gas behavior. In combustion science, more complex equations of state are employed when dealing with real gas. One of them is the Soave-Redlich-Kwong equation of state [3]

$$P = \frac{RT}{V_m - b_c} - \frac{a_c \alpha}{V_m \left(V_m + b_c\right)}, \tag{1.10}$$

with

$$a_c = 0.42747 \frac{R^2 T_c^2}{P_c}, \tag{1.11}$$

$$\alpha = \left(1 + \left(0.480 + 1.574\omega - 0.176\omega^2\right)\left(1 - \sqrt{T_r}\right)\right)^2, \tag{1.12}$$

and

$$b_c = 0.08664 \frac{RT_c}{P_c}, \tag{1.13}$$

where T_c and P_c are the critical temperature and pressure ; ω=-log $(P_r^{sat}) - 1$ at T_r=0.7 is the acentric factor ; P_r^{sat}=P^{sat}/P_c is the reduced saturation vapor pressure ; and T_r=T/T_c is the reduced temperature.

Note that for a mixture of real gases, the following mixing rules apply [4]

$$a_c = \sum_i \sum_j x_i x_j \sqrt{a_{c,i} a_{c,j}}, \tag{1.14}$$

and

$$b_c = \sum_i x_i b_{c,j}. \tag{1.15}$$

A second widely used equation of state for real gas is the Peng-Robinson equation [5]

$$P = \frac{RT}{V_m - b_c} - \frac{a_c \alpha}{V_m \left(V_m + b_c\right) + b_c \left(V_m - b_c\right)}, \tag{1.16}$$

with

$$a_c = 0.457235 \frac{R^2 T_c^2}{P_c}, \tag{1.17}$$

$$\alpha = \left(1 + \left(0.37464 + 1.54226\omega - 0.26992\omega^2\right)\left(1 - \sqrt{T_r}\right)\right)^2, \tag{1.18}$$

and

$$b_c = 0.077796 \frac{RT_c}{P_c}. \tag{1.19}$$

Note that the mixing rules given in Equation 1.14 and Equation 1.15 also apply for the Peng-Robinson equation of state.

It is noted that accounting for real gas effects is not limited to the modification of the equation of state. It also requires to modify other thermodynamic properties such as the enthalpy and the entropy.

1.3 Thermodynamic systems

1.3.1 General definition

A thermodynamic system corresponds to the part of the Universe where the transformation is taking place. It is thus possible to distinguish the system and its surrounding or environment as illustrated in Figure 1.3.

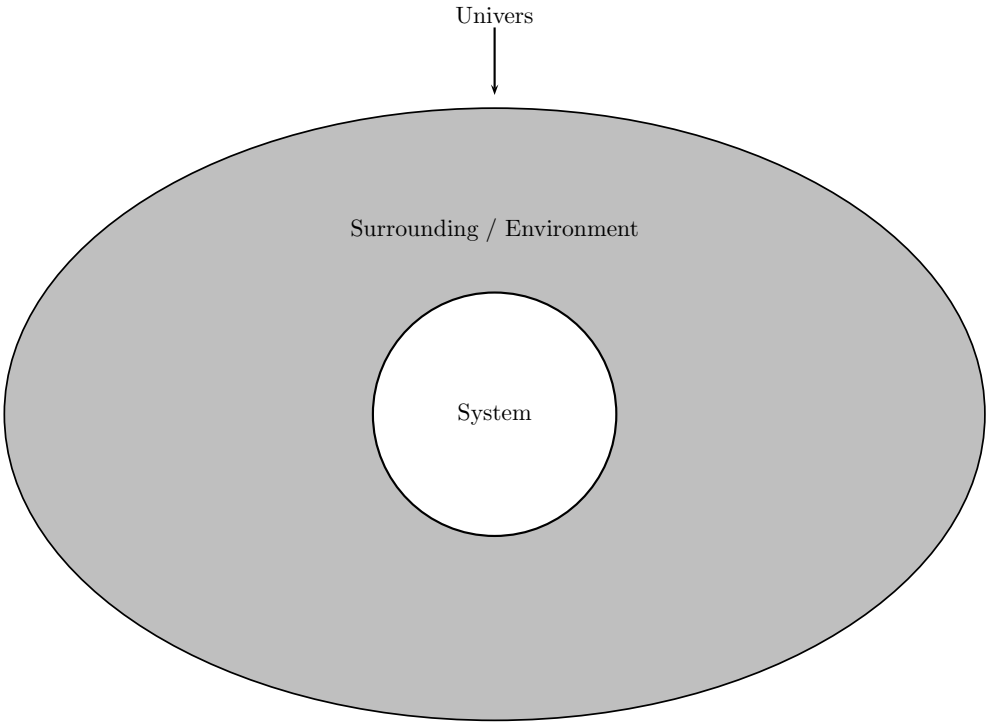


Figure 1.3: Schematic of a thermodynamic system and its environment.

1.3.2 Different types of systems

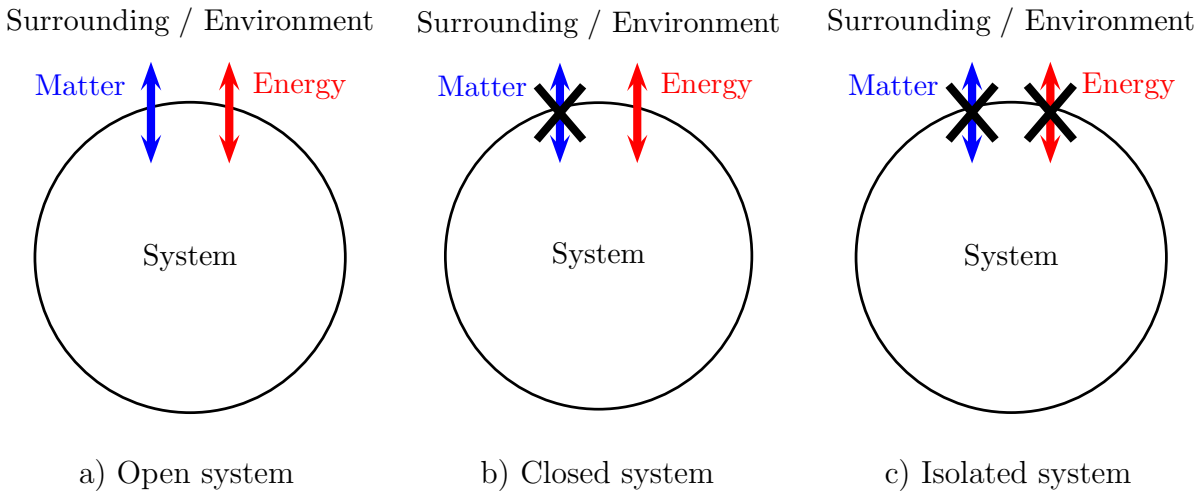
Three types of thermodynamic systems can be defined

— Open systems : exchanges of matter and energy with the environment are possible

— Closed systems : exchange of energy with the environment is possible

— Isolated systems : no exchange with the environment is possible

Figure 1.4 illustrates these three types of thermodynamic systems.



a) Open system b) Closed system c) Isolated system

Figure 1.4: Schematics of the different types of thermodynamic systems.

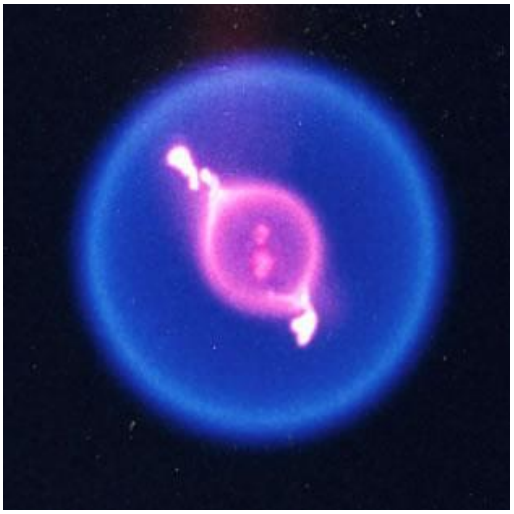


Figure 1.5: Droplet of a liquid hydrocarbon fuel burning in a micro-gravity environment. Picture from NASA.

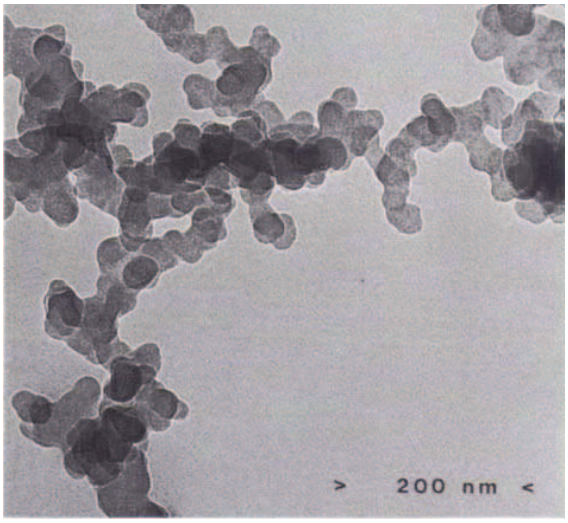
1.3.3 Nature of the system

Whatever the type of thermodynamic system is, it is possible to distinguish systems of two different natures :

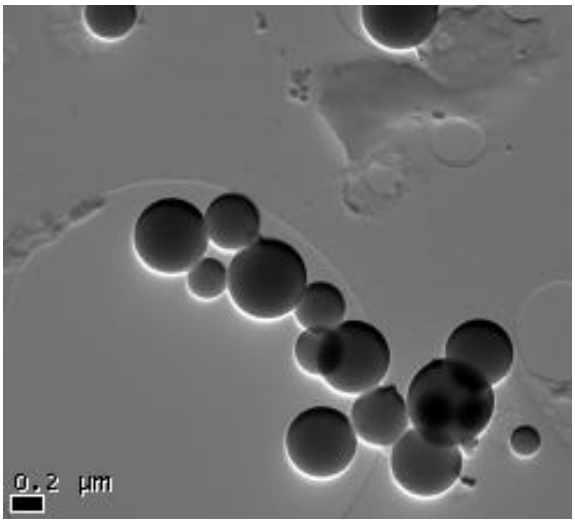
— Homogeneous system for which there is only one phase (gas, liquid, solid)

— Heterogeneous system for which there are two or more phases

In the present course, we focus mostly on homogeneous gaseous system. It is noted however, that combustion systems often involve several phases. For example, liquid fuels are used in most engines. Figure 1.5 shows a droplet of a liquid hydrocarbon fuel burning in a micro-gravity environment. In addition, some of the combustion products are solid or liquid. This is the case of soot which are formed during hydrocarbon combustion, see Figure 1.6 a). This is also the case during the combustion of fuels which contain silicon such as silane (SiH_4), see Figure 1.6 b).



a) Soot particles [6]



b) Silica particles [7]

Figure 1.6: Example of solid or liquid particles formed during combustion processes imaged using a Transmission Electronic Microscope.

1.3.4 Types of variable of the system

A thermodynamic system is characterized by a number of variables which quantify the properties of the system. Two types of variable can be distinguished

— Intensive variable

— Extensive variable

The intensive variables/properties do not depend on the size of the system. The temperature, the index of refraction, and the density are examples of intensive properties.
The extensive variables/properties depend on the size of the system. The mass, and the volume are examples of extensive properties.

1.3.5 Concept of state function

Thermodynamic systems are characterized by a number of state functions which only depend on the current state of the system and are independent of the path taken between the previous and the current state. Functions which describe the path between two states are referred to as "path functions" [2].

State functions have particular mathematical properties. Let $F(x, y)$ be a state function. The variation of F between any initial state and any final state is given by

$$\Delta F = \int_i^f F = F_f - F_i. \tag{1.20}$$

In addition, the total derivative of a state function is exact which implies

$$F_{yx} = \frac{\partial F_x}{\partial y} = \frac{\partial F}{\partial y \partial x} = \frac{\partial F}{\partial x \partial y} = \frac{\partial F_y}{\partial x} = F_{xy}. \tag{1.21}$$

Such a property is useful to establish important relationship between thermodynamic variables some of which being experimentally accessible while other are not.

1.4 Thermodynamics laws and important quantities

1.4.1 Laws of thermodynamics

1.4.1.1 First law of thermodynamics

The first law of thermodynamics reads :

The energy of an isolated system is conserved. It can be neither created nor destroyed.

It has the following consequences

— Work cannot be collected without a counterpart

— Perpetual motion is impossible

— The different forms of energy are equivalent (work and heat are both energy)

1.4.1.2 Second law of thermodynamics

The second law of thermodynamics reads :

An isolated system evolves spontaneously from a state of non-equilibrium toward a state of equilibrium.

It has the following consequences

— A system evolves toward a state of highest disorder

— A spontaneous reaction is characterized by a creation of entropy

1.4.1.3 Third law of thermodynamics

The third law of thermodynamics reads :

At T=0 K, all energy of thermal motion has been quenched, and in a perfect crystal all the atoms or ions are in a regular, uniform array. The entropy is zero (S=0) because there is only one way of arranging the molecule, only one micro-state is accessible : the ground state.

1.4.2 Important thermodynamics quantities

1.4.2.1 Work, heat, and energy

Work (w) is defined as motion against an opposing force [2]. It is equivalent to raising a weight in the environment of the thermodynamic system. As illustrated in Figure 1.7, the expansion of a gas that pushes a flexible wall (or a piston) is an example of work. This is the physical mechanism that is taking place in an internal combustion engine as the expanding combusting gas mixture is pushing the engine piston back in the cylinder.

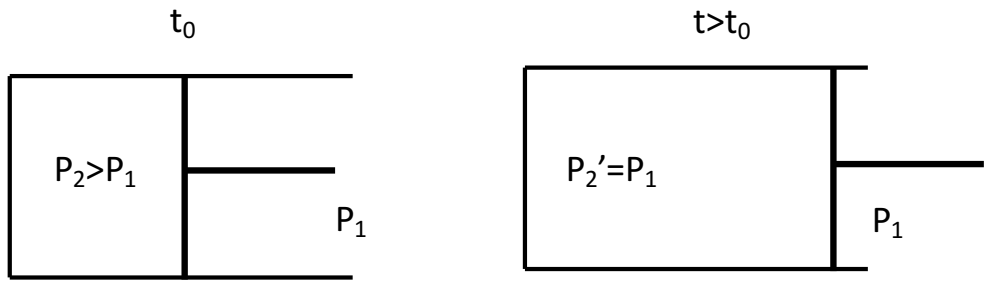


Figure 1.7: Work done by the expansion of a gas pushing a piston.

The energy of a system corresponds to its capacity to do work. It is noted that energy can be transfer to a system as work but also as heat. Increasing the temperature of a system enables to extract more work from it. This is a consequence of the first law of thermodynamics : the different forms of energy are equivalent ; work and heat are both energy.

Heat (q) is defined as a transfer of energy which originates from a difference in temperature between the system and its surrounding/environment. Considering an ideal gas, temperature can be considered as proportional to the average kinetic energy of the random microscopic motions of the gas molecule.

Two types of heat transfer processes can be defined

— the exothermic processes, that releases energy as heat into its surroundings

— the endothermic processes, that acquires energy as heat from its surroundings

The combustion reactions are overall exothermic. However, a number of elementary reactions that are taking place during the combustion process are endothermal. To illustrate this aspect, let us consider the following global reaction

$$CH_4 + 2O_2 = CO_2 + 2H_2O, \tag{1.22}$$

$\Delta_r H^\circ = -804.29\ kJ/mol$.

The negative sign for the enthalpy of reaction indicates that this reaction provides/releases energy to its environment. It is thus an exothermic process.

Now, let us consider the following elementary reaction

$$CH_4 = CH_3 + H, \tag{1.23}$$

$\Delta_r H^\circ = +438.56\ kJ/mol$.

This reaction corresponds to a decomposition reaction. It is taking place at high temperature and is one of the initial steps of methane combustion. The positive sign indicates that this reaction is acquiring energy from its surrounding and is thus an endothermal process. Energy is required to break one of the C-H bounds in CH_4 and as no other chemical bound is formed during the reaction, no energy is released.

1.4.2.2 Internal energy

The total energy of a thermodynamic system is called *internal energy* and is noted U. The internal energy is a state function and as such, its value depends on the value of the state variables like pressure and temperature. It is an extensive property. The change of internal energy is define as

$$\Delta U = q + w \tag{1.24}$$

where q is the heat ; and w is the work.

In case energy is brought to the system, q and/or $w > 0$, then $\Delta U > 0$. In case energy is lost by the system, q and/or $w < 0$, then $\Delta U < 0$.

According to the first law of thermodynamics, the internal energy of an isolated system is constant so that $\Delta U = q = w = 0$.

The total energy of an isolated gas molecule (E_T) is given by

$$E_T = \underbrace{E_{Tr} + E_R + E_V}_{\text{Thermal energy}} + \underbrace{E_{El} + E_N}_{\text{Elementary particle energy}} \tag{1.25}$$

where

- E_{Tr} is the translational energy and is due to the motion of the gravity center of the molecule,
- E_R is the rotational energy and is due to the motion of rotation of the molecule around its gravity center,
- E_V is the vibrational energy and is due to the motion of vibration of the chemical bounds,
- E_{El} is the electronic energy,
- E_N is the nuclear energy.

It is noted that E_{El} and E_N are constant for moderate temperature, i.e. up to several thousands of Kelvin.

According to the equipartition theorem, the mean energy of one degree of freedom of motion is given by

$$E = \frac{1}{2} k_B T \tag{1.26}$$

where k_B is the Boltzmann's constant and is equal to : $k_B = 1.381E^{23}JK^1$.

The translational motion is three-dimensional and thus has three degrees of freedom. The translational energy is given by

$$E_{Tr} = \frac{3}{2} k_B T. \tag{1.27}$$

The expression for the rotational and vibrational energies of a molecule depend on the number of atom and of its geometry.

For a diatomic molecule and a linear polyatomic molecule (more than two atoms), there are only two degrees of freedom (see Figure 1.8) and the rotational energy is given by

$$E_R\left(Di\right) = E_R\left(PolyLin\right) = k_B T. \tag{1.28}$$

For a nonlinear polyatomic molecule, there are three degrees of freedom (see Figure 1.8) and the rotational energy is given by

$$E_R\left(PolyNLin\right) = \frac{3}{2} k_B T. \tag{1.29}$$

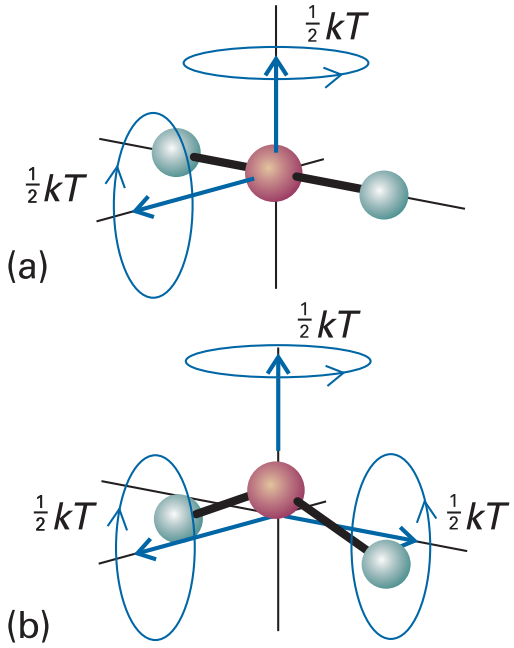


Figure 1.8: The rotational modes of molecules and the corresponding average energies at a temperature T. In a) a linear molecule is considered whereas in b) a nonlinear molecule is considered. Taken from [2].

For a diatomic molecule, the vibrational energy is given by

$$E_V\left(Di\right) = \underbrace{E_k}_{\text{kinetic energy}} + \underbrace{E_p}_{\text{Potential energy}} = k_B T \tag{1.30}$$

where E_k and E_p are determined using the harmonic oscillator model in which the force of the spring corresponds to the force of the chemical bound.

For a linear polyatomic molecule, the vibrational energy is given by

$$E_V\left(PolyLin\right) = \left(3N - 5\right) k_B T, \tag{1.31}$$

whereas for a nonlinear polyatomic molecule, it is given by

$$E_V\left(PolyNLin\right) = \left(3N - 6\right) k_B T, \tag{1.32}$$

where N is the number of atom in the molecule.

It is noted that atoms do not have rotational and vibrational energies.

1.4.2.3 Enthalpy

The internal energy is a useful quantity when considering a constant volume process. However, when considering a constant pressure process, another quantity is more convenient, the enthalpy (H). In order to maintain the pressure constant, expansion work needs to be transferred back to the system's surrounding. The enthalpy is an extensive property and is defined as

$$H = U + PV. \tag{1.33}$$

Considering an infinitesimal change in the state of the system, the change of enthalpy is given by

$$H + dH = U + dU + \left(P + dP\right)\left(V + dV\right) = U + dU + PV + PdV + VdP + dPdV. \tag{1.34}$$

The term $dPdV$ can be neglected and Equation 1.34 reduces to

$$H + dH = H + dU + PdV + VdP. \tag{1.35}$$

Substituting $dU = dq + dw$ into Equation 1.35 leads to

$$dH = dq + dw + PdV + VdP. \tag{1.36}$$

Considering that only expansion work is done, $dw = -PdV$, and that the process is taking place at constant pressure, we obtain

$$dH = dq, \tag{1.37}$$

which indicates that the change of enthalpy of the system corresponds to the energy supplied as heat at constant pressure.

1.4.2.4 Heat capacity

It is useful to express the internal energy of a molecule using the following forms

$$U_m = U_m(0) + \frac{N_f}{2} k_B T, \tag{1.38}$$

where N_f is the number of degrees of freedom ; and $U_m(0)$ is the molar internal energy at T = 0 K. At such a temperature, the translational motion is frozen and only the internal energy arising from the internal structure of the atoms contributes.

From Equation 1.38, it is seen that the internal energy of a molecule increases with temperature. The rate of increase of the internal energy at a given temperature is referred to as the heat capacity. The rate of increase of the internal energy depends on the conditions under which increase of temperature is taking place. Considering a gas contained in a close vessel of constant volume, the constant volume heat capacity (C_V) can be defined as

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V. \tag{1.39}$$

The heat capacity is an extensive property. However, it is useful to define a molar heat capacity as

$$C_{V,m} = \frac{C_V}{n}, \tag{1.40}$$

which is an intensive capacity and has units of $JK^{-1}mol^{-1}$.

It is also useful to define a specific heat capacity (or specific heat) as

$$C_{V,s} = \frac{C_V}{m}, \tag{1.41}$$

which has units of $JK^{-1}g^{-1}$. The symbol m represents the mass of the sample.
For a monoatomic gas, the molar constant volume heat capacity is equal to

$$C_{V,m} = \frac{\partial}{\partial T}\left(U_m\left(0\right) + \frac{3}{2}RT\right) = \frac{3}{2}R = 12.47\ JK^{-1}mol^{-1}, \tag{1.42}$$

given that $R = k_B \times N_A$ where N_A is the Avogadro constant and is equal to $N_A = 6.022E^{23}\ mol^{-1}$

Considering a gas contained in a vessel maintained at constant pressure, the constant pressure heat capacity (C_P) can be defined as

$$C_P = \left(\frac{\partial H}{\partial T}\right)_P. \tag{1.43}$$

The relationship between the constant volume and the constant pressure heat capacity is referred to as the Mayer's relation and is given by

$$C_p - C_V = nR, \tag{1.44}$$

where n is the number of mole ; and R is the perfect gas constant.

The heat capacity links the change of internal energy or enthalpy of a system to the change of temperature

$$dU = C_V dT. \tag{1.45}$$

The higher the heat capacity, the more energy is required to heat the system

$$\frac{dU}{C_V} = \frac{q}{C_V} = dT. \tag{1.46}$$

As a consequence, much more energy is required to heat a combustible mixture, for example a hydrogen-oxygen mixture, diluted with CO_2 than a similar mixture diluted with Ar.

Under certain conditions, the heat capacity may be considered as infinite. This is the case for example during a change of phase. Consider the vaporization of water

$$H_2O_{Liq} = H_2O_{Vap}, \tag{1.47}$$

which is taking place at 373 K, the boiling point of water. At this temperature, the energy supplied to the system is entirely used to sustain the endothermal phase transition and the temperature of the system is staying constant. Recalling Equation 1.46, this means that C_V is infinite.

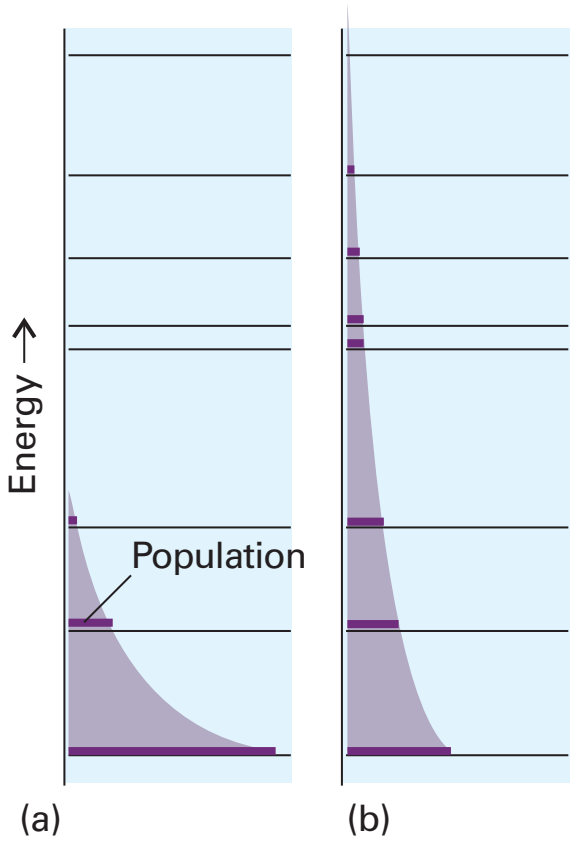


Figure 1.9: Distribution of the population of molecules over levels of energy as a function of temperature. (a) and (b) illustrate the low temperature case and high temperature case, respectively. Taken from [2].

1.4.2.5 Entropy

The entropy (S) is defined as

$$dS = \frac{dq_{rev}}{T}, \tag{1.48}$$

where rev indicates a reversible thermodynamic process.

The entropy indicates the direction of spontaneous changes since for such transformations, there is a creation of entropy. This is the second law of thermodynamics.

The second law of thermodynamics can be interpreted at the molecular level by considering that the total population of molecules or atoms is distributed over a number of energy levels. These energy levels are referred to as "micro-states". As the temperature is increased, the population of high-energy levels is increased. This is illustrated in Figure 1.9.
The second law of thermodynamics can be summarized by the equation of Boltzmann

$$S = k_B \ln W, \tag{1.49}$$

where W is the number of micro-states.

1.4.2.6 Adiabatic changes

An adiabatic, reversible change is characterized by

$$dq_{rev} = 0. \tag{1.50}$$

Let us consider an adiabatic, reversible expansion process for an ideal gas. The change of internal energy is given by

$$dU = dq + dw. \tag{1.51}$$

The reversible work done is given by

$$dw_{rev} = -PdV. \tag{1.52}$$

The change of enthalpy is expressed as

$$dH = dU + PdV + VdP = dq + dw + PdV + VdP. \tag{1.53}$$

Substituting Equation 1.50 and Equation 1.52 into Equation 1.51 and Equation 1.53 leads respectively to

$$dU = 0 - PdV = -PdV, \tag{1.54}$$

and

$$dH = 0 - PdV + PdV + VdP = VdP. \tag{1.55}$$

Recalling Equation 1.39 and Equation 1.43, we can write

$$dU = nC_V dT = -PdV, \tag{1.56}$$

and

$$dH = nC_P dT = VdP. \tag{1.57}$$

Dividing Equation 1.57 by Equation 1.56 gives

$$\gamma = \frac{C_P}{C_V} = -\frac{\frac{dP}{P}}{\frac{dV}{V}}, \tag{1.58}$$

where γ is the isentropic coefficient.

Equation 1.58 may also be written as

$$\int \gamma \frac{dV}{V} = \int -\frac{dP}{P} + constant, \tag{1.59}$$

which leads to

$$PV^{\gamma} = constant. \tag{1.60}$$

Considering an expansion between a volume V_1 and a volume V_2 which brings the pressure from P_1 to P_2, the following equation can be written

$$\int_{V_1}^{V_2} \gamma \frac{dV}{V} = \int_{P_1}^{P_2} -\frac{dP}{P}, \tag{1.61}$$

which leads to

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^{\gamma}. \tag{1.62}$$

Equation 1.62 is referred to as the isentropic relationship between pressure and volume. Using the perfect gas law, similar isentropic relationship can be written for temperature and density as

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} = \left(\frac{\rho_2}{\rho_1}\right)^{\gamma-1}. \tag{1.63}$$

1.4.2.7 The Helmholtz and Gibbs energies

Two other thermodynamic functions can be introduced,
the Helmholtz energy (A) defined as

$$A = U - TS, \tag{1.64}$$

and the Gibbs energy (G) defined as

$$G = H - TS. \tag{1.65}$$

The condition for spontaneous transformations are
for constant temperature and constant volume conditions

$$dA_{T,V} \leqslant 0, \tag{1.66}$$

and, for constant temperature and constant pressure conditions

$$dG_{T,P} \leqslant 0. \tag{1.67}$$

1.5 Thermochemistry

Thermochemistry corresponds to the study of the energy transferred as heat during the chemical reactions [2].

1.5.1 Standard enthalpy changes

In order to study the thermochemistry of reactive systems in a standardized way, it is useful to define standard conditions. The standard enthalpy change is noted as

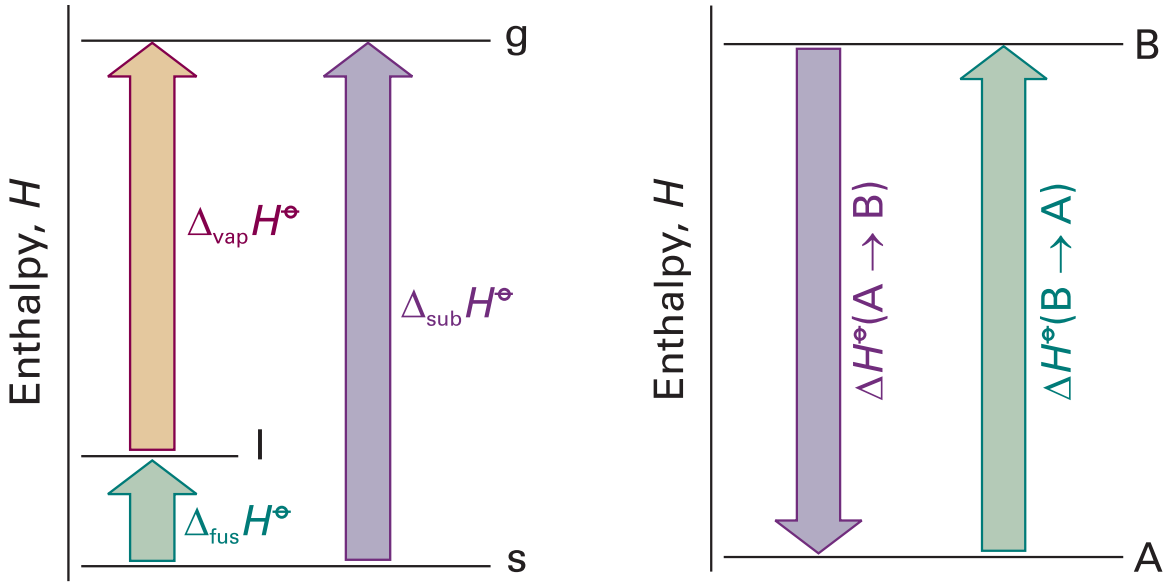
$$\Delta H^{\circ}, \tag{1.68}$$

and is defined as [2]
"the change in enthalpy for a process in which the initial and final substances are in their standard states".

The standard state is defined as [2]

"The standard state of a substance at a specified temperature is its pure form at 100 kPa."

Whereas the standard enthalpy can be given at any temperature, it is usual to use 298.15 K as a reference temperature to report thermodynamic data.



a) Additivity of ΔH° b) Opposite sign of ΔH° for opposite processes

Figure 1.10: Illustration of the properties of the enthalpy of phase transition. Taken from [2].

For a phase transition such as water vaporization,

$$H_2O_{(l)} \rightarrow H_2O_{(g)}, \tag{1.69}$$

the standard enthalpy change is noted as

$$\Delta_{vap}H^{\circ}. \tag{1.70}$$

Because enthalpy is a state function, the enthalpy change does not depend on the path taken. This indicates that the enthalpy change for the sublimation process, direct transition from the solid to the gas phase, is equal to the sum of the enthalpies for fusion and for

vaporization. This concept is summarized as

$$\begin{aligned} H_2O_{(s)} &\rightarrow H_2O_{(l)} \quad \Delta_{fus}H^\circ \\ H_2O_{(l)} &\rightarrow H_2O_{(g)} \quad \Delta_{vap}H^\circ \\ \text{Overall}: \; H_2O_{(s)} &\rightarrow H_2O_{(g)} \quad \Delta_{sub}H^\circ = \Delta_{fus}H^\circ + \Delta_{vap}H^\circ \end{aligned} \tag{1.71}$$

and illustrated in Figure 1.10 a).

In addition, the standard enthalpies for a phase transition and its reverse have opposite sign

$$\Delta_{A\rightarrow B}H^\circ = -\Delta_{B\rightarrow A}H^\circ \tag{1.72}$$

as illustrated in Figure 1.10 b).

1.5.2 Standard enthalpy of chemical reaction

Consider the global combustion reaction of methane (CH_4) in oxygen. For an equivalence ratio of 1, i.e. a stoichiometric mixture, the chemical reaction can be written

$$CH_4(g) + 2O_2(g) = CO_2(g) + 2H_2O(l). \tag{1.73}$$

The standard enthalpy of this reaction can be calculated by using the standard enthalpies of formation of the chemical species involved such as

$$\Delta_r H^\circ = \Delta_f H^\circ\left(CO_2(g)\right) + 2\Delta_f H^\circ\left(H_2O(l)\right) - \Delta_f H^\circ\left(CH_4(g)\right) - 2\Delta_f H^\circ\left(O_2(g)\right). \tag{1.74}$$

Equation 1.74 can be generalized as

$$\Delta_r H^\circ = \sum_{j=1}^{N_j} \nu_j \Delta_f H^\circ(P_j) - \sum_{i=1}^{N_i} \nu_i \Delta_f H^\circ(R_i), \tag{1.75}$$

where ν represents the stoichiometric coefficient; R and P represents the reactants and the products, respectively; N is the number of species; i refers to reactants; and j refers to the products.

It is noted that the enthalpy of formation of elements is taken equal to zero. Thus

$$\Delta_f H^\circ(H_2(g)) = \Delta_f H^\circ(O_2(g)) = \Delta_f H^\circ(N_2(g)) = 0. \tag{1.76}$$

According to the Hess's law [2]

"The standard enthalpy of an overall reaction is the sum of the standard enthalpies of the individual reactions into which a reaction may be divided."

The individual reactions do not need to actually take ; they can be hypothetical, as long as they respect the law of mass conservation, i.e. they are balanced.

The Hess's law is useful to obtain information about the enthalpy of a reaction which may be difficult to obtain directly. As an example, let us consider the combustion reaction of propene (C_3H_6)

$$R_1: \quad C_3H_6(g) + \frac{9}{2}O_2(g) = 3CO_2(g) + 3H_2O(l). \tag{1.77}$$

Reaction R_1 may be decomposed into the following reactions

the hydrogenation of propene

$$R_2: \quad C_3H_6(g) + H_2(g) = C_3H_8(g) \quad \Delta_r H^\circ(R_2) = -124\ kJ/mol, \tag{1.78}$$

the combustion of propane (C_3H_8)

$$R_3: \quad C_3H_8(g) + 5O_2(g) = 3CO_2(g) + 4H_2O(l) \quad \Delta_r H^\circ(R_3) = -2200\ kJ/mol, \tag{1.79}$$

the decomposition of water

$$R_4: \quad H_2O(l) = H_2(g) + \frac{1}{2}O_2(g) \quad \Delta_r H^\circ(R_4) = +286\ kJ/mol. \tag{1.80}$$

The enthalpy of the combustion reaction of propene is given by

$$\Delta_r H^\circ(R_1) = \Delta_r H^\circ(R_2) + \Delta_r H^\circ(R_3) + \Delta_r H^\circ(R_4) = -124 - 2200 + 286 = -2038\ kJ/mol. \tag{1.81}$$

It can be noted that the combustion of propane is more exothermic than the combustion of propene.

According to the Kirchhoffs law, the temperature dependence of reaction enthalpy follows

$$\Delta_r H^\circ(T_2) = \Delta_r H^\circ(T_1) + \int_{T_1}^{T_2} \Delta_r C_p^\circ dT, \tag{1.82}$$

where

$$\Delta_r C_p^\circ = \sum_{j=1}^{N_j} \nu_j C_p^\circ(P_j) - \sum_{i=1}^{N_i} \nu_i C_p^\circ(R_i). \tag{1.83}$$

Equation 1.82 assumes that no phase transition has been taking place during the reaction.

1.6 Chemical equilibrium

Chemical reactions tend to evolve towards an equilibrium in which both reactants and products are present but have no further tendency to undergo change. A reaction may be referred to as "complete" if the concentration of the reactants is negligibly low compared to that of the products. In many cases however, the reaction is not complete and the remaining concentration of the reactants is not negligible [2].

1.6.1 The condition for equilibrium

In order to formulate the relevant equations for chemical equilibrium, it is useful to define the chemical potential of species i (μ_i) as

$$\mu_i = \left(\frac{\partial G}{\partial n_i}\right)_{P,T,n_{j\neq i}}. \tag{1.84}$$

For a binary mixture of A and B, the Giggs energy is given by

$$G = n_A\mu_A + n_B\mu_B, \tag{1.85}$$

and as a consequence

$$dG = VdP - SdT + \mu_A dn_A + \mu_B dn_B = VdP - SdT + \sum_{i=1}^{N} \mu_i dn_i. \tag{1.86}$$

Let us consider the following reaction which transforms the species A into B

$$A = B. \tag{1.87}$$

For this reaction, the reaction Gibbs energy is

$$\Delta_r G = \left(\frac{\partial G}{\partial \xi}\right)_{P,T}, \tag{1.88}$$

where ξ is the extent of reaction and is expressed in mole.

At constant pressure and constant temperature, dG is given by

$$dG = \mu_A dn_A + \mu_B dn_B = -\mu_A d\xi + \mu_B d\xi = (\mu_B - \mu_A)\, d\xi. \tag{1.89}$$

By combining Equation 1.88 and Equation 1.89, we obtain

$$\Delta_r G = \left(\frac{\partial G}{\partial \xi}\right)_{P,T} = \mu_B - \mu_A. \tag{1.90}$$

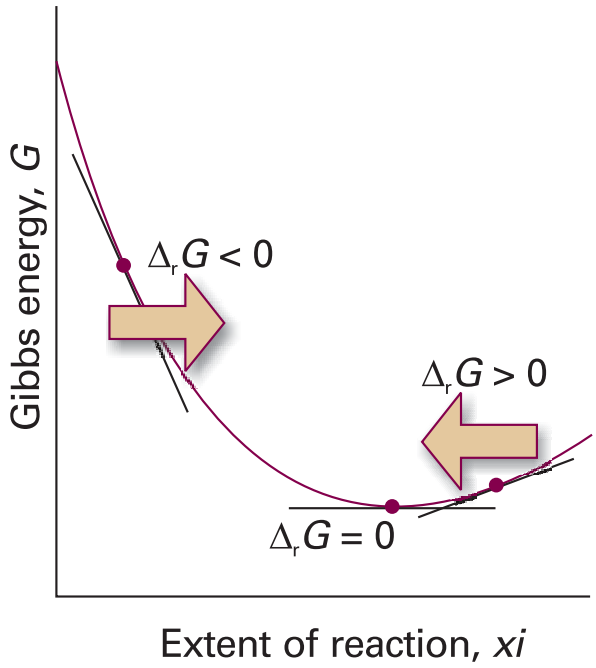


Figure 1.11: Evolution of the reaction Gibbs energy as a function of the extent of reaction. Taken from [2].

As illustrated in Figure 1.11, the condition for equilibrium is

$$\Delta_r G = 0, \tag{1.91}$$

which implies that, at equilibrium

$$\mu_A = \mu_B. \tag{1.92}$$

Equation 1.92 indicates that, under the particular conditions we have considered, the composition at equilibrium is such that the chemical potential of the species A is equal to the chemical potential of the species B.

1.6.2 Description of equilibrium

We first consider a simple reaction such as

$$A = B. \tag{1.93}$$

The chemical potential for a species i is expressed as

$$\mu_i = \mu_i^\circ + RT \ln \frac{P_i}{P^\circ}. \tag{1.94}$$

Substituting Equation 1.94 into Equation 1.90, the reaction Gibbs energy becomes

$$\Delta_r G = \left(\mu_B^\circ + RT \ln \frac{P_B}{P^\circ}\right) - \left(\mu_A^\circ + RT \ln \frac{P_A}{P^\circ}\right) = \Delta_r G^\circ + RT \ln \frac{P_B}{P^\circ}\frac{P^\circ}{P_A}. \tag{1.95}$$

We can define Q, the reaction quotient as

$$Q = \frac{P_B}{P^\circ}\frac{P^\circ}{P_A} = \frac{P_B}{P_A}. \tag{1.96}$$

At equilibrium, we know that $\Delta_r G$=0, so that, using Equation 1.95, we obtain an explicit expression for K, the equilibrium constant (at equilibrium only, Q can be written as K)

$$K = \exp\left(\frac{-\Delta_r G^\circ}{RT}\right). \tag{1.97}$$

For a perfect gas, Equation 1.97 can be generalized as

$$K = \prod_{k=1}^{N_k} a_k^{\nu_k} = \exp\left(\frac{\sum_{i=1}^{N_i} \nu_i \Delta_f G^\circ(i) - \sum_{j=1}^{N_j} \nu_j \Delta_f G^\circ(j)}{RT}\right), \tag{1.98}$$

where k represents any species of the reaction; i represents the reactants, j represents the products; $\Delta_f G^\circ$ is the standard Gibbs energy of formation; and a_k is the chemical activity of the species k defined as

$$a_k = \frac{P_k}{P^\circ}. \tag{1.99}$$

It is noted that for real gas, the definition of the activity is different and requires the use of the fugacity instead of the pressure.

	H_2O	H_2	O_2
Initial amount	n	0	0
Equilibrium amount	$(1-\alpha)n$	αn	$\frac{1}{2}\alpha n$
Mole faction	$\frac{1-\alpha}{1+\frac{1}{2}\alpha}$	$\frac{\alpha}{1+\frac{1}{2}\alpha}$	$\frac{\frac{1}{2}\alpha}{1+\frac{1}{2}\alpha}$
Partial pressure	$P\frac{1-\alpha}{1+\frac{1}{2}\alpha}$	$P\frac{\alpha}{1+\frac{1}{2}\alpha}$	$P\frac{\frac{1}{2}\alpha}{1+\frac{1}{2}\alpha}$

Table 1.1: Evolution of the mixture composition as a function of α for the water vapor decomposition reaction. Taken from [2].

To illustrate the practical use of the equilibrium constant, let us consider the decomposition of water vapor at high temperature [2]. We are seeking at determining the degree of dissociation of water (α) at 2300 K and 100 kPa. The chemical reaction is

$$H_2O(g) = H_2(g) + \frac{1}{2}O_2(g) \quad \Delta_r G^\circ(2300K) = 118.08\ kJ/mol. \tag{1.100}$$

Using Equation 1.100, we can establish Table 1.1 which gives the evolution of the mixture composition as a function of α.

Using Equation 1.99 and the data from Table 1.1, we can write

$$K = \frac{\frac{P_{H2}}{P^\circ}\left(\frac{P_{O2}}{P^\circ}\right)^{1/2}}{\frac{P_{H2O}}{P^\circ}} = \frac{\alpha^{3/2}}{(1-\alpha)(2+\alpha)^{1/2}}\left(\frac{P}{P^\circ}\right)^{1/2}. \tag{1.101}$$

With the assumption that $\alpha \ll 1$, we can use Equation 1.97 to obtain

$$K = \frac{\alpha^{3/2}}{2^{1/2}}\left(\frac{P}{P^\circ}\right)^{1/2} = \exp\left(\frac{-\Delta_r G^\circ}{RT}\right), \tag{1.102}$$

which leads to

$$\alpha = \left(\left(\frac{2P^\circ}{P}\right)^{1/2}\exp\left(\frac{-\Delta_r G^\circ}{RT}\right)\right)^{2/3} = \left(2^{1/2}\exp\left(\frac{-118080}{8.314\times 2300}\right)\right)^{2/3} = 0.0205. \tag{1.103}$$

1.6.3 Pressure and temperature dependence of equilibrium

1.6.3.1 Pressure dependence

Recalling Equation 1.97, it is seen that the equilibrium constant is a function of $\Delta_r G^\circ$ which is defined for a single reference pressure. Thus is can be concluded that K is independent of

pressure

$$\left(\frac{\partial K}{\partial P}\right)_T = 0. \tag{1.104}$$

Nevertheless, it does not mean that the composition at equilibrium is not affected by pressure. To illustrate this aspect, let us consider the following reaction

$$A = 2B. \tag{1.105}$$

At equilibrium the quantity of A is (1 - α)n and the quantity of B is 2αn, where α is the dissociation coefficient of A into 2B. The equilibrium constant for this reaction is

$$K = \frac{\alpha^2}{1-\alpha^2}\frac{P}{P^\circ}. \tag{1.106}$$

From Equation 1.106, it is possible to obtain an explicit expression for α

$$\alpha = \left(\frac{1}{1+\frac{4}{K}\frac{P}{P^\circ}}\right)^{1/2}. \tag{1.107}$$

As pressure appears in the expression of α, the composition at equilibrium depends on pressure.

1.6.3.2 Temperature dependence

According to Le Chatelier's principle [2], for

— exothermic reactions : increased temperature favors the reactants

— endothermic reactions : increased temperature favors the products

This is expressed mathematically by the van't Hoff equation

$$\frac{d\ln K}{d(1/T)} = -\frac{\Delta_r H^\circ}{R}. \tag{1.108}$$

Knowing the equilibrium constant at a temperature T_1 and the enthalpy of reaction, it possible to calculate the equilibrium constant at a temperature T_2 using

$$\ln K_2 - \ln K_1 = -\frac{1}{R}\int_{1/T_1}^{1/T_2} \Delta_r H^\circ d(1/T). \tag{1.109}$$

If $\Delta_r H^\circ$ is constant over the range of temperature $T_1 - T_2$, Equation 1.109 becomes

$$\ln K_2 - \ln K_1 = -\frac{\Delta_r H^\circ}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right). \tag{1.110}$$

1.7 Thermodynamics properties in combustion

1.7.1 Adiabatic flame temperature calculation

Consider the following global combustion reaction

$$aA + bB = cC + dD, \tag{1.111}$$

which transform the reactants, A and B, at initial temperature T_1 into the products, C and D, at temperature T_2. The temperature T_2 is the adiabatic flame temperature (T_f).

Noting the reactants by j and the products by i, the enthalpy of the reaction is given by

$$\Delta_r H = \sum_{i=1}^{N} \nu_i \left[(H^\circ_{T2} - H^\circ_{T0}) + \left(\Delta H^\circ_f\right)_{T0} \right]_i - \sum_{i=j}^{K} \nu_j \left[(H^\circ_{T1} - H^\circ_{T0}) + \left(\Delta H^\circ_f\right)_{T0} \right]_j, \tag{1.112}$$

where T_0 is the reference temperature taken equal to 298.15 K; N is the total number of products; and K is the total number of reactants.

Since we consider an adiabatic process, $\Delta_r H$=0, we obtain for the reaction shown in Equation 1.111

$$\begin{aligned} c\left[(H^\circ_{T2} - H^\circ_{T0})\right]_C + d\left[(H^\circ_{T2} - H^\circ_{T0})\right]_D = & \\ & - \left[\left(\Delta H^\circ_f\right)_{T0}\right]_C - \left[\left(\Delta H^\circ_f\right)_{T0}\right]_D \\ & + a\left[(H^\circ_{T1} - H^\circ_{T0}) + \left(\Delta H^\circ_f\right)_{T0}\right]_A \\ & + b\left[(H^\circ_{T1} - H^\circ_{T0}) + \left(\Delta H^\circ_f\right)_{T0}\right]_B. \end{aligned} \tag{1.113}$$

By using tabulated values of $(H^\circ_T - H^\circ_{T0})$, it is possible to graphically determine the adiabatic flame temperature.

To further illustrate such an approach, a simple example is presented for hydrogen-oxygen mixtures. The global chemical reaction is

$$H_2 + \frac{1}{2}O_2 = H_2O. \tag{1.114}$$

The initial temperature is taken equal to T_1=200 K and the combustion is considered to be complete, i.e. the only combustion product is water vapor. The enthalpy of reaction can be

written as

$$\Delta_r H = \left[(H^\circ_{T2} - H^\circ_{T0}) + \left(\Delta H^\circ_f\right)_{T0}\right]_{H2O} - \left[(H^\circ_{T1} - H^\circ_{T0}) + \left(\Delta H^\circ_f\right)_{T0}\right]_{H2} - \frac{1}{2}\left[(H^\circ_{T1} - H^\circ_{T0}) + \left(\Delta H^\circ_f\right)_{T0}\right]_{O2}. \quad (1.115)$$

By definition, $\left(\Delta H^\circ_f\,(H_2)\right)_{T0}$=$\left(\Delta H^\circ_f\,(O_2)\right)_{T0}$=0, and $\Delta_r H$=0. Equation 1.115 reduces to

$$[(H^\circ_{T2} - H^\circ_{T0})]_{H2O} = -\left[\left(\Delta H^\circ_f\right)_{T0}\right]_{H2O} + [(H^\circ_{T1} - H^\circ_{T0})]_{H2} + \frac{1}{2}[(H^\circ_{T1} - H^\circ_{T0})]_{O2}. \quad (1.116)$$

Temperature (K)	H-H_{298} (kJ/mol)
4000	183.6
4100	189.4
4200	195.2
4300	201
4400	206.9
4500	212.8
4600	218.7
4700	224.5
4800	230.5
4900	236.4
5000	242.3

Table 1.2: Evolution of the enthalpy of water as a function of temperature.

From thermodynamic database, we know that

— $\left[\left(\Delta H^\circ_f\right)_{T0}\right]_{H2O}$=-243.83 kJ/mol

— $[(H^\circ_{T1} - H^\circ_{T0})]_{H2}$=-2.77 kJ/mol

— $\frac{1}{2}[(H^\circ_{T1} - H^\circ_{T0})]_{O2}$=-2.87 kJ/mol

Thus, equation Equation 1.116 becomes

$$[(H^\circ_{T2} - H^\circ_{T0})]_{H2O} = 237.6\ kJ/mol. \quad (1.117)$$

Table 1.2 gives the evolution of $[(H^{\circ}_{T2} - H^{\circ}_{T0})]_{H2O}$ as a function of temperature. Using these values, it is possible to graphically deduce the flame temperature as shown in Figure 1.12. The flame temperature obtained using the approach just described is 4934 K. The actual flame temperature of the considered mixture is on the order of 3100 K. The difference between the calculated and the actual flame temperature can be explained by the fact that at very high temperature, water is not stable and decomposes into H, O, and OH which reduces the flame temperature.

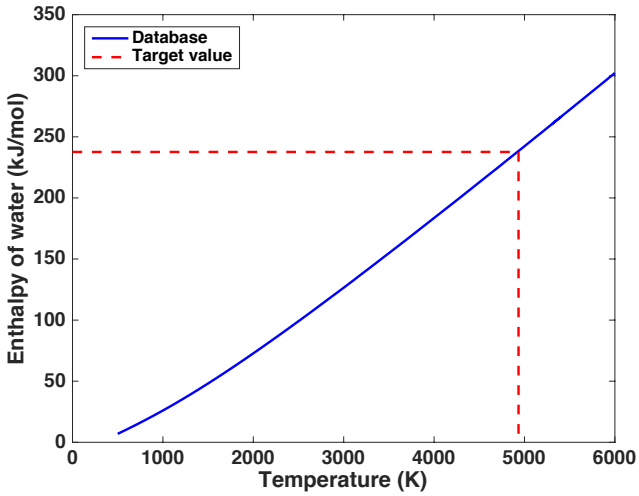


Figure 1.12: Graphical determination of the adiabatic flame temperature for a stoichiometric hydrogen-oxygen mixture.

1.7.2 Use of thermodynamic data for combustion modeling

1.7.2.1 Declaring thermodynamic data

In combustion modeling, the thermodynamic quantities required to determine the properties of the system at equilibrium or the chemical evolution of a system during a combustion process are the heat capacity at constant pressure, the enthalpy, and the entropy. The evolution of these properties as a function of temperature is described for each species by a series of 14 polynomial coefficients. Seven of these coefficients are used to describe the evolution in the low-temperature range (typically from 100-200 K up to 1000-1300 K), and the seven other

coefficients are used to describe the evolution in the high-temperature range (typically from 1000-1300 K and up to 3000-6000 K). Figure 1.13 illustrates the format used in Chemkin to declare the polynomial coefficients of a chemical species.

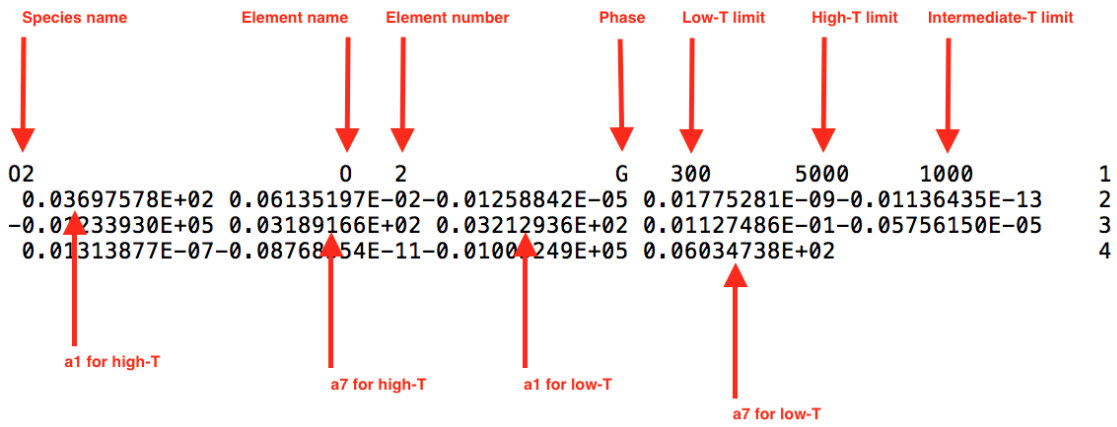


Figure 1.13: Format used in Chemkin to declare the thermodynamic properties polynomial coefficients.

The following formula are used to calculate

the heat capacity at constant pressure

$$\frac{C_P}{R} = a_1 + a_2 T + a_3 T^2 + a_4 T^3 + a_5 T^4, \tag{1.118}$$

the standard enthalpy

$$\frac{H^\circ}{RT} = a_1 + \frac{a_2}{2} T + \frac{a_3}{3} T^2 + \frac{a_4}{4} T^3 + \frac{a_5}{5} T^4 + \frac{a_6}{T}, \tag{1.119}$$

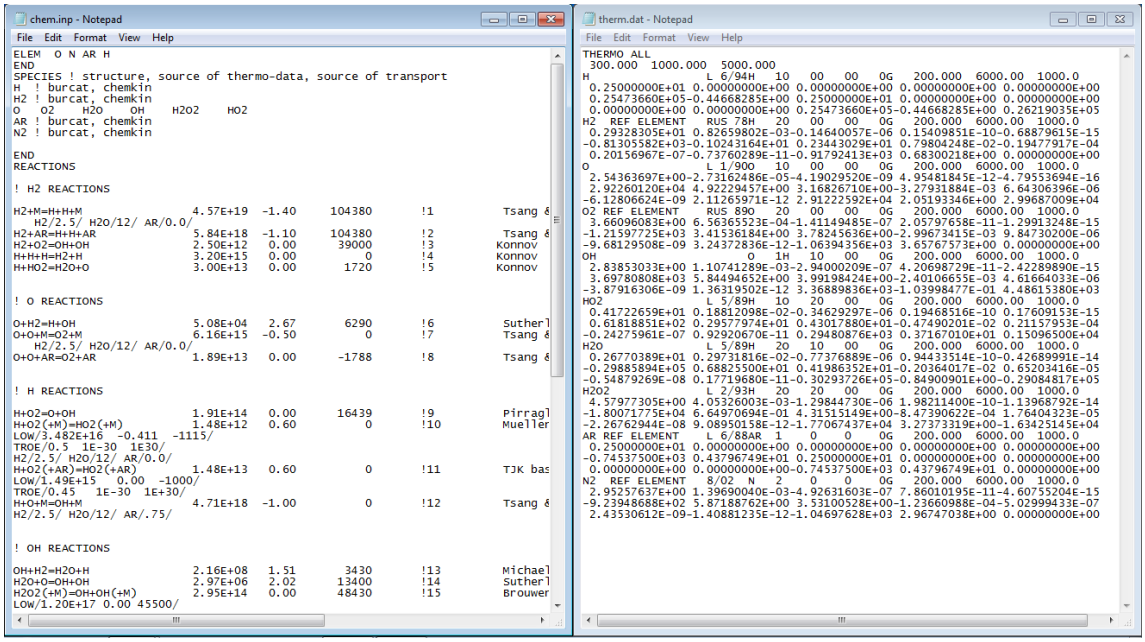
and the standard entropy

$$\frac{S^\circ}{R} = a_1 \ln T + a_2 T + \frac{a_3}{2} T^2 + \frac{a_4}{3} T^3 + \frac{a_5}{4} T^4 + a_7. \tag{1.120}$$

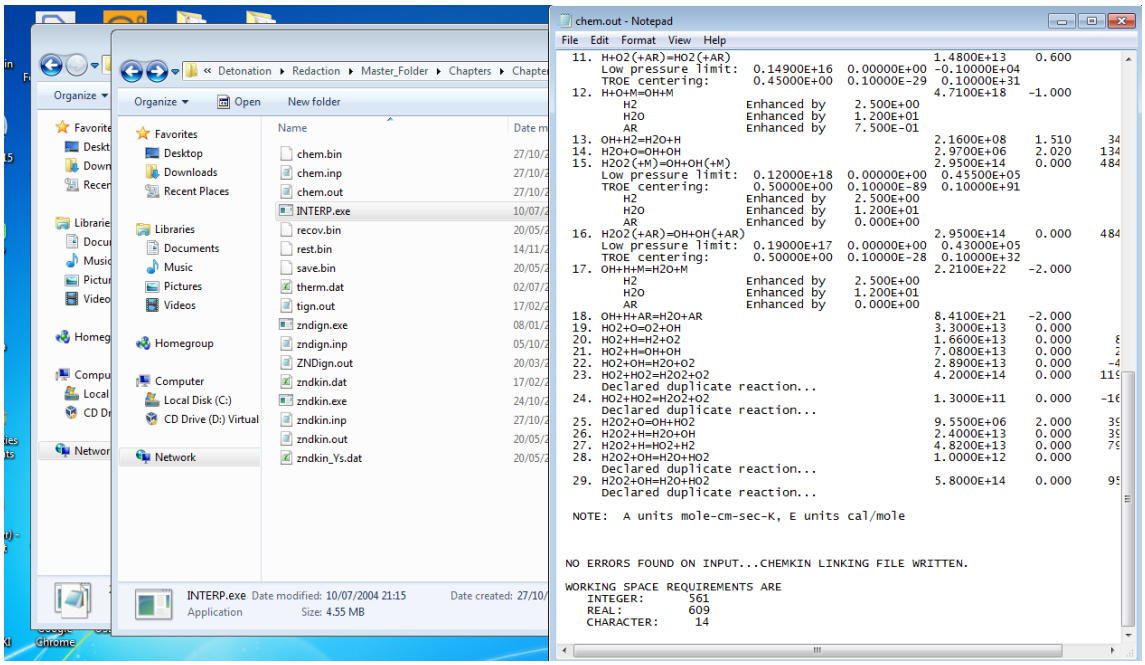
1.7.2.2 Examples of application in combustion research

There are a number of software such as Chemkin [8], Cantera [9], Cosilab [10], etc, dedicated to combustion chemical kinetics modeling. In the following, the use of Chemkin for

calculating some equilibrium combustion properties is demonstrated.



a) Input files



b) Output file

Figure 1.14: Reaction model, thermodynamic files, and "interpreter" used in Chemkin.

The first step toward calculating combustion properties with Chemkin is to provide/edit two files containing the reaction model and the thermodynamic data. These input files, shown in Figure 1.14 a), constitutes the so-called reaction mechanism or chemical kinetics model. The files need to be edited with a specific format referred to as the "Chemkin format". The correctness of these input files has to be verified by running the Chemkin interpreter usually called "interp.exe". This process leads to the edition of an output file which indicates the formatting errors. If no error is found, a binary chem file will be edited. This is shown in Figure 1.14 b).

Three examples of equilibrium properties calculation are presented : (i) the adiabatic flame temperature ; (ii) the maximum explosion pressure ; (iii) the equilibrium composition.

As we have seen early in this first chapter, the adiabatic flame temperature can be calculated using thermodynamic tables. Nevertheless, the result we have obtain is not physical because we have considered complete conversion of the reactants into the most stable products. In order to obtain a more realistic flame temperature, it is required to consider the equilibrium for all the species. For example, considering H_2-O_2 mixtures, water vapor will not be the only products and other species such as H, O, OH need to be considered to calculate the flame temperature. For H_2-O_2 mixtures, the approach based on thermodynamic tables can still be applied. However, for hydrogen-air or hydrocarbon-air mixtures, tens or even hundreds of species have to be considered which makes the flame temperature calculation very difficult.

Using the Equil code [11] of the Chemkin II package, we can calculate the adiabatic flame temperature at constant pressure. We consider a stoichiometric H_2-O_2 mixture at initial temperature of 300 K and initial pressure of 101 kPa. Table 1.3 shows the flame temperature obtained when considering different number of species. Including H, O, and OH in addition to H_2, O_2, H_2O leads to a decrease of the flame temperature of 432 K. Further including HO_2, H_2O_2 does not lead to a further decrease because of the very low stability of these species at

high temperature.

Species included	Flame temperature (K)
H_2, O_2, H_2O	3510.9
H_2, O_2, H_2O H, O, OH	3079.0
H_2, O_2, H_2O H, O, OH HO_2, H_2O_2	3078.9

Table 1.3: Adiabatic flame temperature of a stoichiometric H_2-O_2 mixture as a function of the number of species included. Conditions : T_1= 300 K ; P_1=101 kPa.

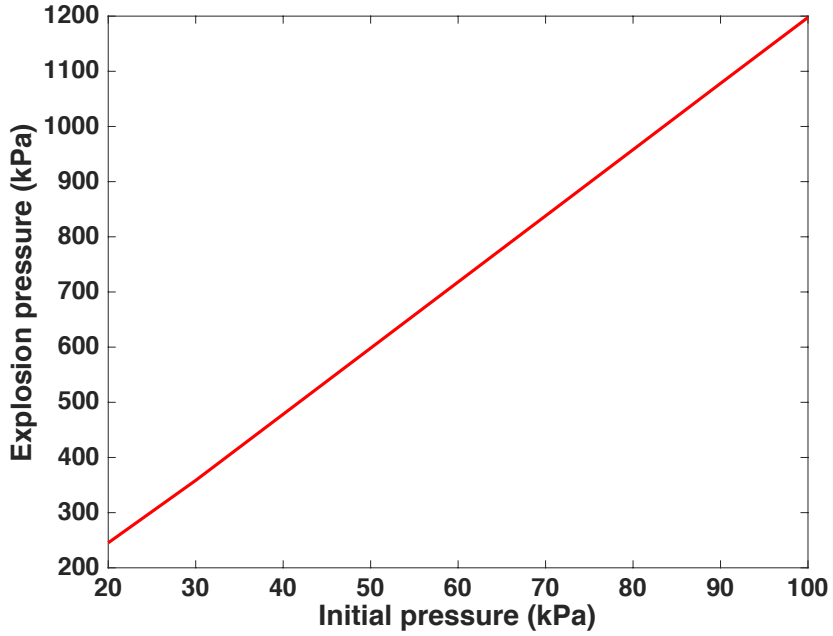


Figure 1.15: Evolution of the constant volume explosion pressure of a stoichiometric H_2-N_2O mixture as a function of initial pressure. The initial temperature is fixed at 300 K.

Now let us consider the following problem : a student needs to design a combustion vessel to perform flame speed measurements of a stoichiometric H_2-N_2O mixture over the range of pressure 20-100 kPa. The initial temperature is fixed at 300 K. The Equil code is used to calculate the constant volume explosion pressure. Figure 1.15 shows the evolution of the explosion pressure as a function of initial pressure. For an initial pressure of 100 kPa, the explosion pressure is on the order of 1.2 MPa. As a consequence, the combustion vessel should be designed to resist a maximum pressure of 1.5 MPa.

Let us consider a third case for which equilibrium calculations are relevant. In order to reduce the risk of fire in various industrial processes or commercial transport activities, flame retardants are commonly used. A promising flame retardant is CF_3I [12]. To determine its effectiveness at reducing the flame speed of some hydrocarbon-air mixtures, laboratory experiments need to be performed. Nevertheless, the combustion of CF_3I leads to the formation of HF which is a highly toxic compound, i.e. lethal concentration on the order of few 1000's ppm. Before performing the experiments, it is required to estimate the concentration of HF formed. We consider a stoichiometric butane-air mixture in which a quantity of CF_3I equal 10% of the fuel concentration has been added. Figure 1.16 shows the output of the Equil code considering a constant pressure combustion process. The mole fraction of HF at equilibrium is close to 14% which is 50 to 70 times the lethal concentration. As a consequence, special equipment is required to perform experiments with CF_3I-based mixtures.

MIXTURE:	INITIAL STATE:	EQUILIBRIUM STATE:
P (atm)	1.0000E+00	1.0000E+00
T (K)	3.0000E+02	1.6299E+03
V (cm3/gm)	7.7787E+02	5.5034E+03
H (erg/gm)	-3.5216E+09	-3.5216E+09
U (erg/gm)	-4.3098E+09	-9.0980E+09
S (erg/gm-K)	6.5881E+07	1.0271E+08
W (gm/mol)	3.1647E+01	2.4300E+01
Mol Fractions		
H	0.0000E+00	2.8015E-05
H2	0.0000E+00	1.4293E-01
O	0.0000E+00	1.1075E-10
O2	1.9733E-01	2.8974E-11
OH	0.0000E+00	7.9976E-07
H2O	0.0000E+00	8.2205E-02
N2	7.3588E-01	5.6504E-01
CO	0.0000E+00	1.6148E-01
CO2	0.0000E+00	2.9677E-02
CH2O	0.0000E+00	1.0379E-08
HCO	0.0000E+00	4.6035E-09
HOCHO	0.0000E+00	1.8994E-09
OCHO	0.0000E+00	1.0163E-12
CH4	0.0000E+00	6.0266E-09
CH3	0.0000E+00	2.4309E-11
C4H10	6.0720E-02	0.0000E+00
HF	0.0000E+00	1.3982E-02
F	0.0000E+00	1.8295E-10
CHF:O	0.0000E+00	2.4694E-10
CF:O	0.0000E+00	1.1060E-12
I2	0.0000E+00	1.7962E-06
HI	0.0000E+00	1.9477E-03
I	0.0000E+00	2.7095E-03
IO	0.0000E+00	2.0206E-11
IOH	0.0000E+00	1.6240E-08
CF3I	6.0700E-03	1.8510E-25

Figure 1.16: Composition at equilibrium for a C_4H_{10}-Air-CF_3I mixture considering a constant pressure combustion process. The initial temperature is 300 K and the initial pressure is 101 kPa.

Chapter 2

Chemical kinetics

2.1 Introduction

Thermodynamics predicts the most favorable (or normal) direction of change of a system given the present state of the system, i.e. its composition, temperature, and pressure. A transformation in the opposite direction requires to provide work to the system [2].

While thermodynamics predicts the direction of spontaneous transformation of a system, it does not consider the rate at which this transformation is taking place. Because the rate at which a system evolves toward its most favorable state can be very low, some spontaneous transformations can be never realized in practice [2]. As an example, the spontaneous reaction between molecular hydrogen and molecular oxygen, although highly exothermic, has a half reaction time of 10^{25} years at ambient temperature and pressure.

In order to determine if a chemical transformation can take place within a given amount of time, it is required to consider the laws of chemical kinetics which are described in the present chapter with emphasis on combustion and environmental applications. Chemical kinetics encompasses the study of the rate of chemical reactions and of the reaction mechanism

of complex chemical processes such as combustion reactions. The present chapter is limited to the reactions taking place in the gas phase.

2.2 Basic definitions in chemical kinetics

2.2.1 Reaction stoichiometry

In combustion, it is important to properly characterize the quantity of the reactants and dilutants. To do so, the first step is to write the global reaction equation. Let us consider the combustion of methane in pure oxygen. The global reaction equation is given by

$$CH_4 + 2O_2 = CO_2 + 2H_2O. \tag{2.1}$$

In order to establish Equation 2.1, the following procedure may be applied

- On the left hand side, write the reactants
- On the right hand side, write the products
- Equilibrate the reaction by applying the law of mass conservation

It is important to note that, by convention, the products considered in the global reaction equation correspond to the most stable form of the chemical elements which compose the reactants. The actual composition at equilibrium is not taken into account for describing the reaction stoichiometry. As an example, let us consider the combustion of ammonia in oxygen. The most stable product for N atom is N_2 whereas the most stable product for H atom is H_2O. Thus, the global equation is

$$NH_3 + \frac{3}{4}O_2 = \frac{1}{2}N_2 + \frac{3}{2}H_2O. \tag{2.2}$$

Global reaction equation can also be written for combustion in air

$$CH_4 + 2\left(O_2 + 3.76N_2\right) = CO_2 + 2H_2O + 2 \times 3.76N_2. \tag{2.3}$$

In this case, N_2 is considered as an inert and the ratio N_2/O_2 is fixed at 3.76.

To characterize the respective quantities of fuel and oxidant in a mixture, the equivalence ratio (Φ) is often used. It is defined as

$$\Phi = \frac{\left(\frac{X_F}{X_{Ox}}\right)_{mix}}{\left(\frac{X_F}{X_{Ox}}\right)_{sto}}, \tag{2.4}$$

where X_F is the fuel mole fraction; X_{Ox} is the oxidant mole fraction; mix refers to the mixture considered; and sto refers to the stoichiometric mixture which is obtained using the global reaction equation.

Because the combustion properties are usually very sensitive to the equivalence ratio, they are very often reported for a given equivalence ratio or as a function of equivalence ratio.

Experimentally or in numerical simulation, it is required to "prepare" a mixture of a given equivalence ratio. For this purpose, Equation 2.4 along with the global reaction equation are useful tools to calculate the mole fraction of the mixture components.

If the reactive mixture is not diluted, Equation 2.4 can be used directly. Since the fuel and the oxidant are the only mixture components, it comes

$$X_F + X_{Ox} = 1. \tag{2.5}$$

Substituting Equation 2.5 into Equation 2.4 leads to

$$X_F = \frac{\Phi}{\Phi + R_s}, \tag{2.6}$$

and

$$X_{Ox} = \frac{R_s}{\Phi + R_s}, \tag{2.7}$$

where R_s is the stoichiometric ratio and is defined as

$$R_s = \left(\frac{X_{Ox}}{X_F}\right)_{sto}. \tag{2.8}$$

If the reactive mixture is diluted, two cases have to be distinguished : (i) the ratio X_{Dil}/X_{Ox} is fixed ; and (ii) X_{Dil} is fixed.

In case (i), Equation 2.6 and Equation 2.7 can still be used but they constitute intermediate results. The mole fractions of the fuel, oxidant and dilutant are given by

$$X_F = \frac{\frac{\Phi}{\Phi+R_s}}{\frac{\Phi}{\Phi+R_s} + (R_d+1)\frac{R_s}{\Phi+R_s}}, \tag{2.9}$$

$$X_{Ox} = \frac{\frac{R_s}{\Phi+R_s}}{\frac{\Phi}{\Phi+R_s} + (R_d+1)\frac{R_s}{\Phi+R_s}}, \tag{2.10}$$

and

$$X_{Dil} = \frac{R_d\frac{R_s}{\Phi+R_s}}{\frac{\Phi}{\Phi+R_s} + (R_d+1)\frac{R_s}{\Phi+R_s}}, \tag{2.11}$$

where R_d is referred to as dilution ratio and is defined as

$$R_d = \frac{X_{Dil}}{X_{Ox}}. \tag{2.12}$$

In case (ii), the mole fractions of the fuel and oxidant are given by

$$X_F = (1 - X_{Dil})\frac{\Phi}{\Phi + R_s}, \tag{2.13}$$

and

$$X_{Ox} = (1 - X_{Dil})\frac{R_s}{\Phi + R_s}. \tag{2.14}$$

Under certain circumstances, such as the study of combustion in engine, it is usually preferred to characterize the mixture composition using the air excess (ε) which is defined as

$$\varepsilon = \frac{1}{\Phi}. \tag{2.15}$$

2.2.2 Reaction rate

Let us consider the following gas phase chemical reaction

$$\nu_A A + \nu_B B = \nu_C C + \nu_D D. \tag{2.16}$$

The total rate of reaction is defined as the number of molecule (or mole) of reactant which are consumed during a period dt (or as the number of molecule of product which are formed). This definition is expressed as

$$v_A = -\frac{dN_A}{dt} = -\frac{dn_A}{dt}, \tag{2.17}$$

or

$$v_C = \frac{dN_C}{dt} = \frac{dn_C}{dt} \tag{2.18}$$

It is noted that v is always positive so that if dN/dt is negative (for reactant consumption), then a negative sign has to be added before the derivative.
It is also noted that depending on which quantity is used to express the reaction rate (either N of n), the units of v will be different.

The specific rate is defined as the number of molecule (or mole) consumed/formed during a period dt per volume unit. For a constant volume condition, it is expressed as

$$v_A = -\frac{1}{V}\frac{dN_A}{dt} = -\frac{1}{V}\frac{dn_A}{dt} = -\frac{d[A]}{dt}. \tag{2.19}$$

The reaction rate can also be expressed for a unity stoichiometric coefficient. Let us consider the table of advancement shown in Table 2.1.

	$\nu_A A$	+	$\nu_B B$	=	$\nu_C C$	+	$\nu_D D$
t=0	a		b		0		0
t	a - $\nu_A x$		b - $\nu_B x$		$\nu_C x$		$\nu_D x$

Table 2.1: Table of advancement of the chemical reaction between A and B.

Considering a unity volume, we can write

$$-\frac{d[A]}{\nu_A} = -\frac{d\,(a - \nu_A x)}{\nu_A} = dx = -\frac{d\,(b - \nu_B x)}{\nu_B}. \tag{2.20}$$

From Equation 2.20, it comes that

$$v = \frac{dx}{dt} = -\frac{d[A]}{\nu_A dt} = -\frac{d[B]}{\nu_B dt} = \frac{d[C]}{\nu_C dt} = \frac{d[D]}{\nu_D dt}. \tag{2.21}$$

Finally, it is also possible to express the reaction rate as a function of the advancement degree (σ). The advancement degree may be expressed as

$$\sigma = \sigma_A = \frac{\nu_A x}{a} = \sigma_B = \frac{\nu_B x}{b}. \tag{2.22}$$

Using Equation 2.22, the reaction rate can be expressed as

$$v_\sigma = \frac{d\sigma}{dt}, \tag{2.23}$$

where v_σ has unit of 1/s.

2.2.3 Reaction order

For many chemical reactions, the rate of reaction can be expressed as

$$v = k[A]^\alpha [B]^\beta, \tag{2.24}$$

there k is the rate constant ; and α and β are the partial orders of reaction with respect to A and B, respectively.

The sum $\alpha + \beta = \gamma$ corresponds to the global order of the reaction.
Both the global reaction order and the partial orders need not be integers as in

$$v = k[A]^{1/2}[B], \tag{2.25}$$

for which the global order is 3/2.

The reaction order may also be zero. In this case, the reaction rate is independent of the reactant concentrations

$$v = k. \tag{2.26}$$

Finally, the reaction order can be indefinite as in the case of complex reactions such as

$$H_2 + Br_2 = 2HBr, \tag{2.27}$$

for which the reaction rate is described by

$$v = \frac{k[H_2][Br_2]^{3/2}}{[Br_2] + k'[HBr]}. \tag{2.28}$$

The reaction has an indefinite order with respect to Br_2 and HBr and has no overall order [2].

2.3 Mathematical description of reaction rate

2.3.1 Constant volume conditions

2.3.1.1 Zero-order reaction

A zero-order reaction corresponds to a special case since the chemical kinetics is controlled by a physical process such as a vaporization process or an heterogeneous catalytic process.

For a zero-order reaction, the reaction rate can be written as

$$v = k = -\frac{d[A]}{dt}, \tag{2.29}$$

where k is the rate constant and has units of mol.m^{-3}.s^{-1}.
Equation 2.29 can be written under its integrated form

$$\int -d[A] = \int k dt + cst, \tag{2.30}$$

which gives

$$-[A] = kt + cst. \tag{2.31}$$

At t=0, $[A] = [A]_0$ so that

$$[A]_0 - [A] = kt. \tag{2.32}$$

Considering a rate of constant $k = 1$ mol.m^{-3}.s^{-1} and an initial concentration of reactant $[A]_0 = 1$ mol, the evolution of the reaction rate and of the reactant concentration as a function of time can be represented as shown in Figure 2.1. The reaction rate is equal to the rate constant and the reactant concentration is varying linearly with time. The reaction rate

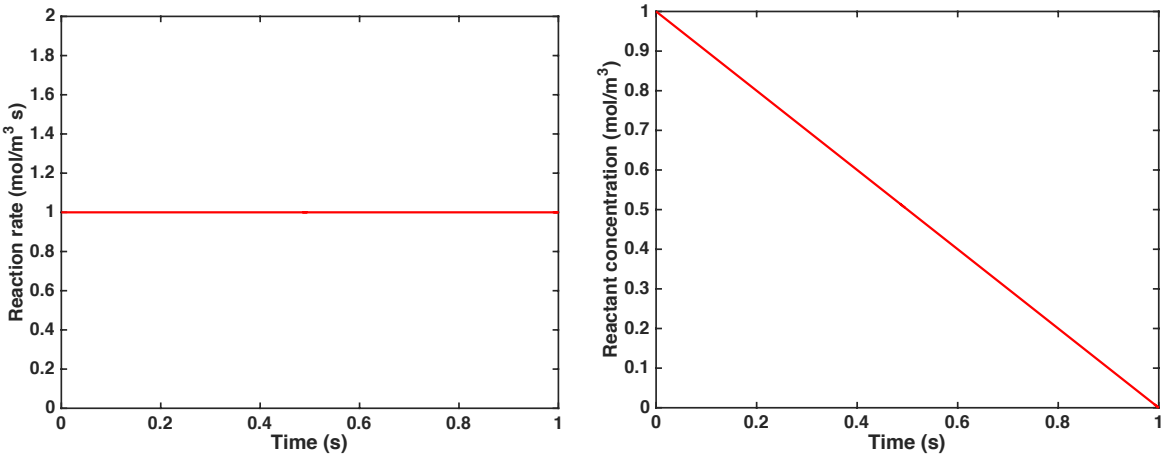
corresponds to the absolute value of slope of $[A]$=f(t).

It is also possible to define a particular time of reaction, the half-reaction time ($t_{1/2}$), which corresponds to the time at which the reactant concentration is half of the initial one. Since at $t = t_{1/2}$, $[A] = [A]_0/2$, it can be written that

$$[A]_0 - \frac{[A]_0}{2} = kt_{1/2}, \tag{2.33}$$

so that

$$t_{1/2} = \frac{[A]_0}{2k}. \tag{2.34}$$



a) Reaction rate b) Reactant concentration

Figure 2.1: Evolution of the reaction rate and of the reactant concentration as a function of time for a zero-order reaction.

2.3.1.2 First-order reaction

Let us consider a first-order reaction which evolution can be described by Table 2.2.

	A	=	P
t=0	a		0
t	a - x		x

Table 2.2: Table of advancement of a first-order reaction.

The rate of reaction can be written as

$$-\frac{d[A]}{dt} = \frac{dx}{dt} = k\,(a-x)\,, \tag{2.35}$$

where k has units of s^{-1}.

Equation 2.35 can be written under its integrated form

$$\int kdt = \int \frac{dx}{a-x} + cst, \tag{2.36}$$

which becomes

$$kt = \ln\frac{1}{a-x} + cst. \tag{2.37}$$

At t=0, $x = 0$ so that

$$cst = \ln a. \tag{2.38}$$

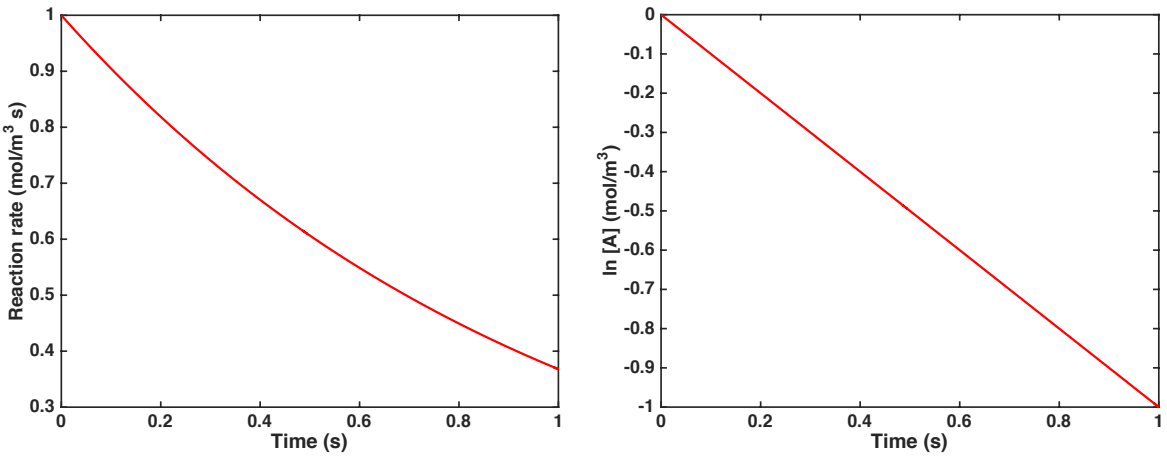
Substituting Equation 2.38 into Equation 2.37, we obtain

$$kt = \ln\frac{a}{a-x}. \tag{2.39}$$

Equation 2.39 may also be written as

$$[A] = [A]_0 \exp(-kt). \tag{2.40}$$

Considering a rate of constant $k = 1\ s^{-1}$ and an initial concentration of reactant $[A]_0 = 1$ mol, the evolution of the reaction rate and of the reactant concentration as a function of time can be represented as shown in Figure 2.2. The reaction rate is maximum at $t = 0$. The Napierian logarithm of the reactant concentration is varying linearly with time. The reaction



a) Reaction rate b) Reactant concentration

Figure 2.2: Evolution of the reaction rate and of the reactant concentration as a function of time for a first-order reaction.

rate corresponds to the absolute value of the slope of ln[A]=f(t).

It is also possible to define the half-reaction time for a first-order reaction. Since at $t = t_{1/2}$, $x = [A]_0/2$, it can be written that

$$kt_{1/2} = \ln \frac{[A]_0}{[A]_0 - [A]_0/2}, \tag{2.41}$$

so that

$$t_{1/2} = \frac{\ln 2}{k}. \tag{2.42}$$

2.3.1.3 Second-order reaction

For a second-order reaction, two cases need to be distinguished : (i) there is only one reactant ; (ii) there are two reactants.

In the one reactant case, the evolution of the reaction can be described by Table 2.3 which is similar to Table 2.2 used for the first-order reaction.

	A	=	P
t=0	a		0
t	a - x		x

Table 2.3: Table of advancement of a second-order reaction with a single reactant.

The rate of reaction can be written as

$$-\frac{d[A]}{dt} = \frac{dx}{dt} = k\left(a-x\right)^2, \tag{2.43}$$

where k has units of $\mathrm{cm^3.mol^{-1}.s^{-1}}$.

Equation 2.43 can be written under its integrated form

$$\int kdt = \int \frac{dx}{\left(a-x\right)^2} + cst, \tag{2.44}$$

which becomes

$$kt = \frac{1}{a-x} + cst. \tag{2.45}$$

At t=0, $x = 0$ so that

$$cst = -\frac{1}{a}. \tag{2.46}$$

Substituting Equation 2.38 into Equation 2.37, we obtain

$$kt = \frac{1}{[A]} - \frac{1}{[A]_0}. \tag{2.47}$$

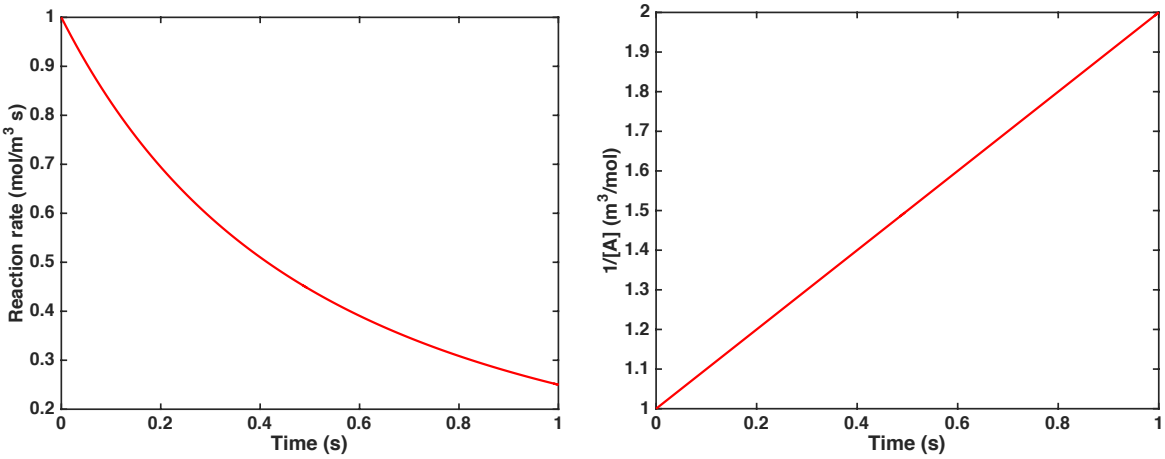
Considering a rate of constant $k = 1$ $\mathrm{cm^3.mol^{-1}.s^{-1}}$ and an initial concentration of reactant $[A]_0 = 1$ mol, the evolution of the reaction rate and of the reactant concentration as a function of time can be represented as shown in Figure 2.3. The reaction rate is maximum at $t = 0$. The inverse of the reactant concentration is varying linearly with time. The reaction rate corresponds to the absolute value of the slope of $1/[A]$=f(t).

It is also possible to define the half-reaction time for a first-order reaction. Since at $t = t_{1/2}$, $x = [A]_0/2$, it can be written that

$$kt_{1/2} = \frac{1}{[A]_0/2} - \frac{1}{[A]_0}, \tag{2.48}$$

so that

$$t_{1/2} = \frac{1}{k[A]_0}. \tag{2.49}$$



a) Reaction rate b) Reactant concentration

Figure 2.3: Evolution of the reaction rate and of the reactant concentration as a function of time for a second-order reaction with a single reactant.

In the two reactants case, the evolution of the reaction can be described by Table 2.4.

	A	+	B	=	P
t=0	a		b		0
t	a - x		b - x		x

Table 2.4: Table of advancement of a second-order reaction with two reactants.

The rate of reaction can be written as

$$-\frac{d[A]}{dt} = \frac{dx}{dt} = k\,(a-x)\,(b-x)\,, \tag{2.50}$$

where k has units of cm^3.mol^{-1}.s^{-1}. It is noted that the partial orders on A and B are both taken equal to 1.

Equation 2.50 can be written under its integrated form

$$\int k dt = \int \frac{dx}{(a-x)\,(b-x)} + cst. \tag{2.51}$$

In order to integrate Equation 2.51, it is needed to write the expression on the right hand side in a different way.

Let us consider that

$$\frac{1}{(a-x)(b-x)} = \frac{A}{(a-x)} + \frac{B}{(b-x)}. \tag{2.52}$$

From Equation 2.52, it is found that

$$\frac{1}{(b-x)} = A + B\frac{(a-x)}{(b-x)}. \tag{2.53}$$

and

$$\frac{1}{(a-x)} = A\frac{(b-x)}{(a-x)} + B. \tag{2.54}$$

For $x = a$, we find that Equation 2.53 becomes

$$A = \frac{1}{(b-a)}. \tag{2.55}$$

For $x = b$, we find that Equation 2.54 becomes

$$B = \frac{1}{(a-b)} = -A. \tag{2.56}$$

Substituting Equation 2.52, Equation 2.55, and Equation 2.56 into Equation 2.51 gives

$$kt = \frac{1}{b-a}\left(\int \frac{dx}{(a-x)} - \int \frac{dx}{(b-x)}\right) + cst, \tag{2.57}$$

which leads to

$$kt = \frac{1}{b-a}\ln\frac{(b-x)}{(a-x)} + cst. \tag{2.58}$$

At t=0, $x = 0$ so that

$$cst = \frac{1}{b-a}\ln\frac{a}{b}. \tag{2.59}$$

Substituting Equation 2.59 into Equation 2.58, we obtain

$$kt = \frac{1}{b-a}\ln\frac{a}{b}\frac{(b-x)}{(a-x)} = \frac{1}{[B]_0 - [A]_0}\ln\frac{[A]_0}{[B]_0}\frac{[B]}{[A]}. \tag{2.60}$$

Equation 2.60 may be re-arranged as

$$\ln\frac{[A]}{[B]} = ([A]_0 - [B]_0)\,kt + \ln\frac{[A]_0}{[B]_0}. \tag{2.61}$$

The evolution of the reactant concentration as a function of time can be represented as shown in Figure 2.4. The Napierian logarithm of the reactant concentration ratio is varying linearly with time. The reaction rate corresponds to the absolute value of the slope of $\ln[A]/[B]$=f(t). The reaction rate is maximum at $t = 0$ because the product $[A] \times [B]$ is maximum.

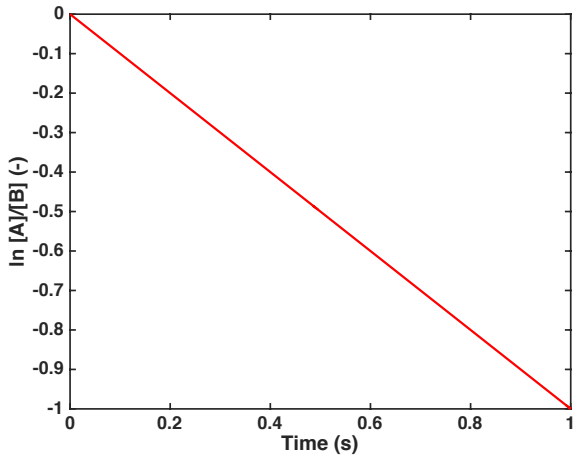


Figure 2.4: Evolution of the reactant concentration as a function of time for a second-order reaction with a two reactants.

Since the chemical reaction involves two reactants, it is not possible to define a single half-reaction time.

2.3.1.4 Third-order reaction

For a third-order reaction, two cases can be distinguished : (i) there is only one reactant ; (ii) there are two reactants.

In the one reactant case, the evolution of the reaction can be described by Table 2.5.

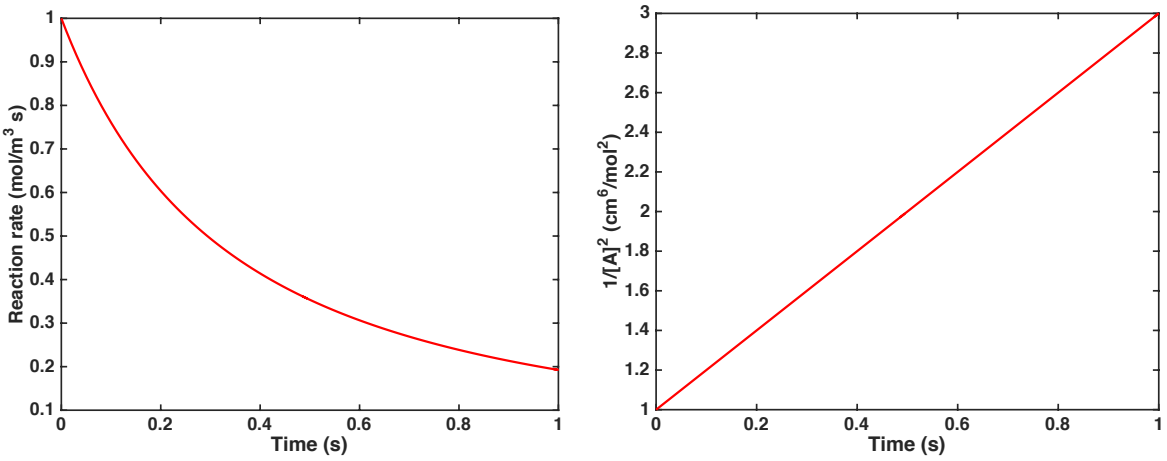
The reaction rate is given by

$$-\frac{d[A]}{dt} = \frac{dx}{dt} = k\,(a-x)^3\,, \tag{2.62}$$

where k has units of $\mathrm{cm}^6.\mathrm{mol}^{-2}.\mathrm{s}^{-1}$.

	A	$=$	P
t=0	a		0
t	a - x		x

Table 2.5: Table of advancement of a third-order reaction with one reactants.



a) Reaction rate b) Reactant concentration

Figure 2.5: Evolution of the reaction rate and of the reactant concentration as a function of time for a third-order reaction with a single reactant.

The integrated rate law reads

$$kt = \frac{1}{2}\left(\frac{1}{([A]_0 - x)^2} - \frac{1}{[A]_0^2}\right). \tag{2.63}$$

Figure 2.5 shows the evolution of the reaction rate and of $1/[A]^2$ as a function of time. The reaction rate is maximum at the beginning of the reaction and $1/[A]^2$=f(t) is a straight line.

The time of half reaction is given by

$$t_{1/2} = \frac{3}{2k[A]_0^2}. \tag{2.64}$$

In the two reactants case, the evolution of the reaction can be described by Table 2.6.

	A	+	$2B$	=	P
t=0	a		b		0
t	a - x		b - $2x$		x

Table 2.6: Table of advancement of a third-order reaction with two reactants.

The reaction rate is given by

$$-\frac{d[A]}{dt} = \frac{dx}{dt} = k\,(a-x)\,(b-x)^2\,, \tag{2.65}$$

where k has units of $\mathrm{cm}^6.\mathrm{mol}^{-2}.\mathrm{s}^{-1}$.

The integrated rate law reads

$$kt = \frac{1}{(2[A]_0 - [B]_0)^2}\left(\frac{2x\,(2[A]_0 - [B]_0)}{[B]_0\,([B]_0 - 2x)} + \ln\left(\frac{[A]_0\,([B]_0 - 2x)}{[B]_0\,([A]_0 - x)}\right)\right). \tag{2.66}$$

2.3.1.5 Fractional order reaction

The reaction order need not be an integer as we have considered so far. In the case of a chemical reaction with fractional order (n), we also can distinguish two cases : (i) there is only one reactant ; (ii) there are two reactants.

In the one reactant case, the reaction rate is given by

$$-\frac{d[A]}{dt} = \frac{dx}{dt} = k\,(a-x)^n\,, \tag{2.67}$$

where k has units of $\mathrm{cm}^{3(n-1)}.\mathrm{mol}^{-(n-1)}.\mathrm{s}^{-1}$.

The integrated rate law reads

$$kt = \frac{1}{n-1}\left(\frac{1}{([A]_0 - x)^{n-1}} - \frac{1}{[A]_0^{n-1}}\right). \tag{2.68}$$

The time of half reaction is given by

$$t_{1/2} = \frac{1}{k\,(n-1)\,[A]_0^{n-1}}\left(2^{n-1}-1\right). \tag{2.69}$$

In the two reactants case, the evolution of the reaction can be described by Table 2.7.

	A	+	sB	=	P
t=0	a		b		0
t	a - x		b - sx		x

Table 2.7: Table of advancement of a reaction with fractional order and two reactants.

The reaction rate is given by

$$-\frac{d[A]}{dt} = \frac{dx}{dt} = k\,(a-x)^{n_A}\,(sa-sx)^{n_B}, \tag{2.70}$$

where s is the stoichiometric coefficient for B, so that b=sa.

Equation 2.70 can be re-written as

$$\frac{dx}{dt} = ks^{n_B}\,(a-x)^{n_A+n_B} = k'\,(a-x)^{n}, \tag{2.71}$$

where k'=ks^{n_B} and n=$n_A + n_B$.

The integrated rate law reads

$$k't = \frac{1}{n-1}\left(\frac{1}{([A]_0-x)^{n-1}} - \frac{1}{[A]_0^{n-1}}\right). \tag{2.72}$$

2.3.2 Variable volume conditions

Let us consider a chemical reaction which is taking place in a reactor of variable volume. The global reaction is

$$A + B = P. \tag{2.73}$$

The reaction rate can be written as

$$v = -\frac{1}{V}\frac{dN_A}{dt} = k\left(\frac{N_A}{V}\right)^{n_A}\left(\frac{N_B}{V}\right)^{n_B}. \tag{2.74}$$

The variation of the concentration of A is given by

$$-\frac{d[A]}{dt} = -\frac{d}{dt}\left(\frac{N_A}{V}\right) = -\frac{1}{V}\frac{dN_A}{dt} + \frac{N_A}{V^2}\frac{dV}{dt}. \tag{2.75}$$

Substituting Equation 2.74 into Equation 2.75 leads to

$$-\frac{d[A]}{dt} = k[A]^{n_A}[B]^{n_B} + \frac{[A]}{V}\frac{dV}{dt}. \tag{2.76}$$

2.3.3 Determination of the reaction order

In order to determine if a chemical reaction follows a simple kinetic law or if a complex chemical mechanism, it is useful to determine the reaction order. Several approaches can be employed to do so.

2.3.3.1 Correlation between concentration and time

A first approach is to try to establish a correlation between the reactant concentration and time.

If a reaction involves only one reactant and is taking place in a constant volume reactor, the following rules apply

- if $[A]$=f(t) is linear, n=0
- if ln$[A]$=f(t) is linear, n=1
- if $1/[A]$=f(t) is linear, n=2
- if $1/[A]^2$=f(t) is linear, n=3

If a reaction involves several reactants and is taking place in a constant volume reactor, the method of the degenerated order may be applied if

- $b \gg a$
- $x < a$

because, in this case, $b \approx cst$.

The rate law can be written as

$$v = k\,(a-x)^{n_A}\,(b-x)^{n_B} \approx k\,(a-x)^{n_A}\,b^{n_B} = k_{app}\,(a-x)^{n_A}\,, \tag{2.77}$$

where

$$k_{app} = k b^{n_B}. \tag{2.78}$$

Equation 2.78 may be re-written as

$$\ln k_{app} = n_B \ln b + \ln k. \tag{2.79}$$

As shown in Figure 2.6, the slope of the curve $\ln K_{app}$=f($\ln b$) corresponds to n_B.

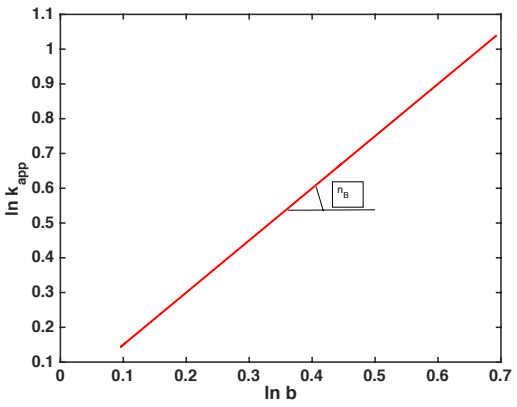


Figure 2.6: Evolution of $\ln k_{app}$ as a function of $\ln b$ for determining the reaction order using the method of the degenerated order.

2.3.3.2 The method of half-reaction time

The method of half-reaction time can be applied only to reactions which involve only one reactant.

If the reaction order is equal to 1, we know from Equation 2.42 that $t_{1/2}$=ln 2/k. Thus, the half-reaction time is independent of the concentration of reactant A.

If the reaction order n is different from 1, we can write the following relationship

$$\ln t_{1/2} = (1-n)\ln a + \ln\left(\frac{2^{n-1}-1}{(n-1)\,k}\right). \tag{2.80}$$

From Equation 2.80, it is seen that the half-time of reaction depends on a and that the slope of ln $t_{1/2}$=f(ln a) corresponds to 1-n.

2.3.3.3 The method of successive characteristic times

The method of successive characteristic times enables to determine if the reaction order is constant during the course reaction. As it is the case for the method of half-reaction time, it can be applied only to reactions which involve only one reactant.

Figure 2.7 shows the positions of three characteristic times of reaction during the course of a chemical reaction monitored by measuring the concentration of the reactant as a function of time.

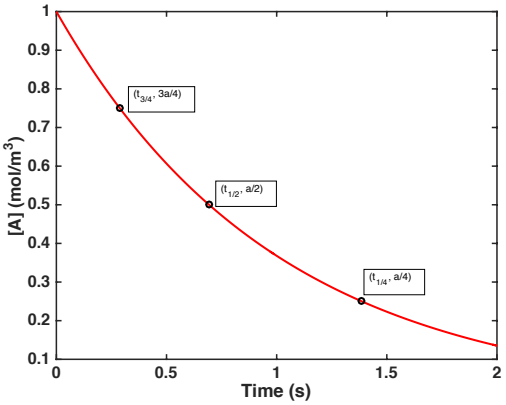


Figure 2.7: Positions of three characteristic times of reaction during the course of a chemical reaction.

If the reaction order is equal to 1, the ratio of two characteristic times is given by

$$\frac{t_{1/p}}{t_{1/q}} = \frac{\ln\left(p/\left(p-1\right)\right)}{\ln\left(q/\left(q-1\right)\right)}, \tag{2.81}$$

where p and q are numbers which define two characteristic times.

If the reaction order is different from 1, the ratio of two characteristic times is given by

$$\frac{t_{1/p}}{t_{1/q}} = \frac{\left(\frac{p}{p-1}\right)^{n-1} - 1}{\left(\frac{q}{q-1}\right)^{n-1} - 1}. \tag{2.82}$$

Table 2.8 gives the characteristic values of the characteristic times ratios as a function of the reaction order.

n	1	2	3
$t_{1/2}/t_{1/4}$	2.41	3	3.86
$t_{1/2}/t_{1/3}$	1.71	2	2.40
$t_{3/4}/t_{1/2}$	2	3	5

Table 2.8: Characteristic values of the characteristic times ratios as a function of the reaction order.

2.3.3.4 The Van't Hoff method

The method of Van't Hoff is based on the measurement of the instantaneous reaction rate. Considering the reaction

$$A + sB = P, \tag{2.83}$$

for which the reaction rate is given by

$$v = k\left(a-x\right)^{n_A}\left(b-sx\right)^{n_B}. \tag{2.84}$$

If the initial concentration of B is maintained constant and the initial concentration of A is changed, the reaction rate becomes

$$v' = k\left(a' - x\right)^{n_A}\left(b - sx\right)^{n_B}. \tag{2.85}$$

The ratio of the reaction rates between two experiments is given by

$$\frac{v}{v'} = \frac{\left(a - x\right)^{n_A}}{\left(a' - x\right)^{n_A}}. \tag{2.86}$$

Equation 2.86 can be re-arranged as

$$n_A = \frac{\ln\left(\frac{v}{v'}\right)}{\ln\left(\frac{a-x}{a'-x}\right)}. \tag{2.87}$$

Figure 2.8 shows the evolution of $\ln\left(v/v'\right)$ as a function of $\ln\left(\left(a - x\right)/\left(a' - x\right)\right)$. If the reaction order with respect to A, n_A, is constant during the reaction, then, a straight line is observed. If the reaction is accelerated by the addition of A, then an increase of $\ln\left(v/v'\right)$ with the increase of $\ln\left(\left(a - x\right)/\left(a' - x\right)\right)$ is observed. Such an evolution is referred to as "self-acceleration". Finally, if the reaction is slowed down by the addition of A, then a decrease of $\ln\left(v/v'\right)$ with the increase of $\ln\left(\left(a - x\right)/\left(a' - x\right)\right)$ is observed. Such an evolution is referred to as "self-inhibition".

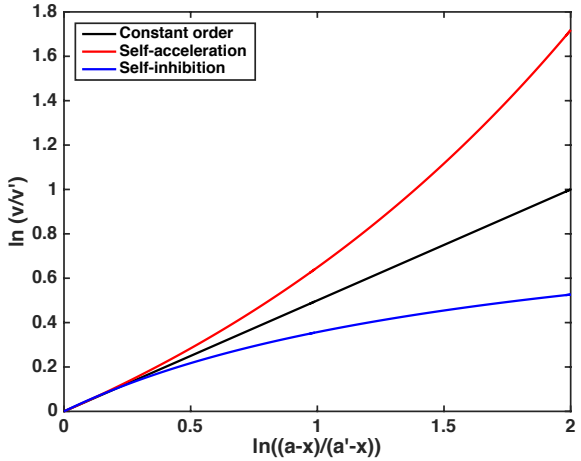


Figure 2.8: Illustration of the method of Van't Hoff : evolution of the reaction rate as a function of reactant concentration.

2.4 The kinetics of complex reactions

2.4.1 Elementary reaction

In general, chemical processes can be divided into a number of elementary chemical steps which are called "elementary reactions". An elementary reaction is defined as a chemical reaction in which the reactants form the products within a single reaction step and with a single transition state. An example of elementary reaction is

$$H + O_2 = OH + O. \tag{2.88}$$

A series of elementary reactions which describe the kinetics of a chemical system is referred to as "reaction mechanism" or "reaction model".

2.4.2 Temperature-dependence of reaction rate

For many reactions, the rate constant (k) is found to increase with temperature. This behavior is illustrated in Figure 2.9 which shows the evolution of k as a function of the reciprocal temperature, i.e. $1/T$.

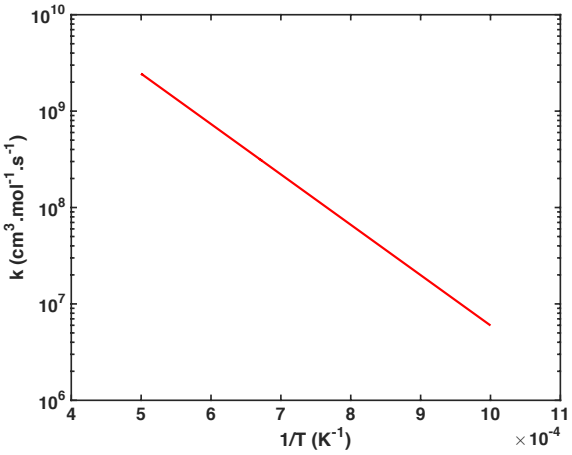


Figure 2.9: Evolution of the rate constant as a function of the reciprocal temperature for a second-order reaction.

The evolution of the rate constant with temperature is described mathematically by the so called "Arrhenius law" which is expressed as

$$k = A \exp\left(\frac{-E_a}{RT}\right), \tag{2.89}$$

where A is the pre-exponential factor or collision frequency factor; and E_a is the activation energy. The pre-exponential factor corresponds to the interception of $\ln k$=f$(1/T)$ with the y axis at $1/T$=0 which corresponds to an "infinite temperature". The activation energy corresponds to the slope of $\ln k$=f$(1/T)$ multiplied by the perfect gas constant R.

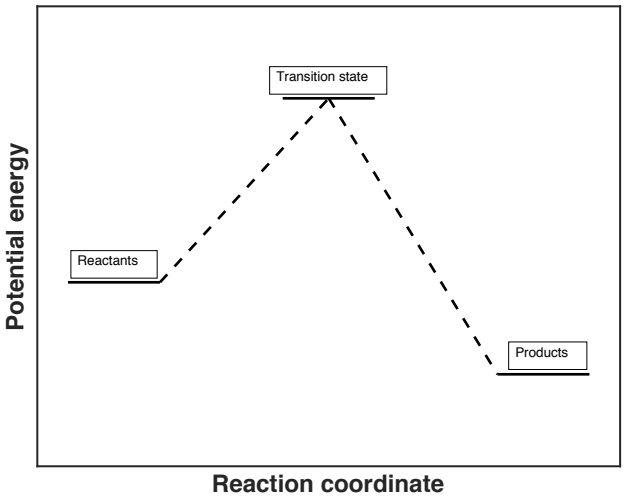


Figure 2.10: Evolution of the potential energy during a chemical reaction.

In order to understand the physical meaning of the activation energy, it is useful to examine the evolution of the potential energy during a chemical reaction as shown in Figure 2.10. The reaction coordinate corresponds to the change of geometry of the molecules as they get closer and closer to each other and then, as the products get further and further apart from each other. As the distance between the reactants decreases, the potential energy increases and reach a maximum as a particular molecule is formed, the transition state, which includes both molecules under a particular geometry. In order for the reaction to proceed, the kinetic energy of the reactants needs to be high enough to overcome the energy required to form the

transition state. Since the distribution of the kinetic energy follows a Boltzmann distribution, only a fraction of the collision between the molecules have enough energy for the reaction to take place [2].

The pre-exponential factor corresponds to the total rate of collision between the molecules. As a consequence, the product $A \times \exp(-Ea/RT)$ represents the rate of collisions which lead to the formation of the product [2].

2.4.3 Rate-determining step

2.4.3.1 Successive elementary reactions

Under certain conditions, a complex chemical process might be described as a succession of elementary reactions such as

$$A \xrightarrow{k_a} I \xrightarrow{k_b} P, \tag{2.90}$$

where A is the reactant ; I is the intermediate species ; P is the product ; k_a and k_b are the rate constants of the first and second reaction, respectively.

The variations of the chemical species concentration are given by

$$\frac{d[A]}{dt} = -k_a[A], \tag{2.91}$$

$$\frac{d[I]}{dt} = k_a[A] - k_b[I], \tag{2.92}$$

and

$$\frac{d[P]}{dt} = k_b[I]. \tag{2.93}$$

The integrated expression for the concentration of A is readily obtained from Equation 2.91. It reads

$$[A] = [A]_0 \exp\left(-k_a t\right). \tag{2.94}$$

The concentration of the intermediate species is given by

$$[I] = \frac{k_a}{k_b - k_a} \left(\exp\left(-k_a t\right) - \exp\left(-k_b t\right)\right) [A]_0. \tag{2.95}$$

Because at all time, $[A] + [I] + [P] = [A]_0$, the concentration of P is expressed as

$$[P] = \left(1 + \frac{k_a \exp(-k_b t) - k_b \exp(-k_a t)}{k_b - k_a}\right) [A]_0. \tag{2.96}$$

Equation 2.94 to Equation 2.96 can be used to calculate the evolution of the species concentration as a function of time. This is shown in Figure 2.11 for $[A]_0$=1 ; k_a=6 ; and k_b=8.

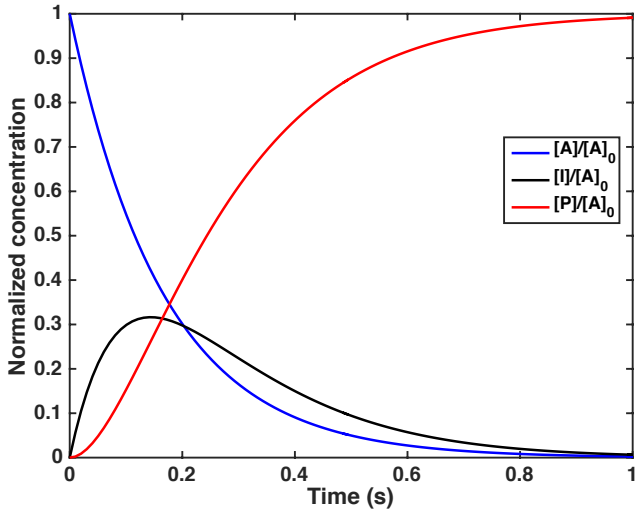


Figure 2.11: Evolution of the species concentration as a function of time for a 2-step reaction model.

In order to determine which is the rate-limiting step among reaction 1 and reaction 2, we can consider that the reaction rate is given by

$$v = \frac{d[P]}{dt} = k_b[I] = \frac{k_a k_b}{k_b - k_a} \left(\exp(-k_a t) - \exp(-k_b t)\right) [A]_0. \tag{2.97}$$

If $k_a \gg k_b$, then $k_b - k_a \simeq -k_a$ and $\exp(-k_a t) - \exp(-k_b t) \simeq -\exp(-k_b t)$, so thatEquation 2.97 reduces to

$$v = [A]_0 k_b \exp(-k_b t). \tag{2.98}$$

If $k_a \ll k_b$, then $k_b - k_a \simeq k_b$ and $\exp(-k_a t) - \exp(-k_b t) \simeq \exp(-k_a t)$, so thatEquation 2.97 reduces to

$$v = [A]_0 k_a \exp(-k_a t). \tag{2.99}$$

From Equation 2.98 and Equation 2.99, it can be concluded that the overall reaction rate of a chemical process which can be described by successive elementary reactions is controlled by the rate of the slowest step.

2.4.3.2 Competitive elementary reactions

Under certain conditions, a complex chemical process might be described by competitive elementary reactions such as

$$A \xrightarrow{k_a} P_1, \tag{2.100}$$

and

$$A \xrightarrow{k_b} P_2, \tag{2.101}$$

where A is the reactant ; P_1 and P_2 are the products ; k_a and k_b are the rate constants of the first and second reaction, respectively.

In order to determine which is the rate-limiting step among reaction 1 and reaction 2, we can consider that the reaction rate is given by

$$v = -\frac{d[A]}{dt} = k_a[A] + k_b[A] = (k_a + k_b)\,[A]. \tag{2.102}$$

If $k_a \gg k_b$, then $k_a + k_b \simeq k_a$, so that Equation 2.102 becomes

$$v = k_a[A]. \tag{2.103}$$

If $k_a \ll k_b$, then $k_a + k_b \simeq k_b$, so that Equation 2.102 becomes

$$v = k_b[A]. \tag{2.104}$$

From Equation 2.103 and Equation 2.104, it can be concluded that the overall reaction rate of a chemical process which can be described by competitive elementary reactions is controlled by the rate of the fastest step.

2.4.3.3 Quasi-steady-state approximation

Under certain conditions, it is possible to reduce the complexity of a complex reaction mechanism by applying the quasi-steady-state approximation ($QSSA$). To illustrate this concept, let us consider again two consecutive reactions such as

$$A \xrightarrow{k_a} I \xrightarrow{k_b} P. \tag{2.105}$$

The variations of the chemical species concentration are given by

$$\frac{d[A]}{dt} = -k_a[A], \tag{2.106}$$

$$\frac{d[I]}{dt} = k_a[A] - k_b[I], \tag{2.107}$$

and

$$\frac{d[P]}{dt} = k_b[I]. \tag{2.108}$$

The concentration of A is given by

$$[A] = [A]_0 \exp\left(-k_a t\right). \tag{2.109}$$

The QSSA is applied to the intermediate species and consists in setting

$$\frac{d[I]}{dt} \simeq 0, \tag{2.110}$$

which leads to

$$[I] = \frac{k_a}{k_b}[A] = \frac{k_a}{k_b}[A]_0 \exp\left(-k_a t\right). \tag{2.111}$$

Substituting Equation 2.111 into Equation 2.108, we obtain

$$\frac{d[P]}{dt} = k_b[I] \simeq k_b \frac{k_a}{k_b}[A] = k_a[A]. \tag{2.112}$$

Integrating Equation 2.112, we obtain

$$[P] = k_a[A] \int_0^t \exp\left(-k_a t\right) dt = \left(1 - \exp\left(-k_a t\right)\right)[A]_0. \tag{2.113}$$

Using Equation 2.94, Equation 2.111, and Equation 2.113, it is possible to calculate the evolution of the species concentration as a function of time. This is shown in Figure 2.12

for $[A]_0$=1 ; k_a=6 ; and k_b=20. The differences between the profiles of $[I]$ and of $[P]$ and the profiles of $[I]_{QSSA}$ and of $[P]_{QSSA}$ depend on the level of validity of the condition $d[I]/dt \simeq 0$. The kinetics implication of this condition can be deduced by considering that in this case, $[I]$ will be small with respect to $[A]_0$ at all time. This leads to

$$\frac{[I]}{[A]_0} = \frac{k_a}{k_b} \exp\left(-k_a t\right) \ll 1, \tag{2.114}$$

which means that $k_b \gg k_a$.

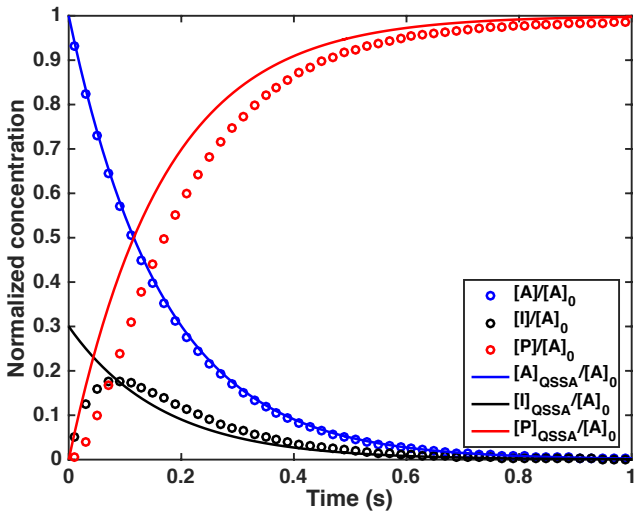


Figure 2.12: Evolution of the species concentration as a function of time for a 2-step reaction model with and without QSSA.

The QSSA can be applied to more complex reaction mechanism such as the one describing the formation of electronically excited OH* radicals which are often used in combustion as a marker of auto-ignition.

Considering the hydrogen-nitrous oxide system, the reactions forming and consuming OH* are

— $R_1 : \ H + O + M = OH* + M$

— $R_2 : \ OH* + M => OH + M$

— $R_3 : \ OH* => OH* + h\nu$

— $R_4 : \ N_2O + H => N_2 + OH*$

Applying the QSSA to OH*, the following expression can be written

$$\frac{d[OH*]}{[dt]} = k_1[H][O][M] - k_{-1}[OH*][M] - k_2[OH*][M] - k_3[OH*] + k_4[N_2O][H] = 0, \quad (2.115)$$

which leads to

$$[OH*]_{QSSA} = \frac{k_1[H][O][M] + k_4[N_2O][H]}{(k_2 + k_{-1})[M] + k_3}. \quad (2.116)$$

It is noted that to obtain Equation 2.116, the reaction R_1 has been considered as reversible whereas the other three reactions have been described as irreversible.

Figure 2.13 compares the OH* profiles obtained with a full detailed reaction model (a system of about 200 differential equations is used to describe the chemistry and is solved numerically) and with the analytic expression Equation 2.116. Qualitative and quantitative agreement is observed.

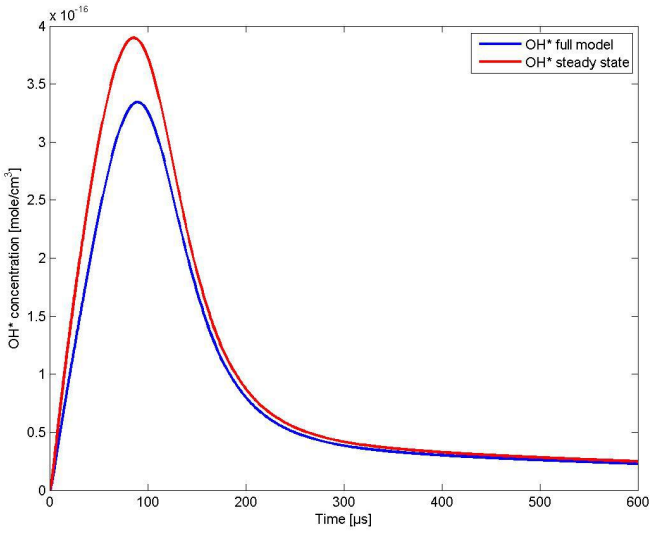


Figure 2.13: Comparison between the OH* profiles obtained with a detailed reaction model and a QSSA expression. Taken from [13].

2.4.4 Chain reactions

A chain reaction mechanism corresponds to an ensemble of elementary reactions which are repeated cyclically. In a chain reaction, a reaction intermediate produced in one step generates an intermediate in a subsequent step, then that intermediate generates another intermediate, and so on [2]. In a chain reaction, the intermediates are referred to as "chain carriers". The different types of reactions which can be encountered in a chain mechanism are :

- Initiation reaction : the initial step of formation of the chain carriers
- Propagation reaction : a chain carrier reacts with the reactant to form a new chain carrier
- Branching reaction : produces more than one chain carrier
- Inhibition reaction : a chain carrier reacts with the reaction product, which slows down the overall reaction rate
- Termination reaction : chain carriers react together to form a stable species

Two kind of chain reactions can be distinguished : (i) the linear chain mechanism ; and (ii) the chain branching mechanism. For a linear chain mechanism, no chain branching reaction are taking place whereas for a chain branching mechanism, a number of chain branching reactions are taking place and have a strong influence on the overall reaction rate.

2.4.4.1 Linear chain mechanism

An example of linear chain mechanism corresponds to the gas phase synthesis of hydrogen bromide (HBr) from hydrogen and bromine

$$H_2(g) + Br_2(g) = 2HBr(g). \tag{2.117}$$

This overall reaction can be described by the following chain mechanism

- Initiation reaction : (1) $Br_2 + M => 2Br + M$
- Propagation reaction : (2) $Br + H_2 => HBr + H$
- Propagation reaction : (3) $H + Br_2 => HBr + Br$
- Inhibition reaction : (4) $HBr + H => Br + H_2$

— Termination reaction : (5) $2Br + M => Br_2 + M$

The rate law of the reaction is given by

$$v = \frac{d[HBr]}{dt} = k_2[Br][H_2] + k_3[H][Br_2] - k_4[HBr][H]. \tag{2.118}$$

Because Equation 2.118 includes the concentration of intermediate species, it cannot be directly used to estimate the rate of reaction. In order to obtain a more useful expression, the QSSA may be applied to the intermediate species, H and Br, which gives

$$\frac{d[H]}{dt} = k_2[Br][H_2] - k_3[H][Br_2] - k_4[HBr][H] = 0, \tag{2.119}$$

and

$$\frac{d[Br]}{dt} = 2k_1[Br_2][M] - k_2[Br][H_2] + k_3[H][Br_2] + k_4[HBr][H] - 2k_5[Br]^2[M] = 0. \tag{2.120}$$

From Equation 2.119, we find

$$[H] = \frac{k_2[Br][H_2]}{k_3[Br_2] + k_4[HBr]}. \tag{2.121}$$

By combining Equation 2.119 and Equation 2.120, we find

$$2k_1[Br_2][M] - 2k_5[Br]^2[M] = 0, \tag{2.122}$$

which leads to

$$[Br] = \left(\frac{k_1}{k_5}[Br_2]\right)^{1/2}. \tag{2.123}$$

Substituting Equation 2.121 and Equation 2.123 into Equation 2.118, we obtain

$$v = \frac{d[HBr]}{dt} = \frac{2k_2\left(\frac{k_1}{k_5}\right)^{1/2}[H_2][Br_2]^{3/2}}{[Br_2] + \frac{k_4}{k_3}[HBr]}, \tag{2.124}$$

which can be re-written as

$$v = \frac{d[HBr]}{dt} = \frac{k_a[H_2][Br_2]^{3/2}}{[Br_2] + k_b[HBr]}, \tag{2.125}$$

where

$$k_a = 2k_2\left(\frac{k_1}{k_5}\right)^{1/2}, \tag{2.126}$$

and

$$k_b = \frac{k_4}{k_3}. \tag{2.127}$$

2.4.4.2 Chain branching mechanism

An example of chain branching mechanism corresponds to the combustion of hydrogen-oxygen mixtures. Under certain conditions, a chain branching explosion may be observed. Such an explosion is characterized by an exponential growth of active radical or chain carrier. It is to be distinguished from thermal explosion which is characterized by a feedback loop between the overall reaction rate and temperature. In this case, the exothermicity of the reaction induces an increase of the temperature which induces an increase of the rate of reaction and so on.

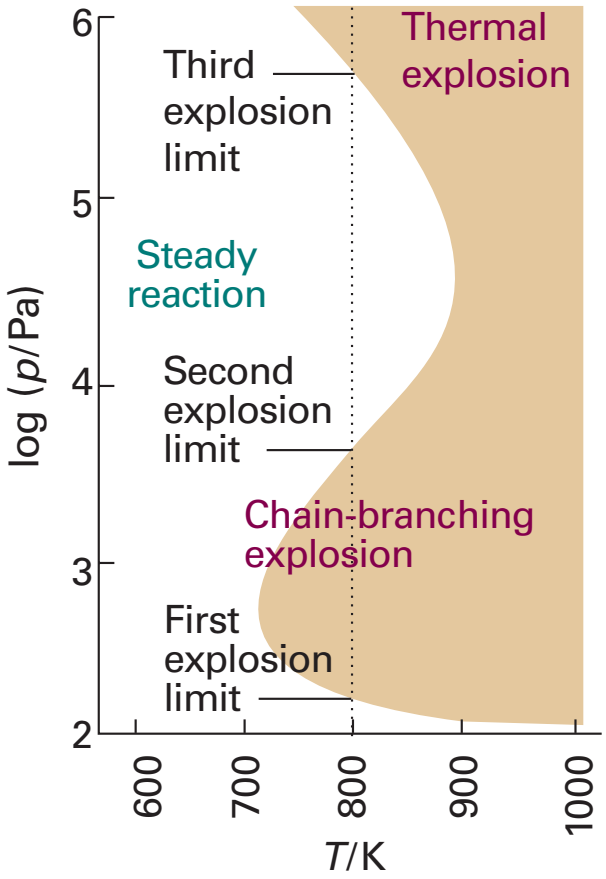


Figure 2.14: Explosion limit diagram for hydrogen-oxygen mixtures. Taken from [2].

The reaction mechanism for H_2-O_2 combustion is described by

— Initiation reaction : (1) $H_2 + O_2 => H + HO_2$

— Propagation reaction : (2) $H_2 + OH => H_2O + H$

— Propagation reaction : (3) $HO_2 + H => 2OH$

— Branching reaction : (4) $H + O_2 => OH + O$

— Branching reaction : (5) $O + H_2 => OH + H$

— Termination reaction : (6) $H + wall => \frac{1}{2}H_2$

— Termination reaction : (7) $H + O_2 + M => HO_2 + M$

This complex mechanism is both temperature and pressure dependent and is responsible for the complex explosion limits observed for H_2-O_2 mixtures as shown in Figure 2.14. At very low pressure, the system is outside the explosion domain and no explosion is observed. As pressure is increased above the first explosion limit, chain branching reactions dominate the reaction kinetics and explosion is observed. Further increasing the pressure above the second explosion limit, the termination reaction R_7 is favored and no explosion is taking place. At pressure above the third explosion limit, reaction R_3 becomes important and explosion is taking place.

The overall rate of reaction is given by

$$v_r = k_2[H_2][OH], \tag{2.128}$$

as a consequence, if $[OH]$ increases exponentially because of the chain branching process, the reaction rate also increases exponentially.

Although the QSSA cannot be strictly applied in the case of chain branching chemical processes, such an approach is useful to qualitatively examine the conditions for which an exponential growth of active radical can occur. Let us consider that the rate of chain carrier formation is given by the rate of H atom formation

$$v_{rad} = \frac{d[H]}{dt} = k_1[H_2][O_2] + k_2[H_2][OH] - k_3[HO_2][H] - k_4[H][O_2] + k_5[O][H_2] - k_6[H] - k_7[H][O_2][M]. \tag{2.129}$$

Applying the QSSA to OH and O gives

$$\frac{d[OH]}{dt} = -k_2[H_2][OH] + 2k_3[HO_2][H] + k_4[H][O_2] + k_5[O][H_2] = 0, \tag{2.130}$$

and

$$\frac{d[O]}{dt} = k_4[H][O_2] - k_5[O][H_2] = 0, \tag{2.131}$$

which leads to

$$[OH] = \frac{2k_3[HO_2][H] + k_4[H][O_2] + k_5[O][H_2]}{k_2[H_2]}, \tag{2.132}$$

and

$$[O] = \frac{k_4[H][O_2]}{k_5[H_2]}. \tag{2.133}$$

Substituting Equation 2.132 and Equation 2.133 into Equation 2.129, we obtain

$$\frac{d[H]}{dt} = v_{ini} + (k_{lin} + k_{br} - k_{ter})\,[H], \tag{2.134}$$

where v_{ini}=$k_1[H_2][O_2]$; k_{lin}=$k_3[HO_2]$; k_{br}=$2k_4[O_2]$; and k_{ter}=$k_6 + k_7[O_2][M]$.

Since we are seeking at examining the conditions for which chain branching explosion is occurring, we may neglect the contribution of R_3 by setting k_{lin}=0. After integration, Equation 2.134 becomes

$$[H] = \frac{v_{ini}}{k_{br} - k_{ter}} \left(\exp\left((k_{br} - k_{ter})\,t\right) - 1\right). \tag{2.135}$$

If $k_{ter} \gg k_{br}$, Equation 2.135 reduces to

$$[H] = \frac{v_{ini}}{k_{ter}}, \tag{2.136}$$

which indicates that no explosion will take place since $[H]$ is constant.

If $k_{br} \gg k_{ter}$, Equation 2.135 reduces to

$$[H] = \frac{v_{ini}}{k_{br}} \left(\exp\left(k_{br}t\right)\right), \tag{2.137}$$

which indicates that explosion will occurs since $[H]$ will increase exponentially.

The critical condition for chain branching explosion to occur corresponds to $k_{br} = k_{ter}$, which gives

$$\frac{2k_4}{k_7[M]} = 1, \tag{2.138}$$

if the process of recombination at the wall is ignored (pure gas phase reaction).

Equation 2.138 corresponds to the classical second explosion limit criterion established by Lewis and von Elbe [14].

2.4.5 Homogeneous catalysis

2.4.5.1 Basic definitions

Catalytic reactions are chemical reactions which are influence by the presence of a chemical compound, the catalyst, which is not included in the overall reaction equation.

A catalyst is a chemical compound which accelerates, orientates, or inhibits a chemical reaction without being consumed. The catalyst is regenerated in the course of the reaction.

There are basically three type of catalytic reactions :

— Homogeneous catalysis for which all the chemical species are in the same fluid phase (gas or liquid)

— Heterogeneous catalysis for which some of the chemical species are in the solid phase and others are in a fluid phase

— Enzymatic catalysis which is a biological process and is close to homogeneous catalysis

2.4.5.2 Thermodynamic conditions

Catalysts do not modify the state functions between the initial and the final states. Thus, the yield of a chemical reaction is not modified. The non-catalyzed reaction can be written as

$$A = P, \tag{2.139}$$

whereas the catalyzed reaction can be written as

$$A + C = P + C. \tag{2.140}$$

The change of Gibbs energy during the reaction is given by

$$(\Delta_r G_1)_{P,T} = (\Delta_r G_2)_{P,T}\,, \tag{2.141}$$

which leads to

$$K_P^1 = \frac{P_P}{P_A} = K_P^2. \tag{2.142}$$

2.4.5.3 Mode of action

Consider the following non-catalyzed chemical process

$R_1 :\ A + B = P.$

The presence of a catalyst (C) modifies the reaction mechanism as follows

— $R_2 :\ A + C = X$

— $R_3 :\ X + B = P + C$

Applying the QSSA to the intermediate species X gives

$$\frac{d[X]}{dt} = k_2[A][C] - k_3[X][B] = 0, \tag{2.143}$$

which leads to

$$[X] = \frac{k_2[A][C]}{k_3[B]}. \tag{2.144}$$

The reaction rate is given by

$$v = \frac{d[P]}{dt} = k_3[X][B] = k_2[A][C], \tag{2.145}$$

which indicates it depends on the concentration of catalyst.

The presence of a catalyst induces a decrease of the activation energy as illustrated by a potential energy diagram shown in Figure 2.15.

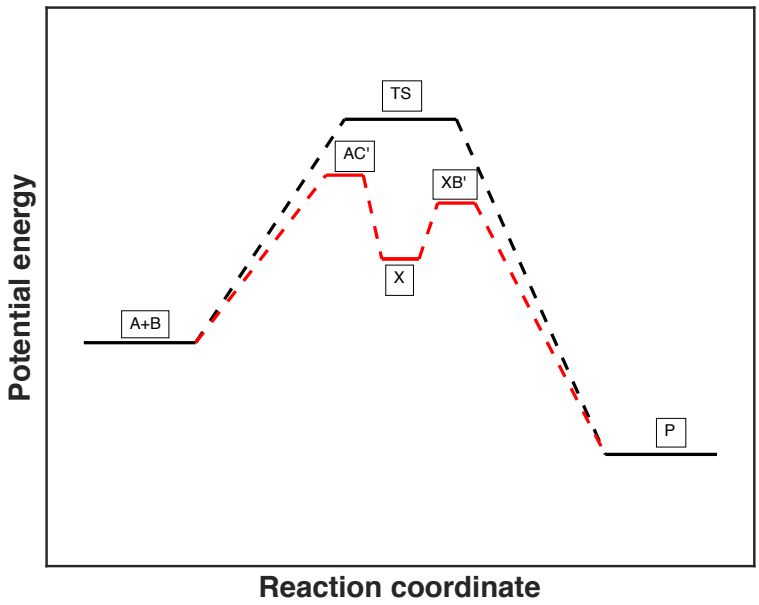


Figure 2.15: Mode of action of a catalyst. Black lines : non-catalyzed process. Red lines : catalyzed process.

2.4.5.4 Atmospheric examples

A well know example of catalytic process is the destruction of the stratospheric ozone by NO and Cl. These chemical processes can be written as

— $NO + O_3 = NO_2 + O_2$

— $NO_2 + O = NO + O_2$

and

— $Cl + O_3 = ClO + O_2$

— $ClO + O = Cl + O_2$

These chemical sequences both result in the following overall reaction

$O_3 + O = 2O_2$

which leads to a depletion of the ozone layer and induces an increase of the UV radiation at the Earth surface.

2.4.5.5 Effect of catalyst on the reaction mechanism

In order to illustrate the effect of catalyst on the reaction mechanism, let us consider the pyrolysis of acetaldehyde (CH_3CHO also referred to as ethanal).

Without catalyst, the reaction mechanism can be written as

- Initiation reaction : (1) $CH_3CHO => CH_3 + CHO$
- Propagation reaction : (2) $CH_3CHO + CH_3 => CH_3CO + CH_4$
- Propagation reaction : (3) $CH_3CO => CH_3 + CO$
- Termination reaction : (4) $2CH_3 => C_2H_6$

where the pyrolysis products are CH_4 and CO.

Applying the QSSA to CH_3CO and CH_3 gives

$$\frac{d[CH_3CO]}{dt} = k_2[CH_3CHO][CH_3] - k_3[CH_3CO] = 0, \tag{2.146}$$

and

$$\frac{d[CH_3]}{dt} = k_1[CH_3CHO] - k_2[CH_3CHO][CH_3] + k_3[CH_3CO] - 2k_4[CH_3]^2 = 0. \tag{2.147}$$

The overall reaction rate is

$$v = \frac{d[CH_4]}{dt} = k_2[CH_3CHO][CH_3]. \tag{2.148}$$

Introducing Equation 2.146 and Equation 2.147 into Equation 2.148 leads to

$$v = \frac{d[CH_4]}{dt} = k_2 \left(\frac{k_1}{2k_4}\right)^{1/2} [CH_3CHO]^{3/2}. \tag{2.149}$$

The reaction of acetaldehyde pyrolysis and be catalyzed by adding I_2. The reaction mechanism becomes

- Initiation reaction : (1) $I_2 + M <=> 2I + M$
- Propagation reaction : (2) $I + CH_3CHO => HI + CH_3CO$
- Propagation reaction : (3) $CH_3CO => CH_3 + CO$
- Catalyst regeneration reaction : (4) $CH_3 + HI => CH_4 + I$

Applying the QSSA to I and HI gives

$$\frac{d[I]}{dt} = 2k_1[I_2][M] - 2k_{-1}[I]^2[M] - k_2[I][CH_3CHO] + k_4[CH_3][HI] = 0, \tag{2.150}$$

and

$$\frac{d[HI]}{dt} = k_2[I][CH_3CHO] - k_4[CH_3][HI] = 0. \tag{2.151}$$

The overall reaction rate is

$$v = \frac{-d[CH_3CHO]}{dt} = k_2[CH_3CHO][I]. \tag{2.152}$$

Introducing Equation 2.150 and Equation 2.151 into Equation 2.152 leads to

$$v = \frac{-d[CH_3CHO]}{dt} = k_2 \left(\frac{k_1}{k_{-1}}\right)^{1/2} [I_2]^{1/2}[CH_3CHO]. \tag{2.153}$$

2.4.5.6 Auto-catalysis

Auto-catalyzed reactions are characterized by an induction period, during which the reaction rate is low, followed by a sudden increase of the reaction rate.

An auto-catalyzed reaction can be simply described as

$$A + B => 2B + P, \tag{2.154}$$

with initial concentration of reactants : $[A]$=a and $[B]$=b.

The reaction rate is given by

$$v = -\frac{d[A]}{dt} = \frac{dx}{dt} = k\,(a - x)\,(b + x)\,. \tag{2.155}$$

Integrating Equation 2.155 with respect to time leads to

$$t = \frac{1}{k\,(a + b)} \ln \frac{a\,(b + x)}{b\,(a - x)}. \tag{2.156}$$

The maximum reaction rate is obtain for

$$v_{max} = \frac{d^2x}{dt^2} = 0 = \frac{d\,(k\,(a - x)\,(b + x))}{dt} = -b - x + a - x, \tag{2.157}$$

which leads to

$$x_{max} = \frac{a-b}{2}. \tag{2.158}$$

Substituting Equation 2.158 into Equation 2.155 leads to

$$v_{max} = \frac{k}{4}(a+b)^2. \tag{2.159}$$

Substituting Equation 2.158 into Equation 2.156 leads to

$$t_{max} = \frac{1}{k(a+b)} \ln \frac{a}{b}. \tag{2.160}$$

Equation 2.156 can also be employed to determine $t_{1/2}$ at which $[A]$=a/2

$$t_{1/2} = \frac{1}{k(a+b)} \ln \frac{2b+a}{b}. \tag{2.161}$$

If $a \gg b$, Equation 2.160 and Equation 2.161 reduce to

$$t_{max} \simeq t_{1/2} \simeq \frac{1}{ka} \ln \frac{a}{b}. \tag{2.162}$$

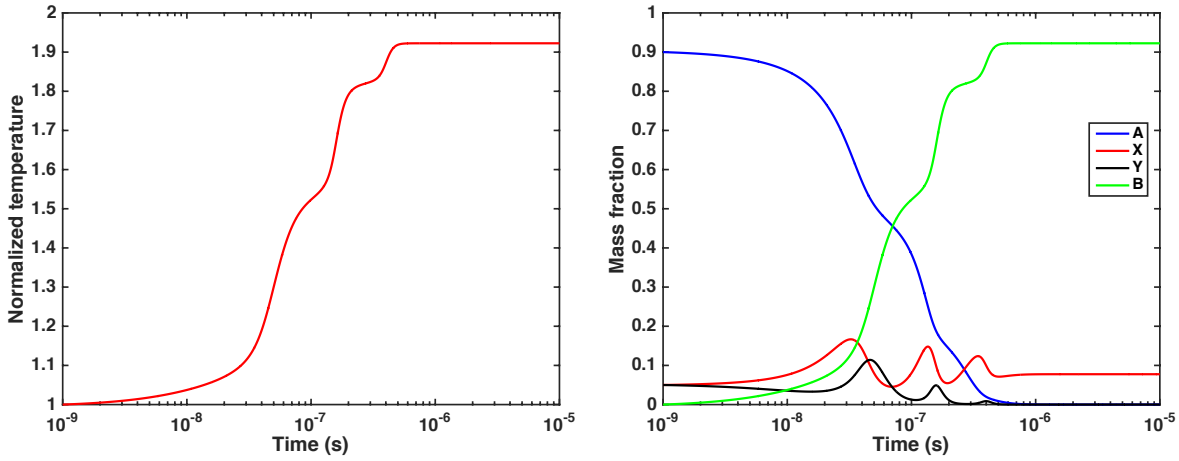
Equation 2.162 indicates that the maximum reaction rate is not reached at initial time but at a time close to $t_{1/2}$.

Some auto-catalyzed reactions may exhibit an oscillating behavior. Let us consider the Lokta-Volterra mechanism described as

- $R_1 : A + X => 2X$
- $R_2 : X + Y => 2Y$
- $R_3 : Y => B$

where R_1 and R_2 are considered non-thermic and R_3 is exothermic. In addition, the reaction R_2 is considered much faster than the two other reactions.

Solving the system of kinetic equations and the energy equation leads to the evolution shown in Figure 2.16. Oscillations of the intermediate species mole fractions are observed and result in a non-monotonous temperature profile.



a) Temperature profile b) Species profiles

Figure 2.16: Temperature and species profiles for a Lokta-Volterra auto-catalyzed oscillating reaction.

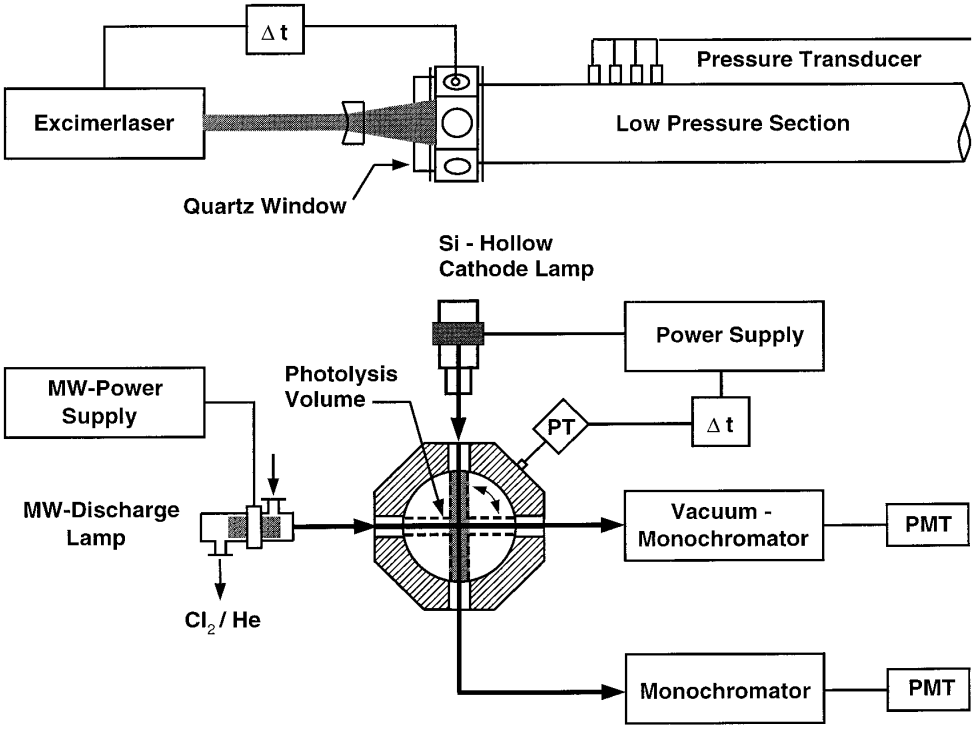


Figure 2.17: Schematic of a shock-tube equipped with a laser flash photolysis system and two absorption-based diagnostics. Taken from [15].

2.4.6 Photochemistry

Photochemical processes correspond to chemical sequences which are initiated by photons. In general, high energy photons, in the UV range, are used to break the chemical bonds of the reactants.

An important example of photochemical process is the formation of tropospheric ozone, or "urban ozone", which is an important source of urban pollution. Ozone production is related to the release into the atmosphere of NO_x, CO, and volatile organic compounds by anthropogenic activities such as energy production by combustion, chemical manufacturing, etc. Ozone formation can be described using the following chemical sequence

- $R_1 : OH + CO => HOCO$
- $R_2 : HOCO + O_2 => HO_2 + CO_2$
- $R_3 : HO_2 + NO => OH + NO_2$
- $R_4 : NO_2 + h\nu => NO + O$
- $R_5 : O + O_2 => O_3$

Reaction R_4 is a photochemical reaction and as a consequence, the formation of ozone is most important during summer.

In combustion, photochemical processes can be used to measure elementary rate constants, see [15], or to initiate ignition of combustible mixtures, see [16]. The laser flash photolysis consists in initiating the formation of chain carriers by a pulse of laser light.

For chemical kinetics applications, this technique presents the advantage of allowing measurements in a range of temperature in which the thermal decomposition of reactants is too slow for typical shock-tube facilities. A typical experimental set-up is shown in Figure 2.17. It includes a laser for photolysing the reactants and two absorption-based diagnostics. As shown in Figure 2.18, the laser pulse induces the rapid formation of silicon atoms by photolysing

$SiCl_4$. The Si atoms are then consumed by a number of reactions.

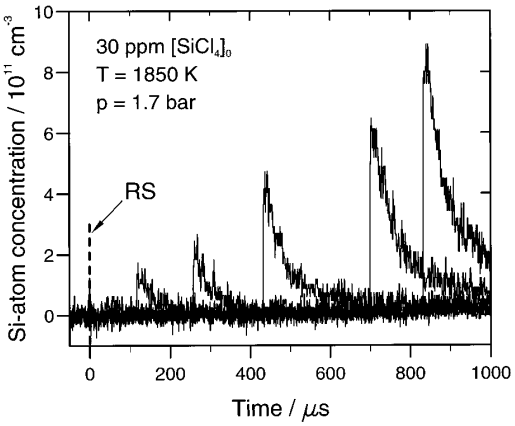


Figure 2.18: Typical silicon atom profiles obtained in a shock-tube equipped with a laser flash photolysis system. Taken from [15].

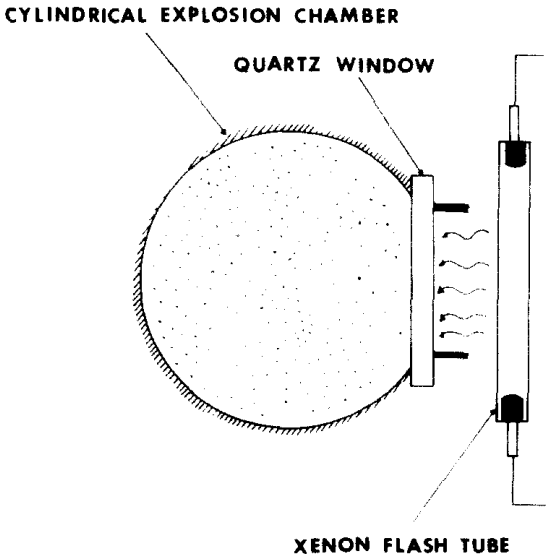


Figure 2.19: Typical experimental set-up employed to initiate an explosion by flash photolysis. Taken from [16].

Figure 2.19 shows a typical experimental set-up employed to initiate an explosion by flash photolysis. A quartz window enables to transmit the UV light into the glass vessel which

contains a H_2-Cl_2 mixture. The photon flux induces the formation of Cl atoms which initiates a chain reaction. Depending on the energy of the light pulse, a super-sonic explosion may be formed as shown in Figure 2.20.

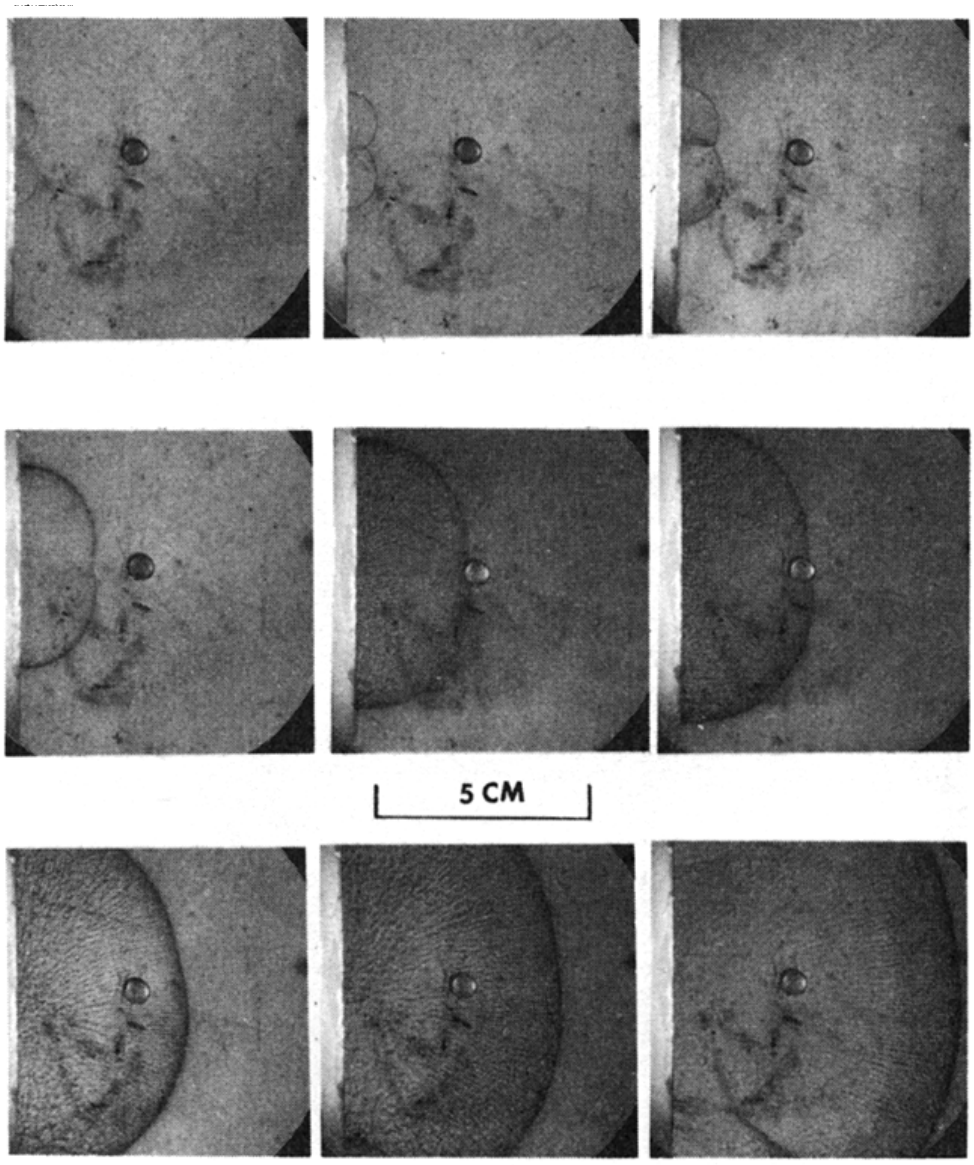


Figure 2.20: Initiation of a detonation in a H_2-Cl_2 mixture by flash photolysis. Taken from [16].

2.5 Kinetics modeling in combustion

2.5.1 Modified-Arrhenius law

The software used for combustion modeling typically describe the rate constant of elementary reactions using a modified version of the Arrhenius law. For some chemical reactions, the Arrhenius law does not describe accurately the evolution of the rate constant with temperature. As a consequence, the following modified version of the Arrhenius law is used

$$k = AT^n \exp\left(\frac{-E_a}{RT}\right), \tag{2.163}$$

where A is the pre-exponential factor ; n is the temperature exponent ; and E_a is the activation energy.

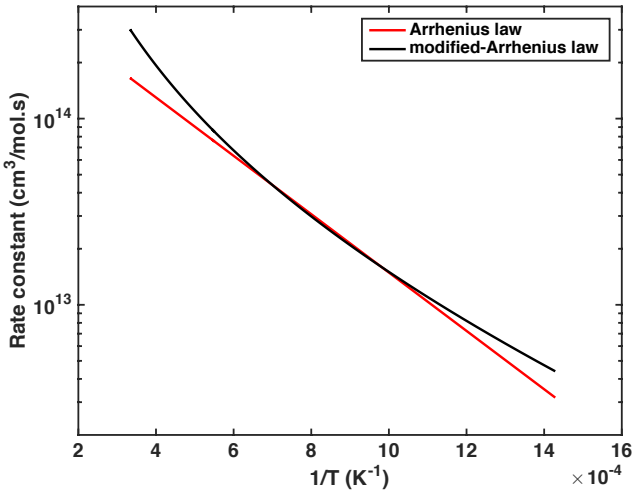


Figure 2.21: Comparison between the Arrhenius and the modified-Arrhenius laws.

The parameters A, n, and E_a are called the Arrhenius parameters or kinetics parameters. They need to be provided for every elementary reaction included in the chemical mechanism under the formalism

$$R = P \qquad A \;\; n \;\; E_a. \tag{2.164}$$

The modified-Arrhenius law is employed when the curvature of the rate constant with temperature is higher than exponential. This is illustrated in Figure 2.21 for a second-order

elementary reaction. The use of a simple Arrhenius law would induce large discrepancies for a range of temperature.

2.5.2 Reversible and irreversible reactions

In chemical kinetics software used in combustion, the reversibility of the reaction has to be included in the description of the reaction. A reversible reaction is noted as

$$R = P, \tag{2.165}$$

whereas an irreversible reaction is noted as

$$R => P. \tag{2.166}$$

If available, the kinetics parameters of the reverse reaction may be provided to the software as

$$\begin{aligned} &R = P \qquad A_f \;\; n_f \;\; E_{a,f} \\ &REV/A_b \;\; n_b \;\; E_{a,b}/ \end{aligned} \tag{2.167}$$

where the sub-scripts f and b refer to the forward and the backward reaction, respectively.

If no kinetics parameters are available for the reverse reaction, its rate constant is calculated using the equilibrium constant (k_{eq}) which links the forward and backward rate constant as

$$k_{eq} = \frac{k_f}{k_b}, \tag{2.168}$$

where

$$k_{eq} = k_p \left(\frac{P}{RT} \right)^{\sum \nu_j}, \tag{2.169}$$

where

$$k_p = \exp\left(\frac{\Delta_r S^\circ}{R} - \frac{\Delta_r H^\circ}{RT} \right), \tag{2.170}$$

where

$$\frac{\Delta_r S^\circ}{R} = \sum_{j=1}^{J} \nu_j \frac{S_j^\circ}{R} \tag{2.171}$$

and

$$\frac{\Delta_r H^\circ}{RT} = \sum_{j=1}^{J} \nu_j \frac{H_j^\circ}{RT} \tag{2.172}$$

In Equation 2.169 to Equation 2.172, j refers to the chemical species involved in the reaction and J to the total number of species.

2.5.3 Second-order reactions

In combustion software, the second-order reactions are simply described using the modified-Arrhenius law. However, for some reactions, the rate constant cannot be described by a single set of kinetics parameters. This is the case when the rate constant appears to be much higher at low temperature that can be expected from extrapolating the high-temperature rate constant. Such a complex evolution is explained by quantum tunneling effect. Basically, the chemical complex formed during the reaction of two reactants is able to tunnel through the potential energy barrier and eventually form the reaction products as illustrated in Figure 2.22.

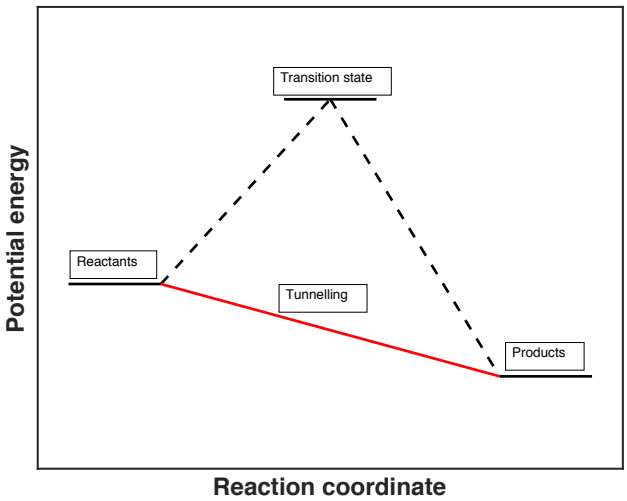


Figure 2.22: Evolution of the potential energy during a chemical reaction with and without quantum tunneling.

In order to account for the tunneling effect, the rate constant is calculated using the following equation

$$k_{tot} = A_1 T^{n_1} \exp\left(\frac{-E_{a,1}}{RT}\right) + A_2 T^{n_2} \exp\left(\frac{-E_{a,2}}{RT}\right), \tag{2.173}$$

where the parameters A_1, n_1, and $E_{a,1}$ describe the rate constant at low-temperature whereas A_2, n_2, and $E_{a,2}$ describe the rate constant at high-temperature.

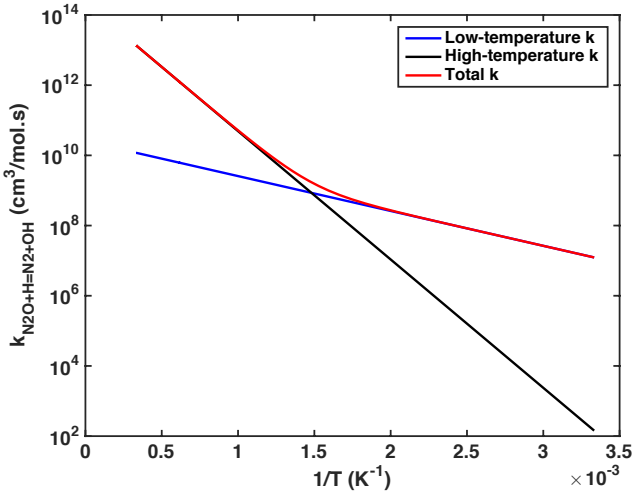


Figure 2.23: Evolution of the rate constant of the reaction N_2O+H=N_2+OH described using the Duplicate formalism.

In combustion software, this formalism is referred to as "Duplicate" and is declared as follows

$$\begin{array}{ll} R = P & A_1 \;\; n_1 \;\; E_{a,1} \\ Dup & \\ R = P & A_2 \;\; n_2 \;\; E_{a,2} \\ Dup & \end{array} \tag{2.174}$$

It is noted that the keyword "Dup" may be replaced by "Duplicate".

The reactions which demonstrate quantum tunneling often involve H atom. An example of such a reaction is

$$N_2O + H = N_2 + OH, \tag{2.175}$$

which is of fundamental important during the of mixtures containing nitrous oxide.

The evolution of the rate constant of the reaction N_2O+H=N_2+OH described using the Duplicate formalism is shown in Figure 2.23. A very important change of activation energy is observed between the low-and the high-temperature range. The limitation of the Duplicate formalism is that the rate constant is over-estimated at the transition temperature.

2.5.4 Pressure-dependent reactions

2.5.4.1 Unimolecular reactions

Chemical reactions are typically interpreted as the result of a collision between two molecules. Consequently, unimolecular reactions pose a problem of interpretation since their rate law is first-order which implies that only one molecule is involved in the reaction. The interpretation of unimolecular reaction can be made using the Lindemann-Hinshelwood mechanism [2].

This mechanism considers that a molecule of reactant A is energetically excited through collision with another molecule, noted M. This first step is described as

$$R_1 : \quad A + M \rightarrow A^* + M, \tag{2.176}$$

where A^* is the energetized molecule which may lose its excess energy through collision

$$R_2 : \quad A^* + M \rightarrow A + M, \tag{2.177}$$

or decompose to form the products

$$R_3 : \quad A^* \rightarrow P. \tag{2.178}$$

The rate law for reactions 1 to 3 are respectively given by

$$\frac{d[A^*]}{dt} = k_1[A][M], \tag{2.179}$$

$$\frac{d[A^*]}{dt} = -k_2[A^*][M], \tag{2.180}$$

and

$$\frac{d[A^*]}{dt} = -k_3[A^*]. \tag{2.181}$$

The QSSA can be applied to A^*

$$\frac{d[A^*]}{dt} = k_1[A][M] - k_2[A^*][M] - k_3[A^*] \simeq 0, \tag{2.182}$$

which leads to

$$[A^*] = \frac{k_1[A][M]}{k_2[M] + k_3}. \tag{2.183}$$

The reaction rate is given by

$$v = \frac{d[P]}{dt} = k_3[A^*] = \frac{k_1 k_3[A][M]}{k_2[M] + k_3}. \tag{2.184}$$

In the limit of low pressure, $k_2[M] \ll k_3$, so that

$$v = \frac{d[P]}{dt} = k_1[A][M], \tag{2.185}$$

which is a second-order law with a pressure-dependence (or $[M]$ dependence). The reaction rate increases linearly with pressure.

In the limit of high pressure, $k_2[M] \gg k_3$, so that

$$v = \frac{d[P]}{dt} = \frac{k_1 k_3}{k_2}[A] = k'[A], \tag{2.186}$$

which is a first-order law. The reaction rate is independent of pressure.

The rate laws obtained in Equation 2.185 and Equation 2.186 are referred to as "low-pressure limit" and "high-pressure limit", respectively.

2.5.4.2 Lindemann and Troe formalism

The pressure dependence of chemical reaction can be described for : (i) unimolecular reactions (decomposition) ; and (ii) tri-molecular reactions (recombination). Example of such reactions are

$$CH_4 + M = CH_3 + H + M, \tag{2.187}$$

for the decomposition, and

$$H + OH + M = H_2O + M, \tag{2.188}$$

for the recombination.

The description of these reactions in combustion software can be made using several approaches.

A first approach is to consider that, under the conditions of the chemical kinetics simulation, the reaction is either within the low-pressure limit range or within the high-pressure limit range. In this case, the reaction is described by

$$CH_4 + M = CH_3 + H + M \qquad A_0\ n_0\ E_{a,0} \tag{2.189}$$

in the low-pressure limit, or by

$$CH_4 = CH_3 + H \qquad A_\infty\ n_\infty\ E_{a,\infty} \tag{2.190}$$

in the high-pressure limit.

A second approach is to take into account the variation of the pressure-dependence of the rate constant with pressure. In this case, the reaction is described as

$$\begin{aligned} &CH_4(+M) = CH_3 + H(+M) \qquad A_\infty\ n_\infty\ E_{a,\infty} \\ &LOW/A_0\ n_0\ E_{a,0}/ \end{aligned} \tag{2.191}$$

The rate constant of the reaction is expressed as

$$k = k_\infty \left(\frac{P_r}{1 + P_r} \right) F, \tag{2.192}$$

where P_r is defined as

$$P_r = \frac{k_0[M]}{k_\infty}, \tag{2.193}$$

and k_0 and k_∞ correspond to

$$k_0 = A_0 T^{n_0} \exp\left(\frac{-E_{a,0}}{RT}\right), \tag{2.194}$$

and

$$k_\infty = A_\infty T^{n_\infty} \exp\left(\frac{-E_{a,\infty}}{RT}\right). \tag{2.195}$$

The parameter F is called the "broadening factor". In the so-called Lindemann formalism, F is simply taken equal to 1. However, it is also possible to consider that F has a temperature- and pressure-dependence. The most widely used formalism to account for these dependencies is called Troe formalism [17]. In this case, F is given by

$$\log F = \left(1 + \left(\frac{\log P_r + c}{n - d\,(\log P_r + c)}\right)^2\right)^{-1} \log F_{cent}, \tag{2.196}$$

where

$$c = -0.4 - 0.67 \log F_{cent}, \tag{2.197}$$

$$n = 0.75 - 1.27 \log F_{cent}, \tag{2.198}$$

$$d = 0.14, \tag{2.199}$$

$$F_{cent} = (1-\alpha)\exp\left(\frac{-T}{T^{***}}\right) + \alpha \exp\left(\frac{-T}{T^{*}}\right) + \exp\left(\frac{-T^{**}}{T}\right). \tag{2.200}$$

The reaction is then described as

$$\begin{aligned} &CH_4(+M) = CH_3 + H(+M) \qquad A_\infty\ n_\infty\ E_{a,\infty} \\ &LOW/A_0\ n_0\ E_{a,0}/ \\ &TROE/\alpha\ T^{***}\ T^{*}\ T^{**}/ \end{aligned} \tag{2.201}$$

To illustrate the kinetics implication of the Lindemann formalism and Troe formalism, let us consider the decomposition of acetaldehyde, CH_3CHO which rate constant was recently

re-evaluated [18]. The reaction is described by

$$\begin{aligned} &CH_3CHO(+M) = CH_3 + CO(+M) \qquad 2.720 \times 10^{+22} \ -1.740\ 86355 \\ &LOW/9.560 \times 10^{82} \ -18.1289\ 106230/ \\ &TROE/0.6\ 1.0\ 1.0\ 3444.1/ \end{aligned} \tag{2.202}$$

where the activation energy is given in cal/mol.

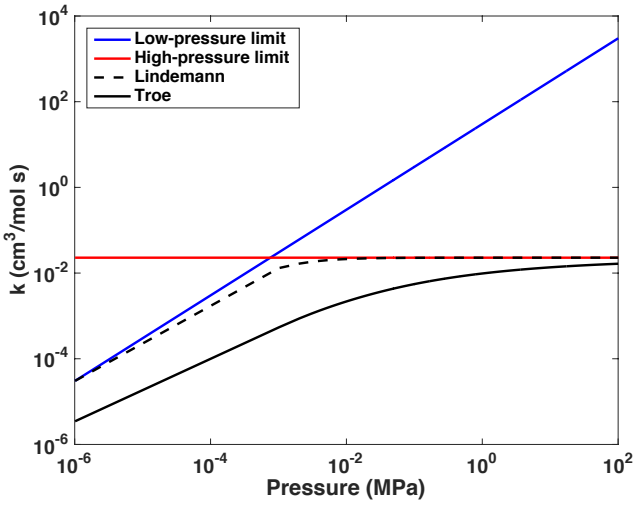


Figure 2.24: Low-pressure limit, high-pressure limit, Lindemann, and Troe representations of the rate constant for the reaction of acetaldehyde decomposition. The rate constants were calculated at T=1300K.

Figure 2.24 shows the rate constants calculated using

— The low-pressure limit

— The high-pressure limit

— The Lindemann formalism

— The Troe formalism

At very low-pressure, the rate constant tends toward the low-pressure limit whereas at very high-pressure, the rate constant tends toward the high-pressure limit. In the intermediate region, the rate constant is lower than either limits. This region is called the fall-off region. The Lindemann formalism predicts a rate constant higher than the Troe formalism. In addition,

the rate constant calculated with the Lindemann formalism is much closer to the limiting rate constants than the rate constant calculated with the Troe formalism.

2.5.4.3 PLOG formalism

Because the Troe formalism cannot reproduce the actual rate constant for some chemical reactions, a new formalism has been introduced recently and is referred to as "PLOG formalism". In this approach, the rate constant at a given pressure (P_i) is given by

$$k(T, P_i) = A_i T^{n_i} \exp\left(\frac{-E_{a,i}}{RT}\right). \tag{2.203}$$

In the combustion software, the reaction is described by

$$\begin{aligned}
&CH_3 + OH = CH_2O + M_2 \qquad 3.50 \times 10^{+05}\ 1.441\ -3.24 \times 10^{3} \\
&PLOG/0.01\ 3.50 \times 10^{+05}\ 1.44\ -3.24 \times 10^{+03}/ \\
&PLOG/0.1\ 8.85 \times 10^{+05}\ 1.33\ -2.98 \times 10^{+003}/ \\
&PLOG/1\ 1.65 \times 10^{+07}\ 0.97\ -2.01 \times 10^{+003}/ \\
&PLOG/10\ 5.37 \times 10^{+09}\ 0.29\ 2.80 \times 10^{+002}/ \\
&PLOG/100\ 9.49 \times 10^{+18}\ -2.20\ 9.77 \times 10^{+003}/
\end{aligned} \tag{2.204}$$

where the first number following the keyword PLOG corresponds to the pressure at which the rate constant parameters are given.

In order to calculate the rate constant at any pressure which is not included in the list, an interpolation is performed between the rate constant given at two pressures as given by

$$\log k(T, P_i) = \log k_i + (\log P - \log P_i)\,\frac{\log k_{i+1} - \log k_i}{\log P_{i+1} - \log P_i}. \tag{2.205}$$

The interpolation process is also illustrated in Figure 2.25.

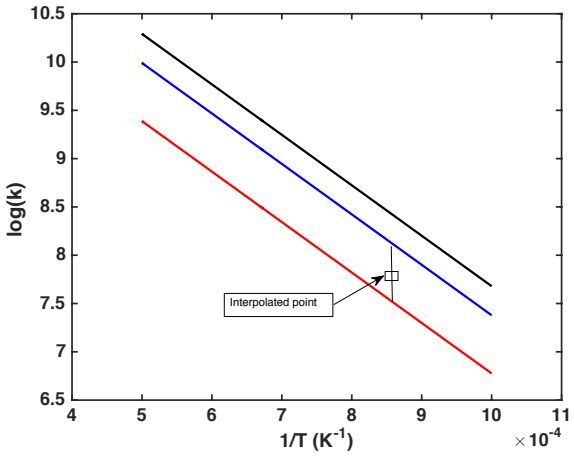


Figure 2.25: Illustration of the interpolation procedure performed when using the PLOG formalism. The solid lines correspond to the calculated rate constant for different pressures.

2.6 Chemical kinetics studies in combustion

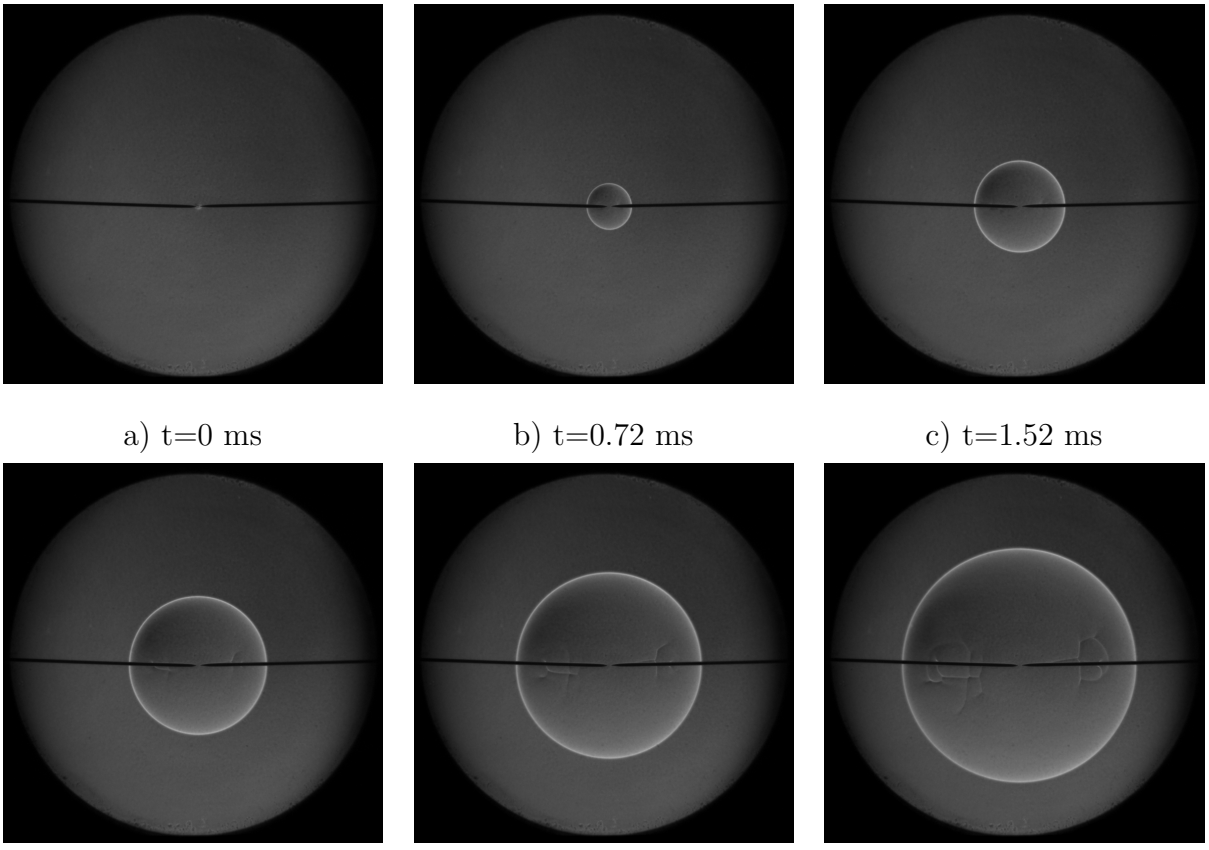
2.6.1 Experimental facilities

A number of experimental facilities are used in combustion science and may be employed to study chemical kinetics or determine rate constant of chemical reaction.

2.6.1.1 Constant volume combustion chamber

Constant volume combustion chambers are used to observe freely propagating flames which can be used to measure the so-called laminar flame speed. It is defined as the normal propagation velocity of fresh gas relative to a fixed, planar flame front and represents a measure of the mixture reactivity and diffusivity [19]. The spherically expanding flame (SEF) technique is the most reliable technique to measure the flame speed. A SEF experiment is performed as follows, (i) the flammable mixture is spark ignited at the center of a constant volume or constant pressure combustion vessel equipped with an optical access ; (ii) the spherical flame propagates outwardly ; (iii) the evolution of the flame radius (R_f) in time (t) is monitored using an optical system and a high-speed camera ; (iv) the flame speed is deduced from the

$R_f(t)$ curve and the expansion ratio. Figure 2.26 shows the propagation of a spherical flame captured using a high-speed camera.



a) t=0 ms b) t=0.72 ms c) t=1.52 ms

d) t=2.32 ms e) t=3.12 ms f) t=3.92 ms

Figure 2.26: Expanding spherical flame propagating in a rich hydrogen-air mixture. Conditions : Φ=2 ; P=80 kPa ; T=295 K.

In order to obtain the laminar flame speed from SEF experiments, it is required to correct the experimental data for stretch. Asymptotic theoretical analysis by Sivashinsky [20], Matalon and Matkowsky [21] and Clavin [22], performed in the limit of high activation energy, demonstrated the linear relation between the stretched and unstretched flame speeds in the low stretch rate regime, leading to the simple expression

$$S_L = S_L^0 - L \cdot K, \tag{2.206}$$

where K is the streth rate.

Karlovitz et al. [23] expressed the stretch rate in terms of the normalized rate of change of an elementary flame front area as

$$K = \frac{1}{A} \cdot \frac{dA}{dt}, \tag{2.207}$$

where A the flame front area.

In the case of a spherical flame, the flame surface is given by

$$A = 4 \cdot \pi \cdot R_f^2, \tag{2.208}$$

which leads to the following expression for the stretch rate [24–27] :

$$K = 2 \cdot \frac{V_S}{R_f}, \tag{2.209}$$

given that the stretched spatial velocity, or the flame speed, V_S, corresponds to the flame radius increase rate

$$V_S = \frac{dR_f}{dt}. \tag{2.210}$$

According to Bradley et al. [28], the relationship between the burning speed and the spatial velocity for a spherical flame propagating in a closed vessel is given by

$$S_L = \frac{1}{\sigma} \cdot \left(V_S + \frac{R_b}{3 \cdot P_b} \cdot \frac{dP_b}{dt} \right), \tag{2.211}$$

where R_b the burnt gas radius, P_b is the burnt gas pressure, and σ is the expansion ratio defined as

$$\sigma = \frac{\rho_u}{\rho_b}, \tag{2.212}$$

where ρ_u and ρ_b are the unburnt and burnt gas densities, respectively.

In the case of a large volume vessel and for measurements limited to the initial period of propagation where the flame radius is small, the pressure derivative vanishes, so that the burning speed and the spatial velocity are linked only through the expansion ratio across the flame front

$$S_L = \frac{V_S}{\sigma}. \tag{2.213}$$

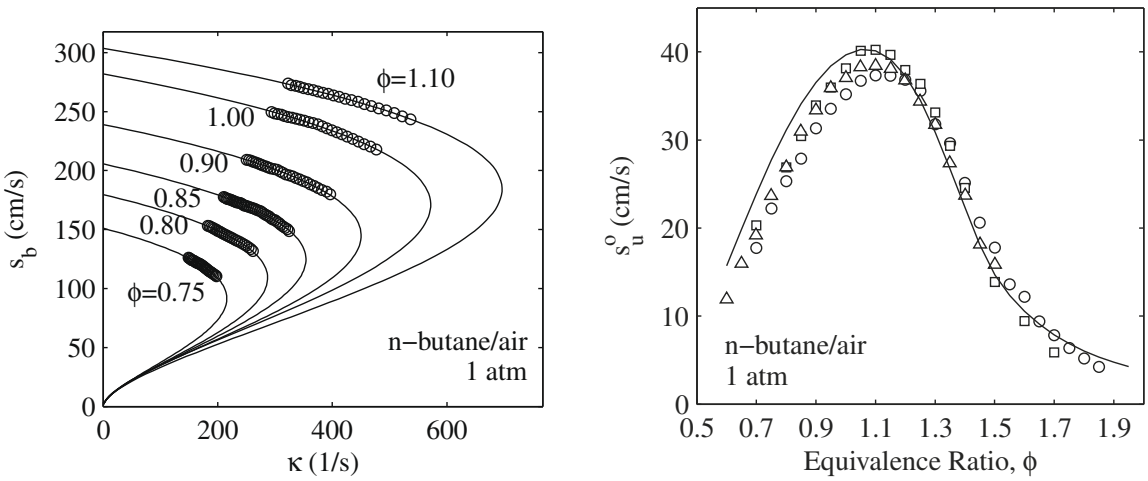
Substituting Equation 2.209 and Equation 2.213 into Equation 2.206, and integrating with respect to time, one obtains the expression of the unstretched spatial flame velocity as a function of time and flame radius

$$V_S^0 \cdot t = R_f + 2\sigma L \cdot ln\,(R_f) + C_{st} \tag{2.214}$$

where C_{st} is an integration constant.

The unstretched spatial velocity and the Markstein length can then be derived by applying a least square fitting procedure to a data set of the flame radius versus time.

It is noted that other equations than Equation 2.206 can be used to extract the unstretched flame speed. Other equations may consider a non-linear relationship between stretch rate and flame speed [29].



a) Non-linear extrapolation

e) S_L^0=f(Φ)

Figure 2.27: Extrapolation process for spherically expanding flame and laminar flame speed as a function of equivalence ratio for butane-air mixtures. Taken from [30].

Figure 2.27 a) illustrates the extrapolation to zero-stretch process for butane-air flames whereas Figure 2.27 b) shows the evolution of the laminar flame speed as a function of equivalence ratio. The solid line in Figure 2.27 b) corresponds to the calculated flame speed.

For this kind of calculation, a steady 1-D simulation is performed. It takes into account both the chemical and the transport properties of the different chemical species included in the reaction mechanism.

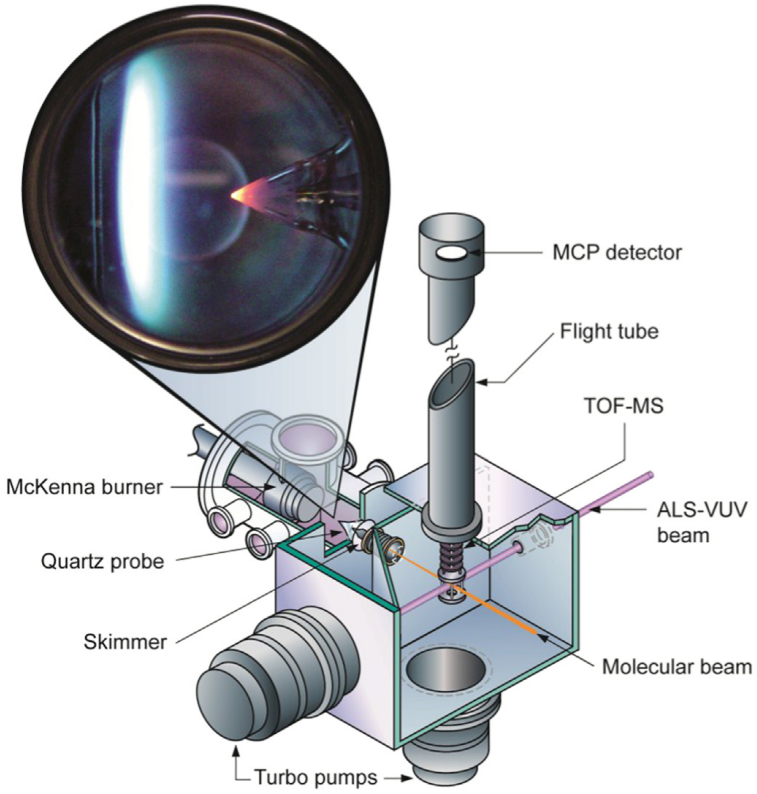


Figure 2.28: Picture of a flat burner. Taken from [19].

2.6.1.2 Flat flame burner

A flame can also be stabilized on a burner. The configuration which is the most used in combustion for chemical kinetics study is the flat flame burner. The premixed mixture is flowing from downstream of the burner's surface and the flame is stabilized above the burner's surface. The gas composition within the flame can be characterized using laser-based diagnostics or using a sample probe along with analytical techniques such as gas chromatography. Figure 2.28 shows an example of a flat flame burner. For such an experimental configuration, distance-resolved species profiles are obtained as shown in Figure 2.29. The distance

corresponds to the distance above the burner's surface. In order to model such experiments, a steady 1-D simulation which takes into account the chemical and the transport properties of the chemical species is employed. Since the flame above a flat flame burner is not adiabatic, it is required to measure the temperature profile using a thermocouple or a spectroscopic diagnostics.

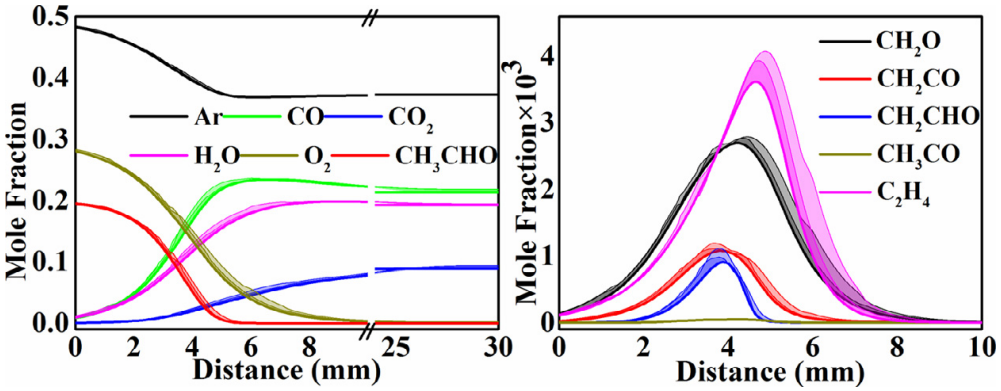


Figure 2.29: Typical example of species profiles obtained with a flat burner. Taken from [31].

2.6.1.3 Jet-stirred reactor

A jet-stirred reactor is a homogeneous gas-phase reactor usually operated at steady-state and constant residence-time, temperature, and pressure [32]. Jet-stirred reactors can be employed to study the oxidation of various fuels in the temperature and pressure range 500-1500 K, and 0.1-4 MPa, respectively. The residence-time is typically between few tens of ms and several seconds. Figure 2.30 shows a typical jet-stirred reactor facility. The reactants are supplied to the reactor through several nozzles which creates strong jets of gas. This delivery system ensures an intense turbulent mixing and minimize the effect of diffusion on the chemical kinetics. The composition of the gas at the reactor outlet can be characterized using a variety of analytical techniques such as gas chromatography, mass spectrometry, and laser-based diagnostics. By changing the temperature step by step, the evolution of the gas-phase composition as a function of temperature can be obtained for a given pressure and initial composition. Figure 2.31 shows an example of the temperature-resolved species profiles obtained

during a jet-stirred reactor experiment. Since in a jet-stirred reactor, both the temperature and the composition are homogeneous within the reactor, the evolution of the chemical species can be modeled using a zero-dimensional, steady-state reactor model. Figure 2.31 also shows the calculated species profiles.

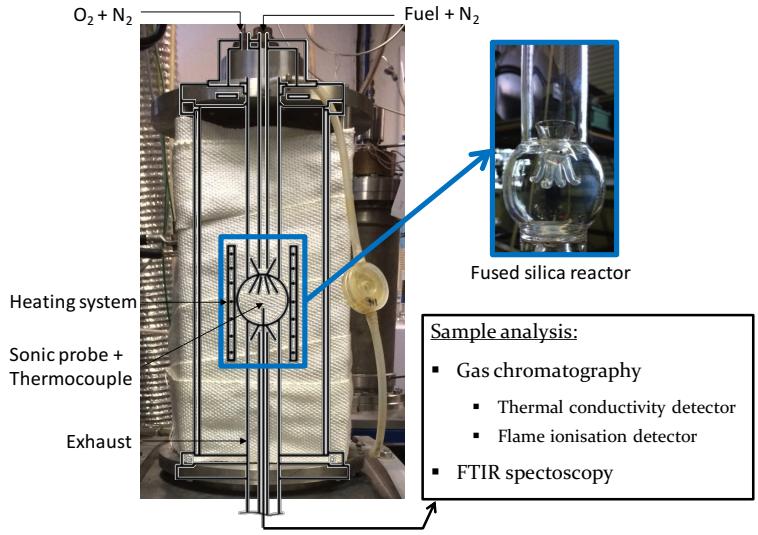


Figure 2.30: A jet-stirred reactor setup. Taken from [33].

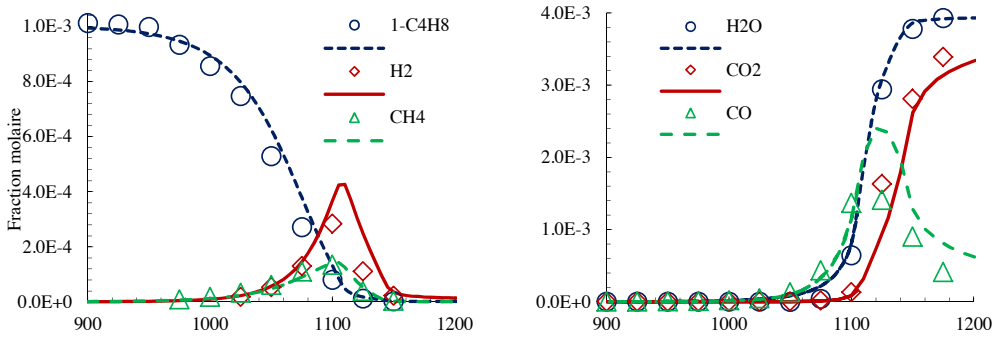


Figure 2.31: Typical example of species profiles obtained with a jet-stirred reactor. Taken from [33].

2.6.1.4 Flow reactor

A flow reactor is a constant diameter cylindrical tube in which the reactive mixture flows continuously along the tube [32]. The reactive mixture may be premixed before its entrance into the reactor or it can be formed by mixing streams of reactants at the reactor entrance. An example of flow reactor set-up is shown in Figure 2.32. Flow reactor can typically be operated at temperature between 500 and 1500 K and pressure between 0.02 and 4 MPa. Residence-time can be varied between few tens of ms and several seconds. A number of analytical techniques or laser-based diagnostics can be used to measure the species concentration [34]. A flow reactor can be operated either at a constant residence-time or with a varying residence-time. In the former case, the temperature is varied and temperature-resolved species profiles are obtained. In the latter case, time-resolved species profiles are be obtained as illustrated in Figure 2.33. In most cases, the flow reactor experiments can be modeled by using a time-dependent 0-D, constant-pressure reactor model. The solid and dotted lines in Figure 2.33 corresponds to the species profiles calculated with a detailed chemical mechanism.

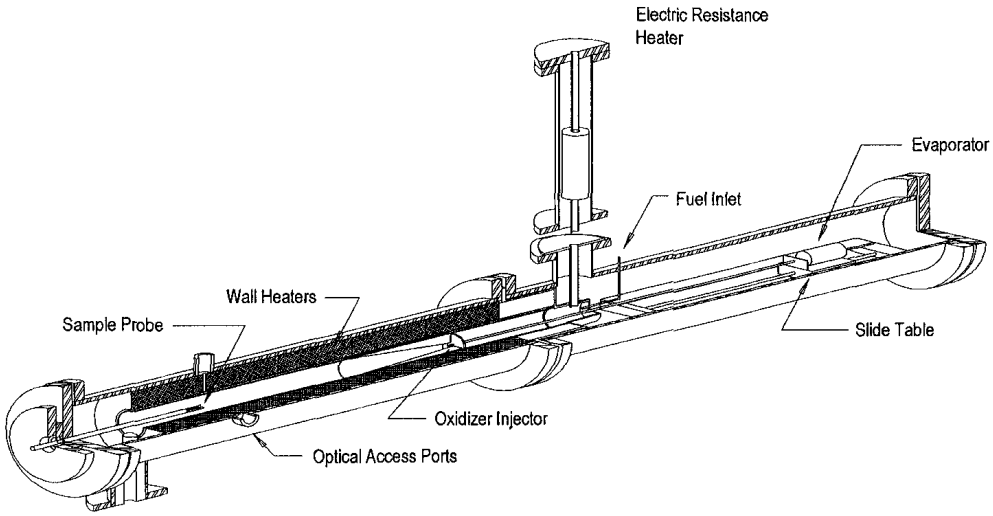


Figure 2.32: Schematic of a flow reactor setup. Taken from [35].

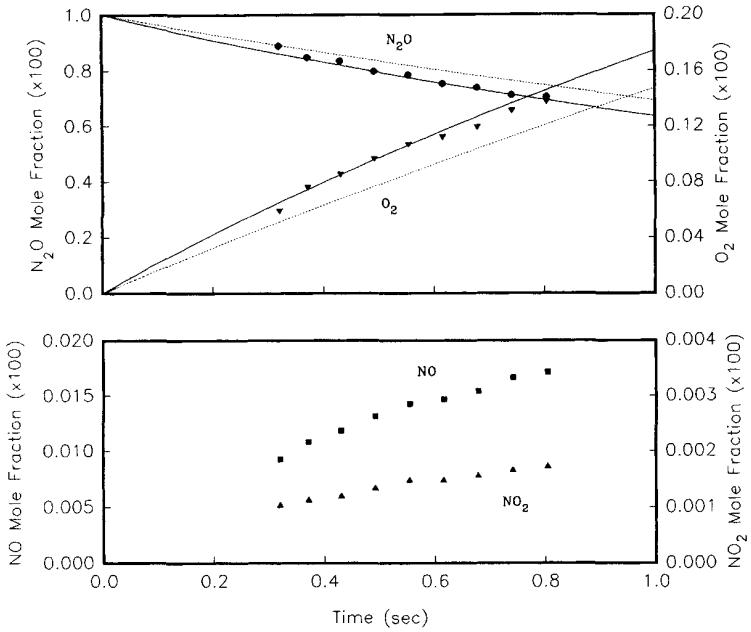


Figure 2.33: Typical example of species profiles obtained with a flow reactor. Taken from [35].

2.6.1.5 Rapid compression machine

A rapid compression machine (RCM) is a laboratory combustion facility which resembles a single-piston engine. The premixed combustible mixture is introduced at low-pressure (<0.1-0.2 MPa) within a long cylindrical tube. Then a moving piston is accelerated using compressed-air. The reduction of the chamber's volume by the piston induces simultaneously an increase of the pressure and temperature. After a certain distance, the piston is stopped using for example a oil bath. A schematic of the RCM facility of CCE is shown in Figure 2.34. The combustion event can be studied in the temperature range 600-1100 K and in the pressure range 0.1-7 MPa. The observation time in a RCM is typically between 2 and 500 ms.

By measuring the evolution of the pressure inside the combustion chamber, the auto-ignition delay-time, defined as the time between the end of compression and the maximum pressure derivative, can be measured as illustrated in Figure 2.35. When equipped with a

sampling system, as shown in Figure 2.36, time-resolved species profiles can be obtained. In this case, only one data point can be obtained per experiment. The sampling is performed at different instant during the induction period, i.e. the period between the end of compression and the ignition. The gas samples are then analyzed using various analytical techniques such as gas chromatography. Examples of species profiles obtained in a RCM are shown in Figure 2.37.

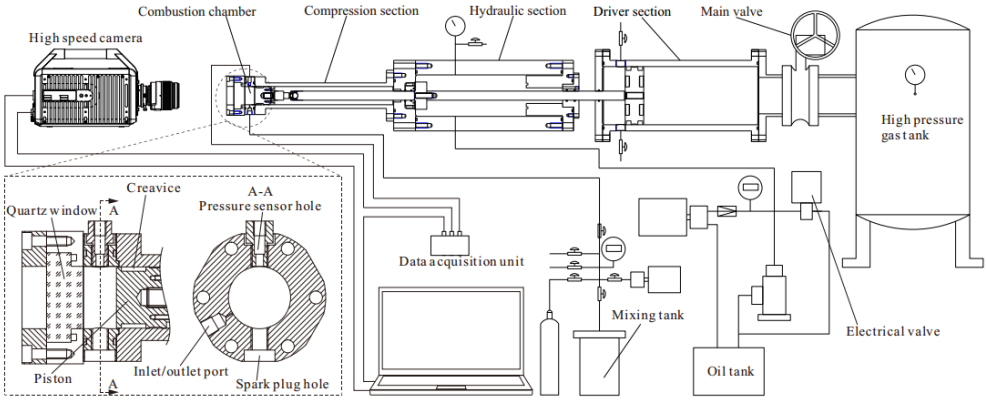


Figure 2.34: Schematic of a rapid compression machine. Taken from [36].

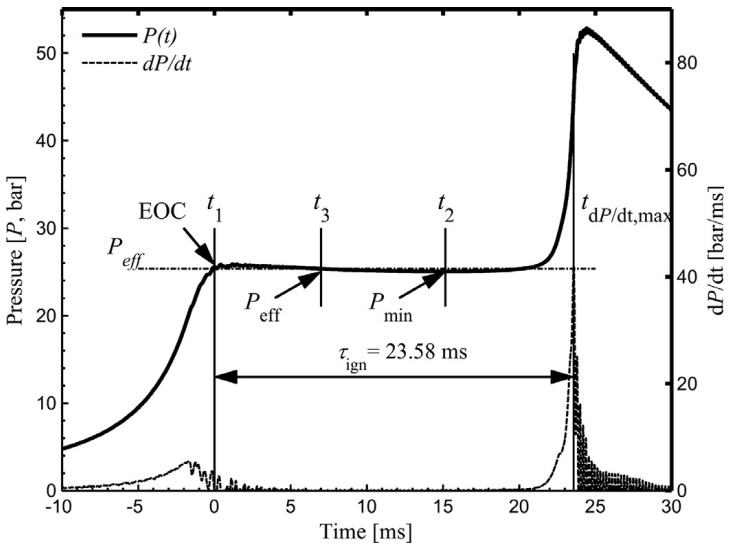


Figure 2.35: Typical example of a pressure trace obtained with a rapid compression machine. Taken from [37].

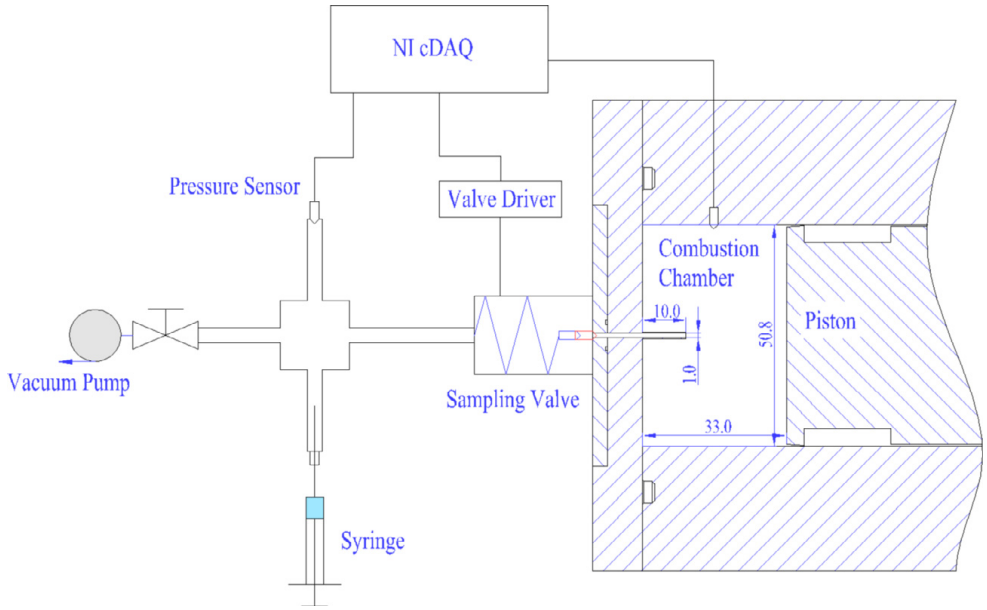


Figure 2.36: Schematic of a rapid compression machine equipped with a sampling system. Taken from [37].

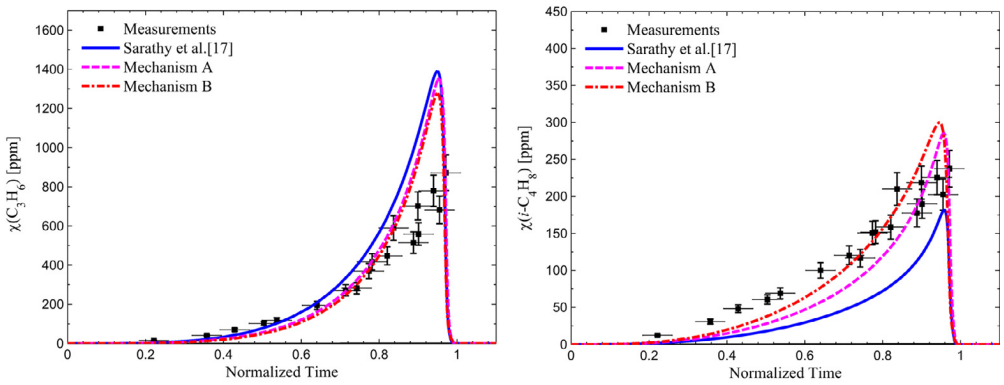


Figure 2.37: Typical example of species profiles obtained with a rapid compression machine. Taken from [37].

2.6.1.6 Shock tube

The shock tube has been invented by Vieille in 1899 [38]. This reactor has been widely used for the past 60 years with applications relevant to aerodynamics, medecine, geo-science, astrophysics and material synthesis. A shock tube allows for bringing quasi-instantaneously

a gaseous mixture at high temperature and pressure. This property has enabled the study of the high-temperature dynamics of a variety of reactive systems [39].

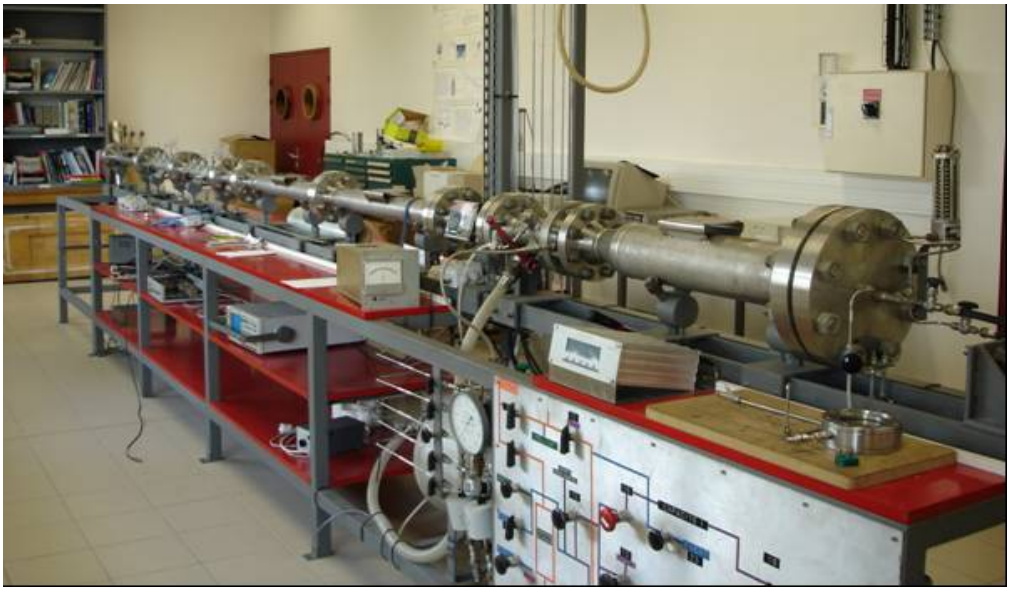


Figure 2.38: Picture of a shock-tube.

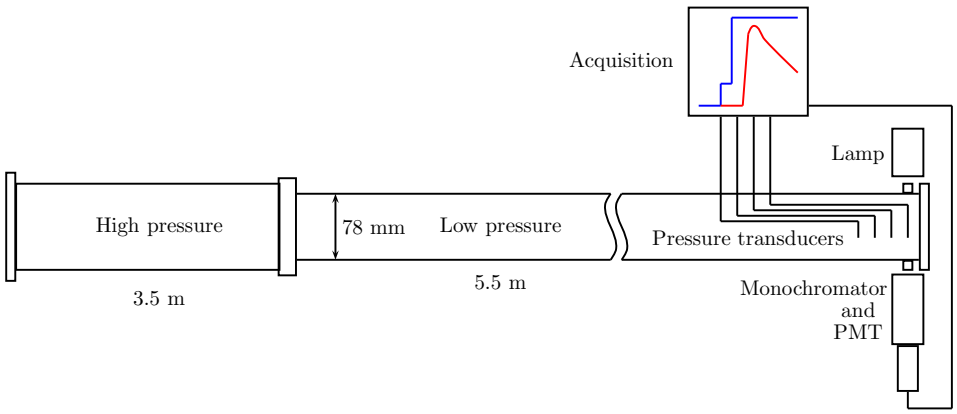


Figure 2.39: Schematic of a shock-tube. Adapted from [39].

As illustrated in Figure 2.38 and Figure 2.39, a shock tube is a cylindrical reactor constituted of a driver and a driven section separated by one or two membranes. The driver section

is filled at high pressure with an inert gas, usually helium, whereas the driven section is filled at low pressure with the reactive mixture to be investigated [39].

Figure 2.40 displays the trajectories of shock and expansions waves formed during a shock tube experiment. Also shown are the temperature and pressure evolutions. [39]. Subscripts refers to the following thermodynamic conditions : (1) initial in the driven section ; (2) behind the incident shock wave ; (3) behind the contact surface ; (4) initial in the driver section ; and (5) behind the reflected shock wave.

At the membrane burst, the driver gas is brought to motion and acts as an uniformly accelerated piston. According to Beckel's mechanism [14], every velocity increment induces the formation of a compression wave. After a given distance, the successive compression waves coalesce and form a plane incident shock wave. The driver gas flows toward the end of the shock tube and is separated from the reactive medium by the contact surface. The distance between the contact surface and the incident shock wave progressively increases. The incident shock wave propagates at a quasi-constant velocity and induces a first increase of temperature and pressure of the reactive mixture. It reflects elastically at the end wall of the shock tube. The reflected shock wave induces a second increase of temperature and pressure and brings back the gas into motion to rest. The desired thermodynamic conditions are obtained behind the reflected shock wave at the observation window, usually located few mm away from the end wall. Later on, the reflected shock wave collides with the contact surface. This collision generates acoustic disturbances which modify the thermodynamic conditions at the observation window. The observation time during a shock tube experiment is defined as the time ellapse between the reflected shock wave passage and the apparition of the acoustic disturbances [40] and ranges between 0.5 and 50 ms depending on the shock tube dimensions, geometry, and driver gas composition [41, 42].

The thermodynamic conditions behind the incident and reflected shock waves can be calculated from the 1-D ideal shock wave theory using the initial conditions and the experimental velocity of the incident shock [39].

The dynamics of the reactive system can be studied using a variety of spectroscopic and analytical, gas chromatograph or mass spectrometry, diagnostics [43].

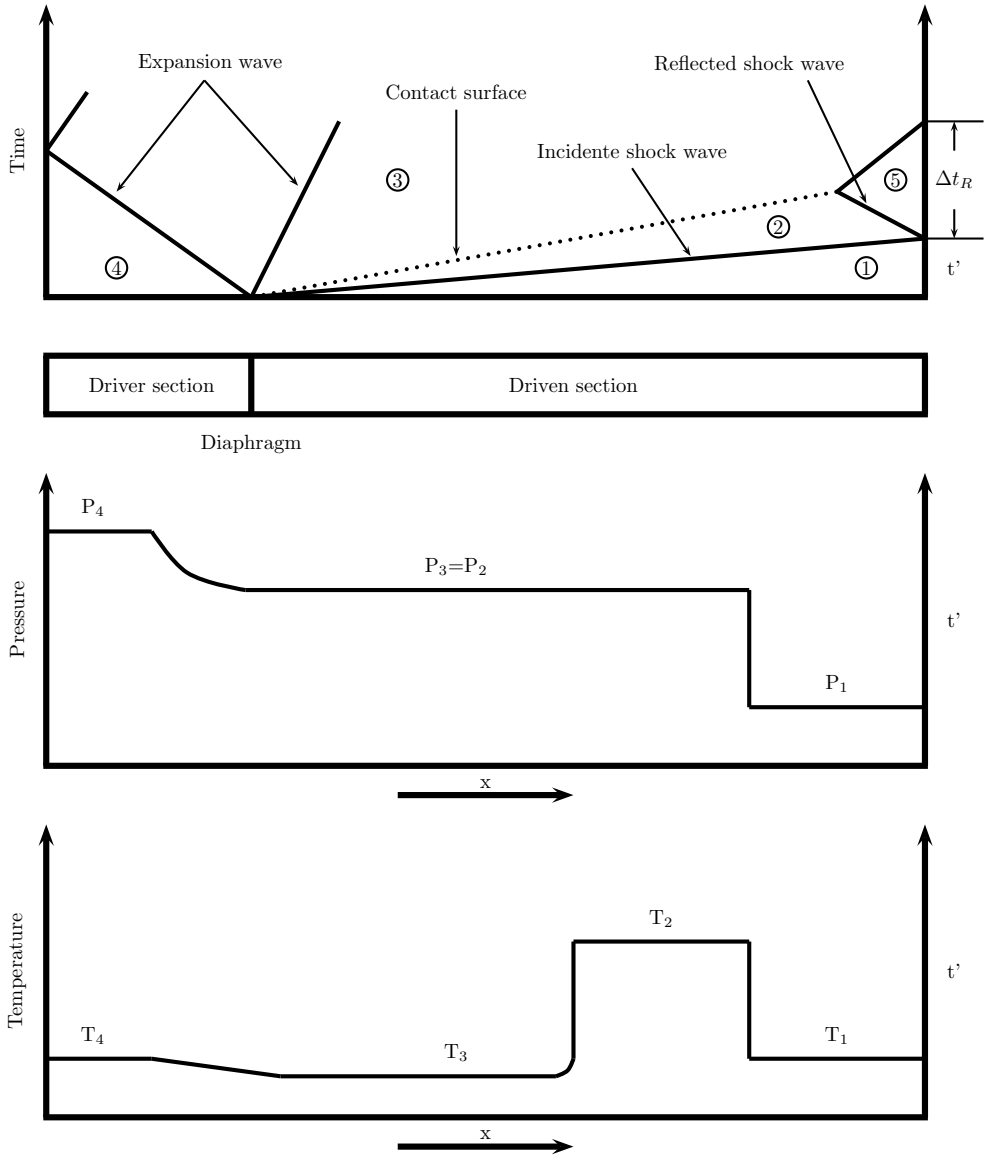
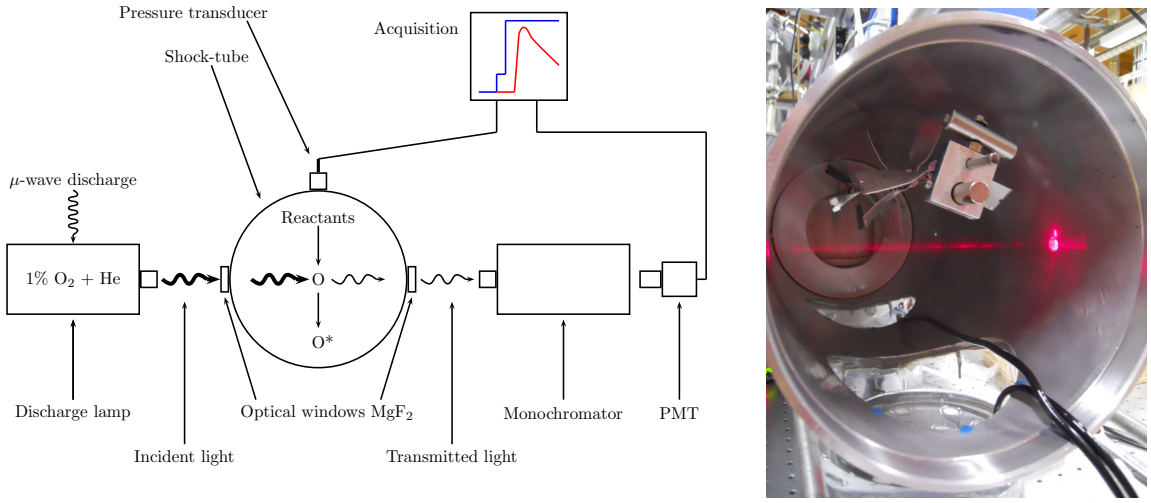


Figure 2.40: Wave trajectories during a shock-tube experiments. Adapted from [39].

2.6.2 Rate constant measurement

In general, the constant volume combustion chamber, the flat flame burner, the flow and jet-stirred reactors, and the RCM are not employed to determine rate constant for elementary reaction. The flame speed, ignition delay-time, and species profiles obtained using these facilities are used to validate a detailed reaction model by comparing the experimental and the calculated quantities.



a) ARAS set-up

b) Laser set-up

Figure 2.41: Typical experimental set-up for rate constant measurement in shock-tube. ARAS corresponds to Atomic Resonance Absorption Spectroscopic. Image in b) taken from the website of the High Temperature Gasdynamics Laboratory.

On the other hand, the shock-tube coupled with the absorption technique is widely used to determine rate constant. The species mole fraction of stable or radical species is detected using a selective light source such as a micro-wave discharge lamp or a laser. Figure 2.41 illustrates such experimental set-up. As shown in Figure 2.42, time-resolved mole fraction profiles are obtained behind the reflected shock wave. The experimental profile is then fitted by a calculated profile. The value of the rate constant of interest is modified until the best

agreement between the experimental and calculated profile is achieved. The process is repeated for a series of experiments performed over a range of temperature. The values obtained for the rate constant are then used to obtain an Arrhenius fit of the rate constant as shown in Figure 2.43 a). Estimating the uncertainty on the rate constant is an important aspect of the rate constant determination. The sources of uncertainty include

- the temperature
- the mixture composition
- the time zero
- the absorption coefficient
- the laser wavelength
- the fitting procedure
- the rate constant of secondary reactions

Typical uncertainty on the rate constant measured in shock-tube ranges between ±10% and ±30%.

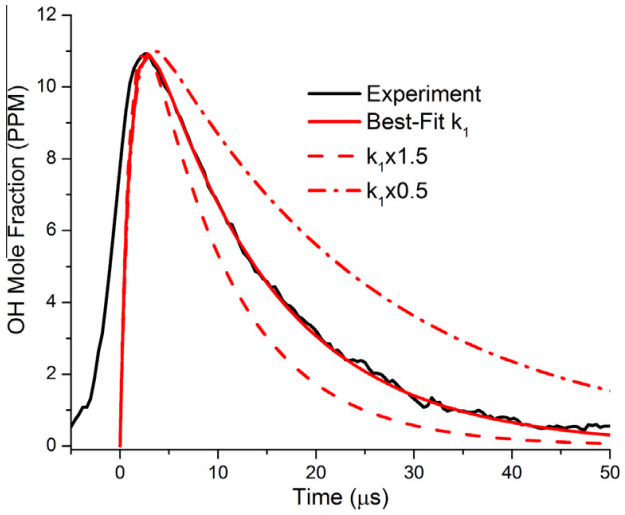


Figure 2.42: Fitting of the OH mole fraction profile for rate constant determination. Taken from [44].

It is noted that the measurement of rate constant in shock-tube often involves the pseudo-first order approach for second-order reactions. The reaction rate for a second-order chemical process is given by

$$v = k[A][B]. \tag{2.215}$$

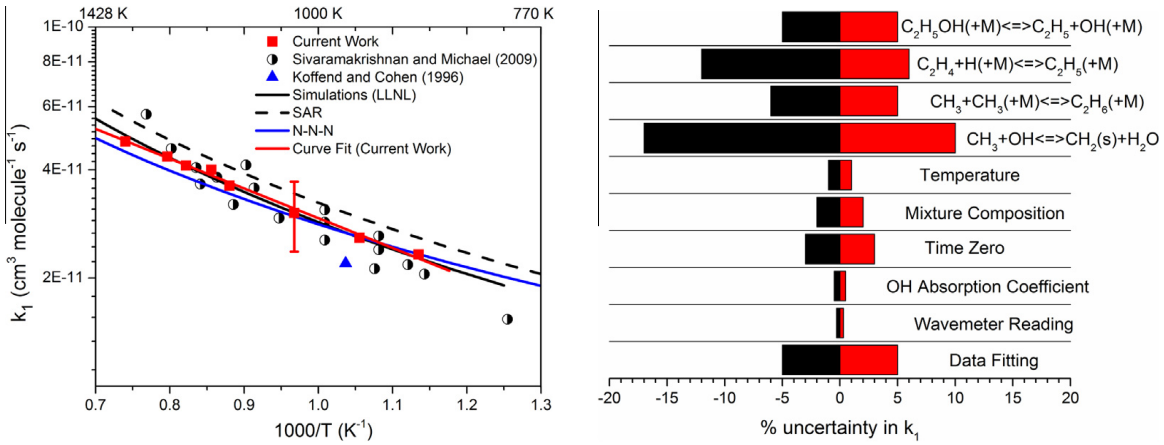
Determining the reaction rate constant would require to measure both $[A]$ and $[B]$ as a function of time. However, if the experiment is performed with a mixture for which $[A] \gg [B]$ and A is not significantly consumed by other chemical processes, $[A]$ can be considered as essentially constant and Equation 2.215 can be re-written as

$$v = k[A][B] = k_{ap}[B], \tag{2.216}$$

where $k_{ap} = k[A]$.

In practice, k_{ap} will be measured in the experiment and k will be deduced from

$$k = \frac{k_{ap}}{[A]}. \tag{2.217}$$



a) Arrhenius plot b) Uncertainty plot

Figure 2.43: Typical Arrhenius and uncertainty plots for rate constant measurement in shock-tube for the reaction C_6H_{14}+OH. Taken from [44].

2.7 Numerical analyses of chemical kinetics

Combustion chemistry software are equipped with a number of utilities which enable analyzing the chemical dynamics of reactive systems. The quantities of interest are : (i) the rate of energy release per reaction ; (ii) the rate of production ; and (iii) the sensitivity coefficient.

2.7.1 Rate of energy release per reaction

The rate of energy release per reaction analysis enables to identify which reactions are responsible for increasing and decreasing the temperature of the system. It is defined as

$$q_i = v_i \Delta_r H_i, \tag{2.218}$$

where q is the rate of energy release of the reaction ; v is the rate of the reaction ; $\Delta_r H$ is the enthalpy of the reaction ; and the subscript i refers to the reaction i.
The quantity q is the rate of energy production/consumption per unit volume.

2.7.2 Rate of production

The rate of production (ROP) enables to identify the most important chemical pathways producing or consuming a given chemical species. It is defined as

$$ROP_j = \sum_{i=1}^{N} \nu_j v_{i,j}, \tag{2.219}$$

where ROP_j is the total ROP for the species j ; N is the total number of species which involve j ; ν_j is the stoichiometric coefficient for the species j ; and $v_{i,j}$ is the rate of the reaction i.
As an example, let us consider the following chemical scheme

- R_1 : R+M=>B+M
- R_2 : R+B=>B+B
- R_3 : R+B+M=>C+M
- R_4 : C+M=>B+B+M
- R_5 : B+B+M=>P+M

where R represents the reactant ; B is a chain branching radical ; C is a stable intermediate species ; and P is the product.

The ROP for the species B is given by

$$ROP_B = \frac{d[B]}{dt} = \sum_{i=1}^{N} \nu_B v_{i,B} = k_1[R][M] + k_2[R][B] - k_3[R][B][M] + 2k_4[C][M] - 2k_5[B][B][M]. \tag{2.220}$$

By calculating the ROP for several species, it is possible to establish chemical pathway diagrams such as those shown in Figure 2.44 and Figure 2.45. As seen in these figures, the chemical pathways for commonly used fuels such as hydrocarbon fuels are extremely complicated.

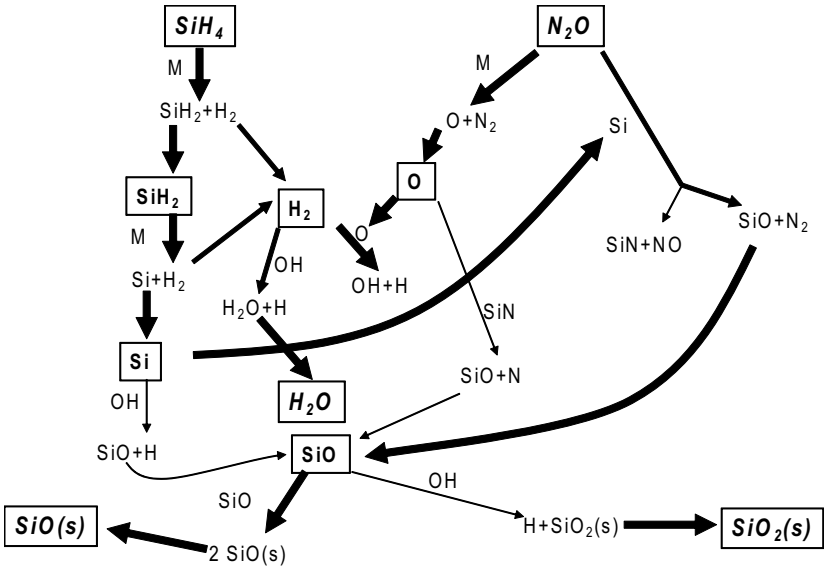


Figure 2.44: Reaction pathway analysis performed at 2001 K and 268 kPa for a SiH_4-N_2O-Ar mixture. Composition : X_{SiH4}=10 ppm ; X_{N2O}=20 ppm. Taken from [45].

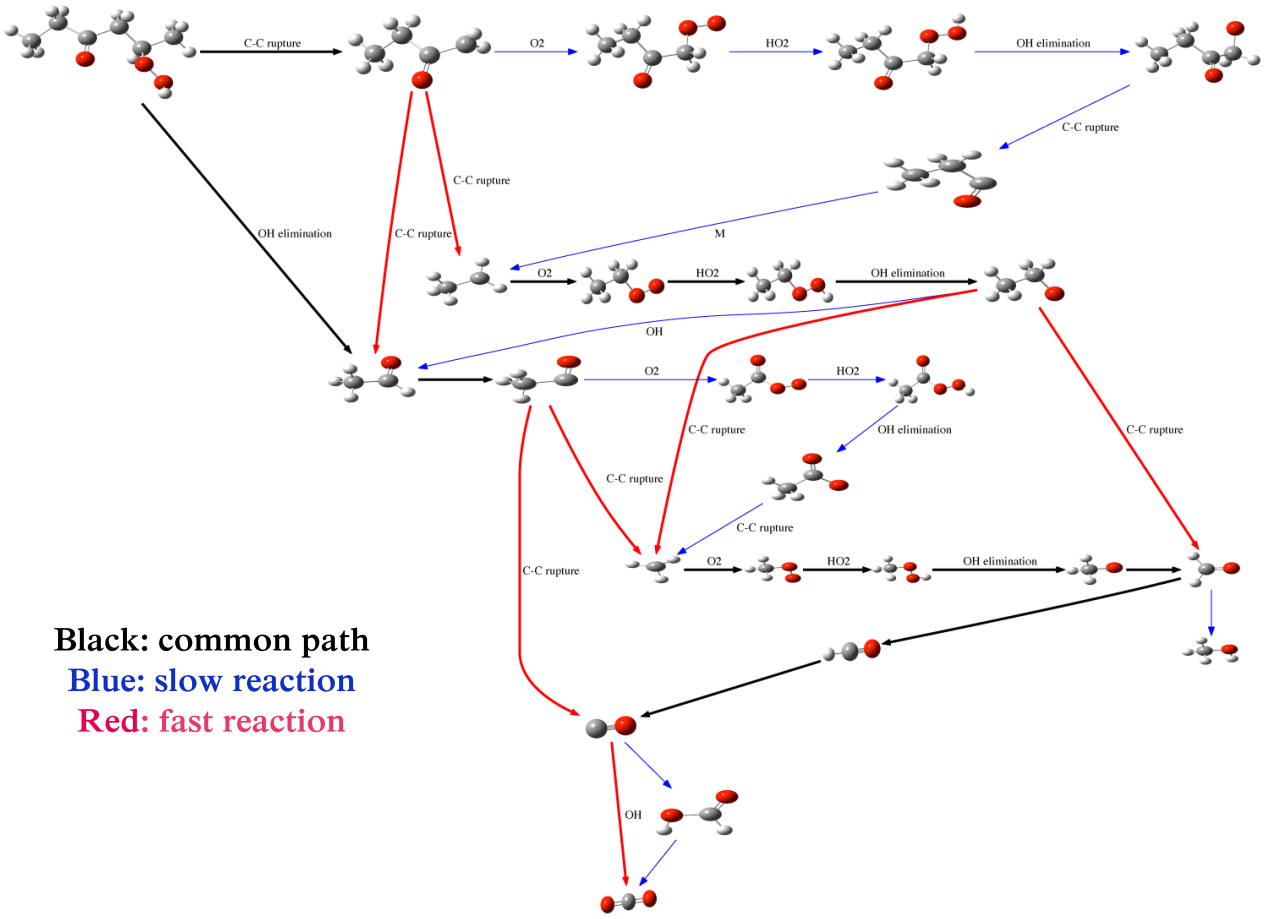


Figure 2.45: Carbon reaction pathways during the second phase of a n-hexaneair mixture oxidation for two heating rates. Taken from [46].

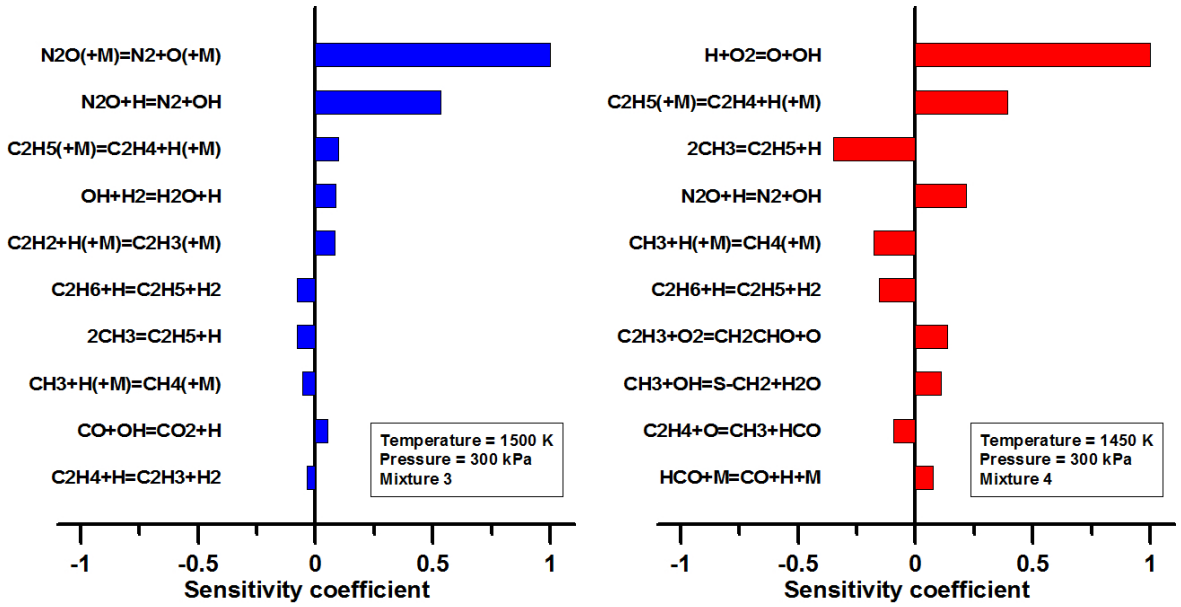
2.7.3 Sensitivity analysis

Sensitivity analyses enable to identify the most important reactions with respect to a given parameter of the chemical system. These parameters can be species concentration but also the temperature. The sensitivity coefficient on the species concentration $[j]$ for reaction i is defined as

$$C_i = \frac{\partial \ln[j]}{\partial \ln k_i} = \frac{k_i}{[j]} \frac{\partial [j]}{\partial k_i}, \tag{2.221}$$

where k_i is the rate constant of reaction i. A positive/negative coefficient indicates that the reaction promotes/inhibits the production of j.

Sensitivity coefficients are typically displayed as bar graphs as shown in Figure 2.46 for C_2H_6-based mixtures.

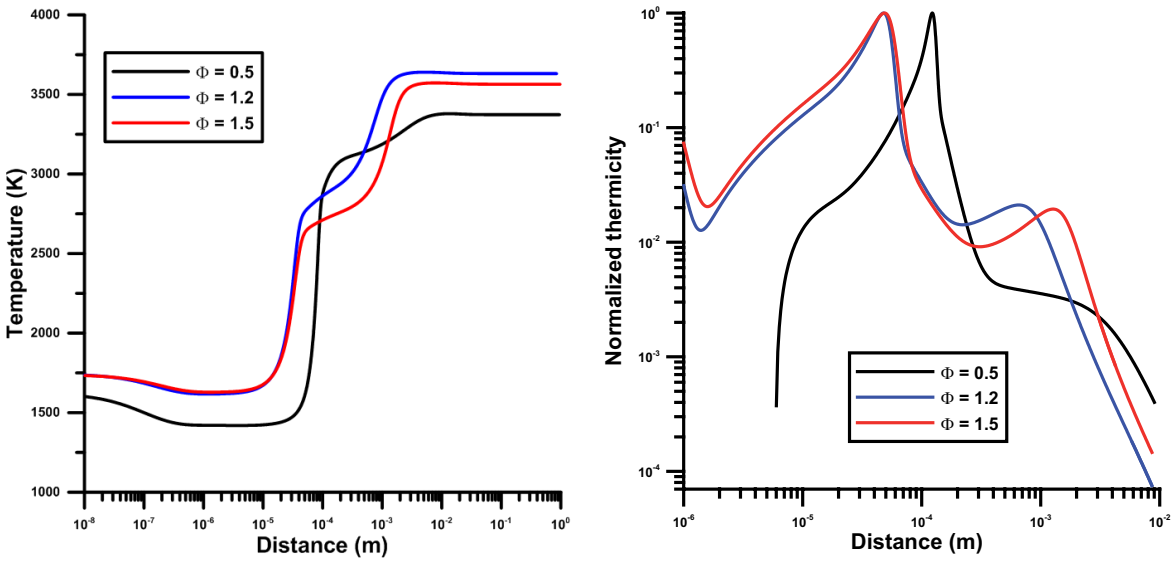


a) Mixture with N_2O b) Mixture with N_2O and O_2

Figure 2.46: Sensitivity coefficient on during the oxidation of ethane by nitrous oxide with and without oxygen. Taken from [47].

2.7.4 Example of application

To illustrate the use of the chemical analysis utilities of combustion software, let us consider the special case of H_2-NO_2/N_2O_4 mixtures. As illustrated in Figure 2.47, for these mixtures, the temperature profile is non-monotonous and a single step or a two steps of energy release can be observed depending on the equivalence ratio.



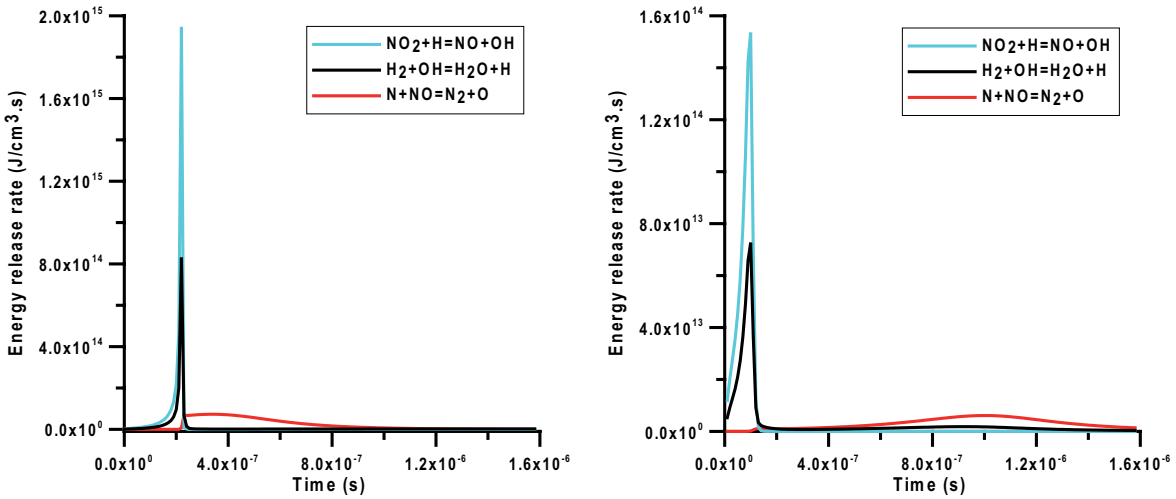
a) Temperature profiles

b) Thermicity profiles

Figure 2.47: Temperature and thermicity profiles in the ZND reaction zone of H_2-NO_2/N_2O_4 mixtures with different equivalence ratios. Initial conditions : Φ=0.5, 1.2, 1.5 ; T_1=293 K ; P_1=100 kPa. Taken from [48].

In order to explain such a complex behavior, it is first needed to identify which reactions are providing the energy to the system by performing an analysis of the rate of energy release per reaction. Figure 2.48 presents the results obtained for two mixtures with Φ=0.5 and Φ=1.5. Only the most important reactions are considered for clarity. The first step of energy release is due to the following linear chain : NO_2+H=NO+OH ; H_2+OH=H_2O+H. Following this first step, the N+NO=N_2+O reaction releases additional energy. In the lean mixture, the N+NO=N_2+O reaction becomes active just after the linear chain has been completed, whereas in the rich mixture, there is a discontinuity between the two steps of energy release.

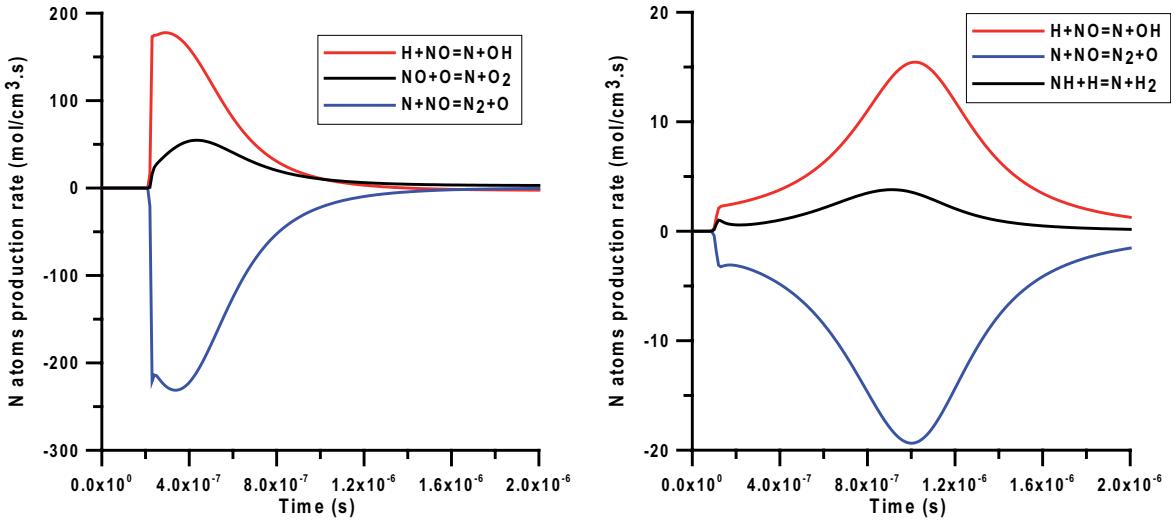
To explain the two distinct behaviours of the energy release, an analysis of the production rates has been performed for N atoms. Figure 2.49 shows the results obtained. For both lean, Φ=0.5, and rich, Φ=1.5, mixtures, the production of N atoms is driven by H+NO=N+OH, which has a high activation energy, and their consumption is driven by the N+NO=N_2+O



a) Lean mixture b) Rich mixture

Figure 2.48: Reaction energy release rate in lean and rich H_2-NO_2/N_2O_4 mixtures. Constant pressure reactor. Left : Φ=0.5 ; T_1=1622 K ; P_1=5.56 MPa. Right : Φ=1.5 ; T_1=1741 K ; P_1=4.49 MPa. Taken from [48].

reaction, which is a zero barrier process. However, the evolution of N atoms production is very different for the two mixtures. For the lean mixture, the production and the consumption are first very high and then decrease progressively. For the rich mixture, the production and the consumption are first very slow and then increase to reach a peak. This difference is due to the evolution of thermodynamic conditions between the two mixtures. In the lean mixtures, the temperature after the first step of energy release is above 3000 K, so that the system has released most of its energy. The temperature is high enough to allow the H+NO=N+OH reaction to proceed. The N atoms are formed and then react with NO, involving a smooth temperature increase to the CJ temperature. In the rich mixture, due to the lack of NO_2, the extent of the chain NO_2+H=NO+OH plus H_2+OH=H_2O+H is limited and the temperature after the first energy release is only around 2600 K. At this temperature, the production of N atoms by the H+NO=N+OH reaction is slow.



a) Lean mixture

b) Rich mixture

Figure 2.49: Rates of production of N atoms in lean and rich H_2-NO_2/N_2O_4 mixtures. Constant pressure reactor. Left : Φ=0.5 ; T_1=1622 K ; P_1=5.56 MPa. Right : Φ=1.5 ; T_1=1741 K ; P_1=4.49 MPa. Taken from [48].

In order to explain the existence of the second peak of energy release, the kinetic process, shown in Figure 2.50, has to be considered. Basically, from a single OH radical, four H atoms and thus four N atoms can be produced. As the rich mixture still contains a large amount of H_2, OH radicals are able to regenerate H atoms through the H_2+OH=H_2O+H reaction. Moreover, the N+NO=N_2+O reaction releases O atoms, which can react with H_2 to form OH and H. Consequently, the production of N atoms, and thus the energy release by N+NO=N_2+O is supported by a chain branching process. This process is characterized by an induction period followed by an exponential growth of active centers. It involves, after a delay, a fast energy release, which corresponds to the second thermicity peak.

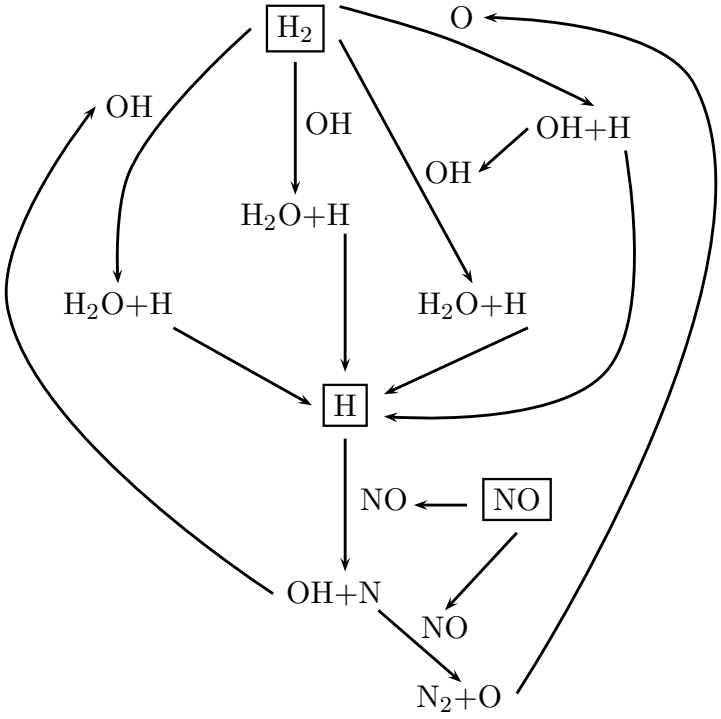


Figure 2.50: Chain branching process for N atoms production in rich hydrogen-nitrogen tetroxide mixture. Initial conditions : Φ=1.5 ; T_1=1741 K ; P_1=4.49 MPa. Taken from [48].

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